



US012408972B2

(12) **United States Patent**
Pearson

(10) **Patent No.:** **US 12,408,972 B2**

(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **METHODS OF STERILIZATION AND
TREATING INFECTION USING
IRREVERSIBLE ELECTROPORATION**

(71) Applicant: **AngioDynamics, Inc.**, Latham, NY
(US)

(72) Inventor: **Robert Pearson**, Sam Jose, CA (US)

(73) Assignee: **AngioDynamics, Inc.**, Latham, NY
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/404,641**

(22) Filed: **Jan. 4, 2024**

(65) **Prior Publication Data**

US 2024/0138903 A1 May 2, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/073,039, filed on
Oct. 16, 2020, now Pat. No. 11,957,405, which is a
continuation of application No. 15/864,421, filed on
Jan. 8, 2018, now abandoned, which is a continuation
of application No. 14/302,678, filed on Jun. 12, 2014,
now Pat. No. 9,895,189.

(60) Provisional application No. 61/834,471, filed on Jun.
13, 2013.

(51) **Int. Cl.**

A61B 18/12 (2006.01)

A61B 18/14 (2006.01)

A61B 18/00 (2006.01)

A61B 34/00 (2016.01)

(52) **U.S. Cl.**

CPC **A61B 18/1477** (2013.01); **A61B**
2018/00577 (2013.01); **A61B 2018/00613**
(2013.01); **A61B 2018/124** (2013.01); **A61B**
2018/1467 (2013.01); **A61B 2034/252**
(2016.02); **A61B 2034/254** (2016.02)

(58) **Field of Classification Search**

CPC **A61B 18/1477**; **A61B 2018/00577**; **A61B**
2018/00613; **A61B 2018/124**; **A61B**
2018/1467

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,258,249 B1 * 7/2001 Simpson **A61L 2/03**
205/687

8,267,884 B1 * 9/2012 Hicks **A61B 18/042**
604/23

2011/0160514 A1 * 6/2011 Long **A61B 18/1477**
606/41

* cited by examiner

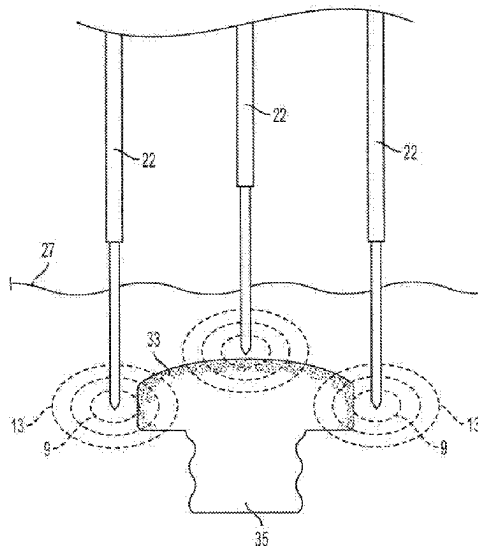
Primary Examiner — Daniel W Fowler

(74) *Attorney, Agent, or Firm* — Kevin P. Radigan, Esq.;
Heslin Rothenberg Farley & Mesiti P.C.

(57) **ABSTRACT**

A method for treating an infection using irreversible elec-
troporation is presented. The method includes providing an
ablation device that has at least one electrode and inserting
the ablation device into a target tissue of a patient. The tissue
at least partially surrounds the implanted medical device.
The outer surface of the medical device is at least partially
covered by infectious cells. The method also involves posi-
tioning at least one electrode in or near the implanted
medical device and delivering electrical pulses to or near the
implanted medical device sufficient to irreversibly elec-
troporate the infectious cells.

19 Claims, 38 Drawing Sheets



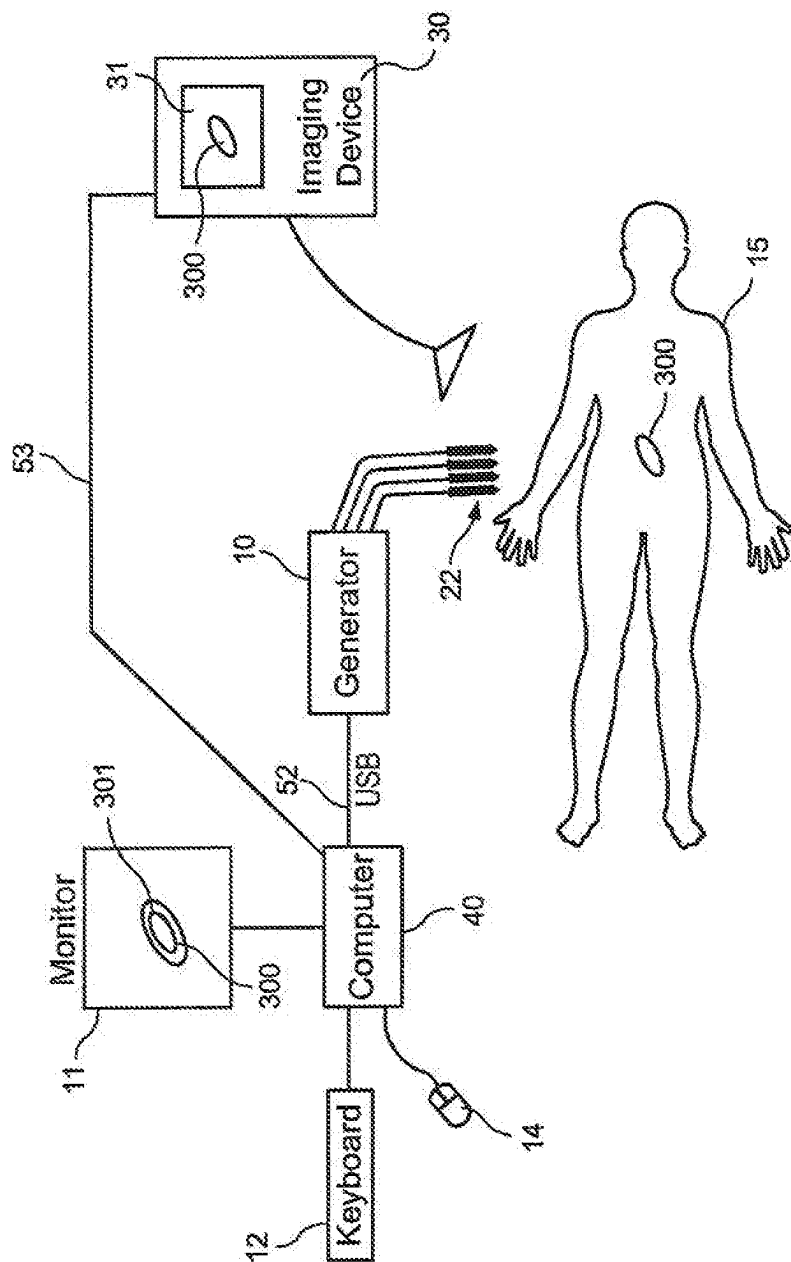


FIG. 1

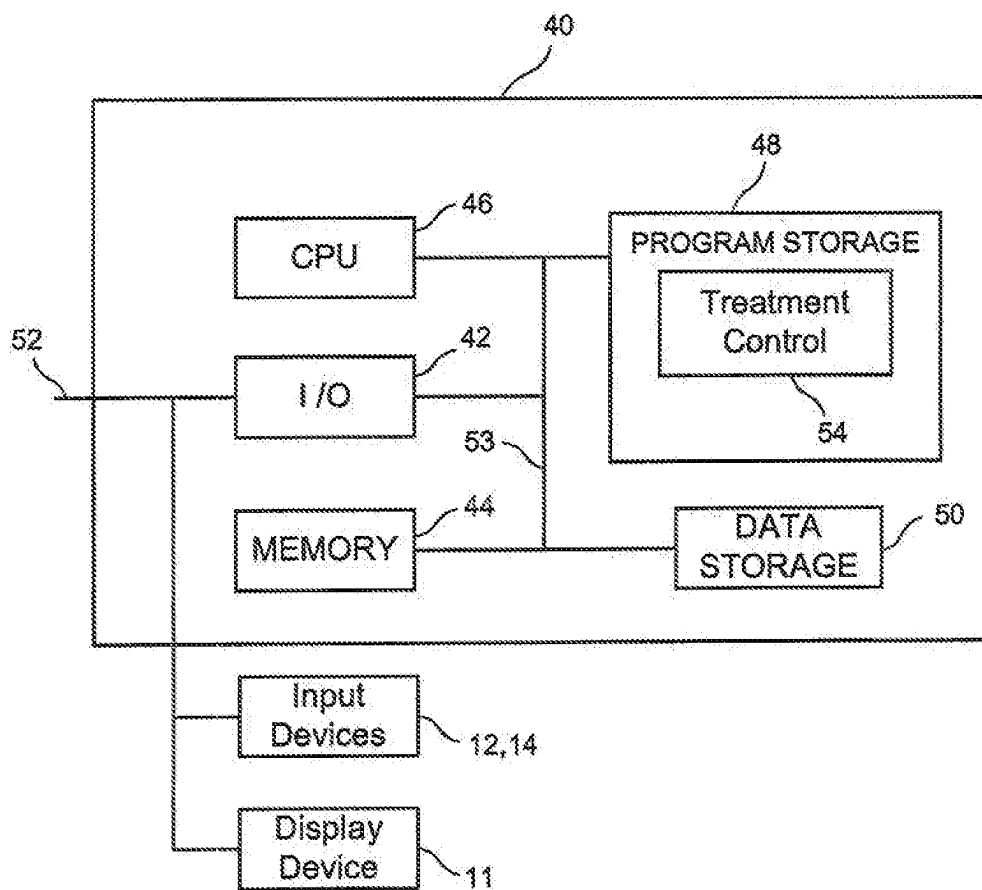


FIG. 2

Figure 1 is a schematic diagram of a medical device interface. The interface is divided into two main sections: a top section labeled "Information" (100) and a bottom section labeled "Clinical data" (110).

The "Information" section (100) contains the following fields:

- Patient ID: 999 (101)
- Name: John Smith (102)
- Age: 60 (103)
- Procedure date: 3/12/2010 4:05 PM (111)
- Physician name: (112)
- Case Notes: (113)

The "Clinical data" section (110) contains the following fields:

- Clinical Indication: (114)
- Lesion zone (cm) table (104):

Length	Width	Depth
2.0	1.0	1.0
4.0	3.0	3.0
- Target zone (cm) table (107):

Length	Width	Depth
4.0	3.0	3.0
- Margin: 1.0 (106)
- ECG Synchronization: (108)
- ECG Synchronization: (109)

The interface also includes a "NanoKnife" logo (115) and a "Settings" button (116).

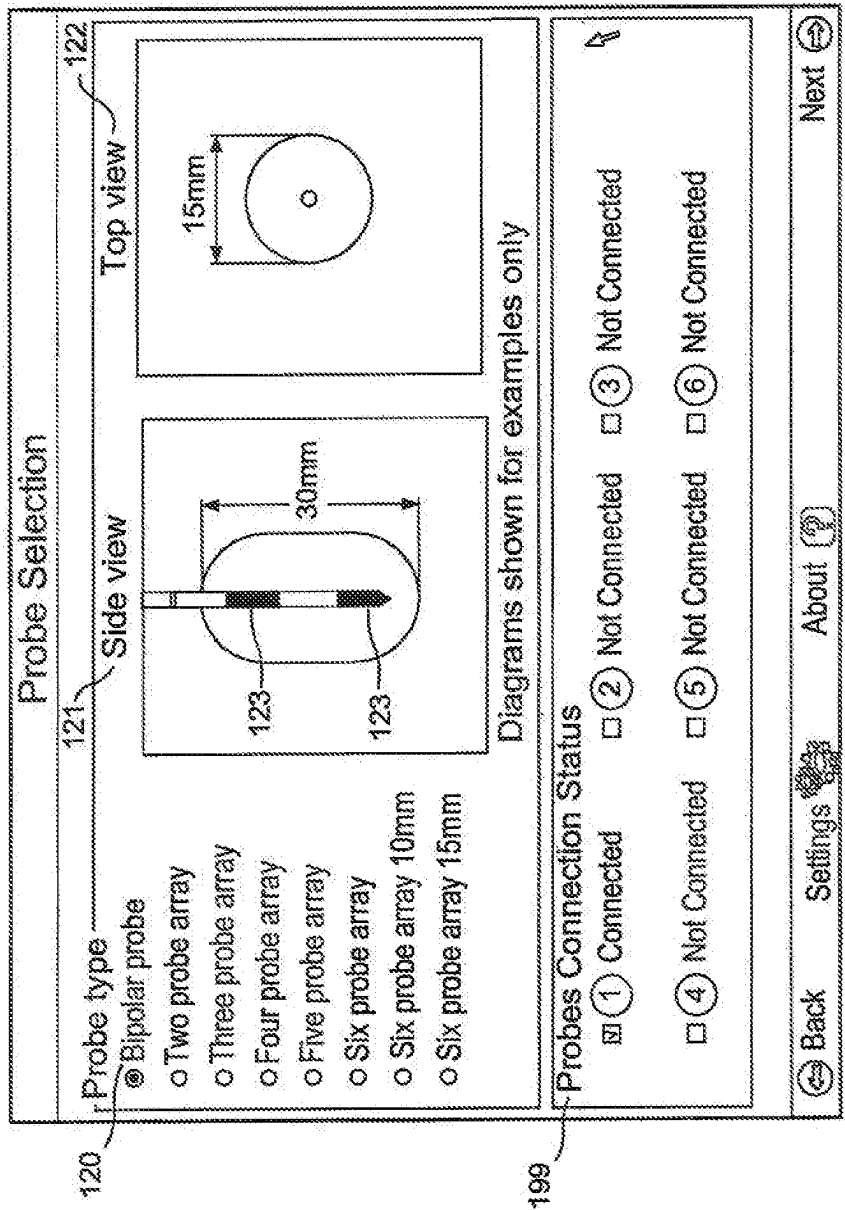


FIG. 4

Probe type

☐ Bipolar probe
 ☒ Two probe array
 ☐ Three probe array
 ☐ Four probe array
 ☐ Five probe array
 ☐ Six probe array
 ☐ Six probe array 10mm
 ☐ Six probe array 15mm

Side view

Top view

Diagrams shown for examples only

Probes Connection Status

☒ ① Connected
 ☒ ② Connected
 ☐ ③ Not Connected
 ☐ ④ Not Connected
 ☐ ⑤ Not Connected
 ☐ ⑥ Not Connected

Back

Settings

About

Next

FIG. 5

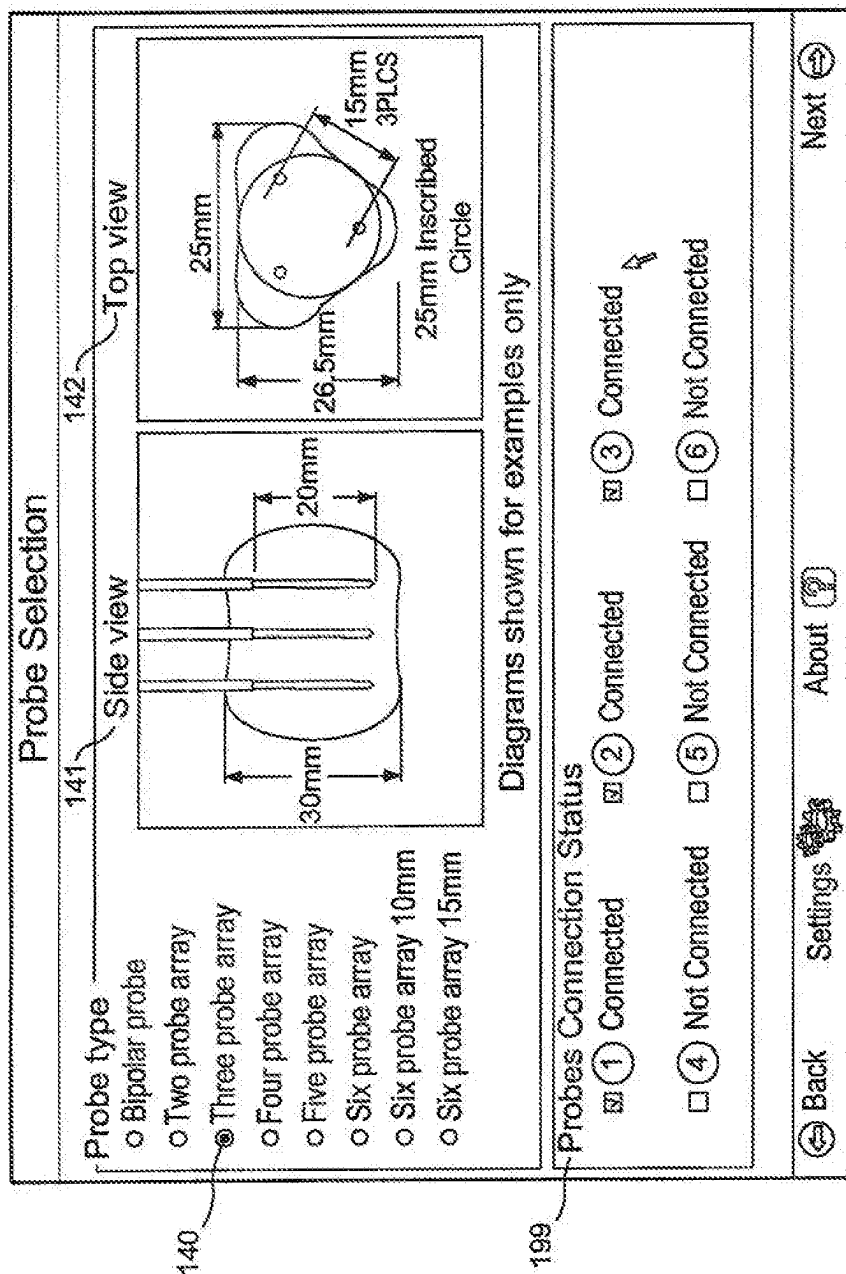


FIG. 6

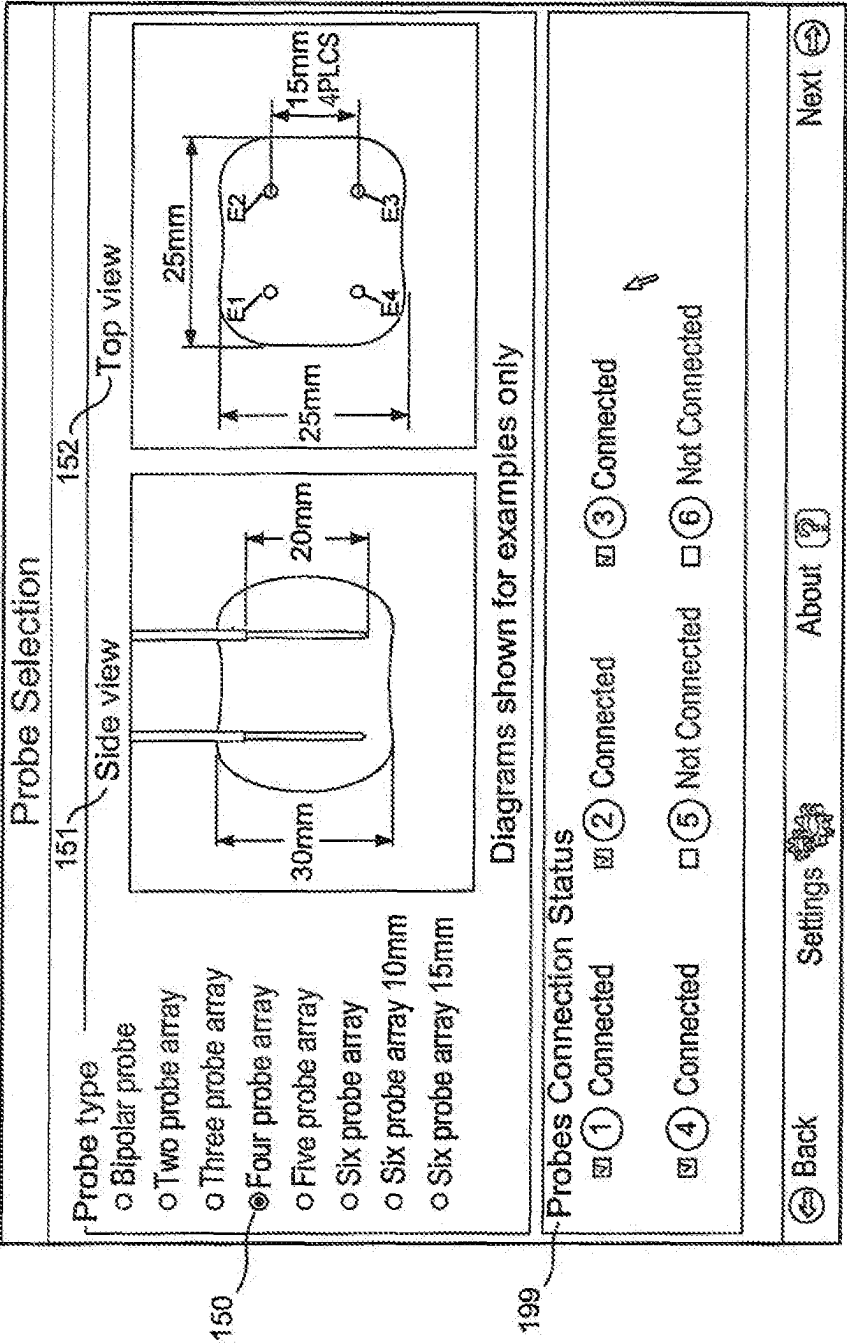


FIG. 7

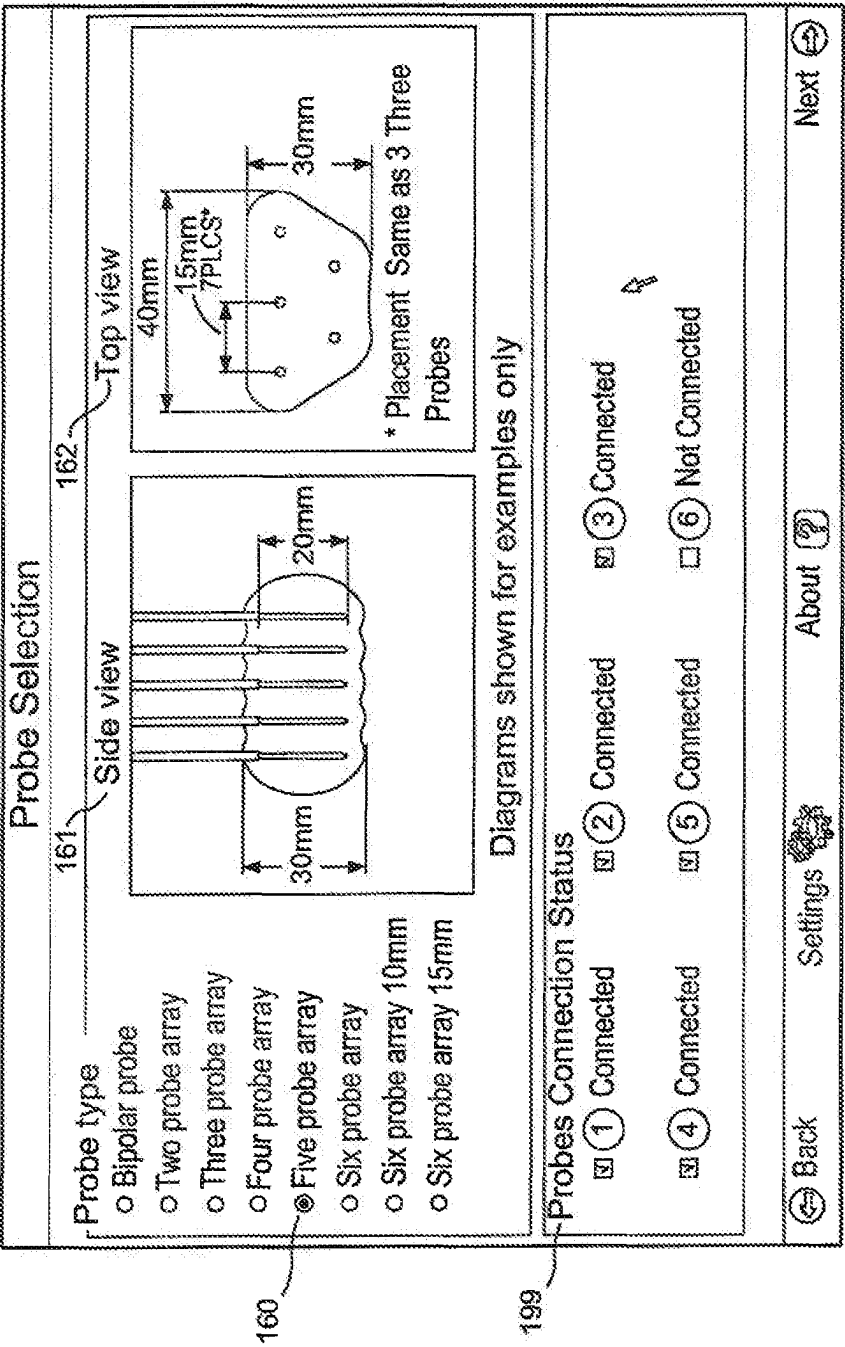


FIG. 8

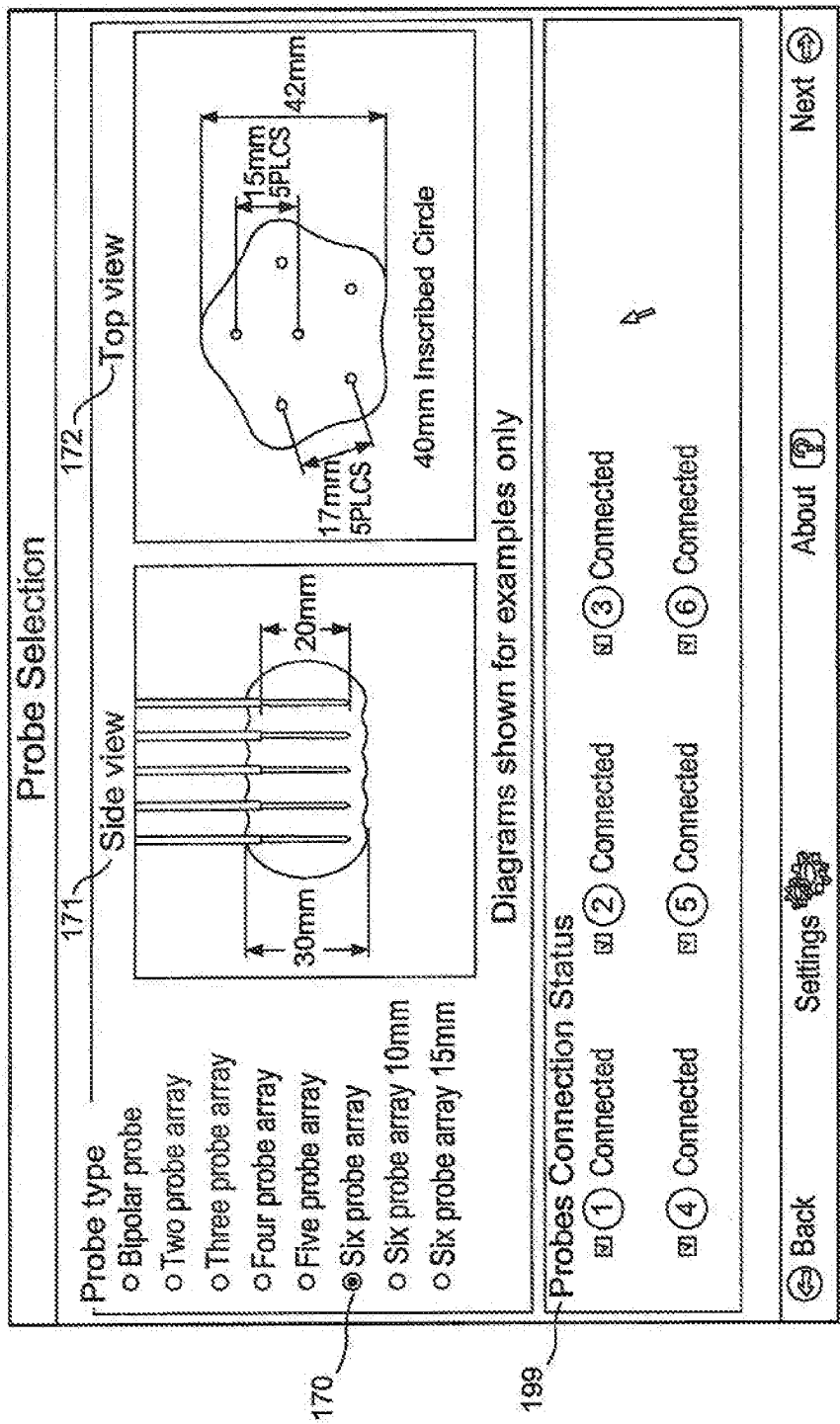


FIG. 9

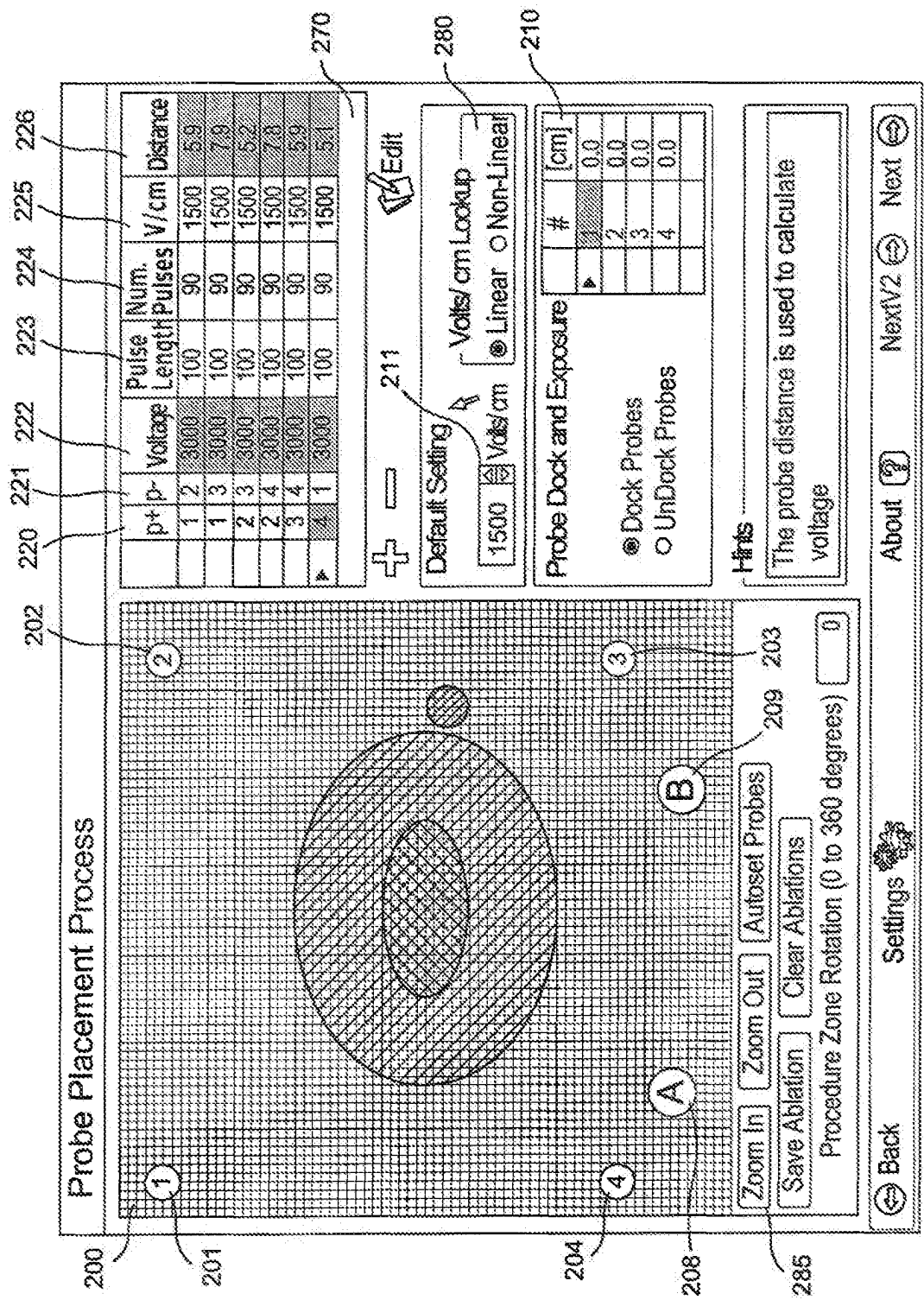


FIG. 10

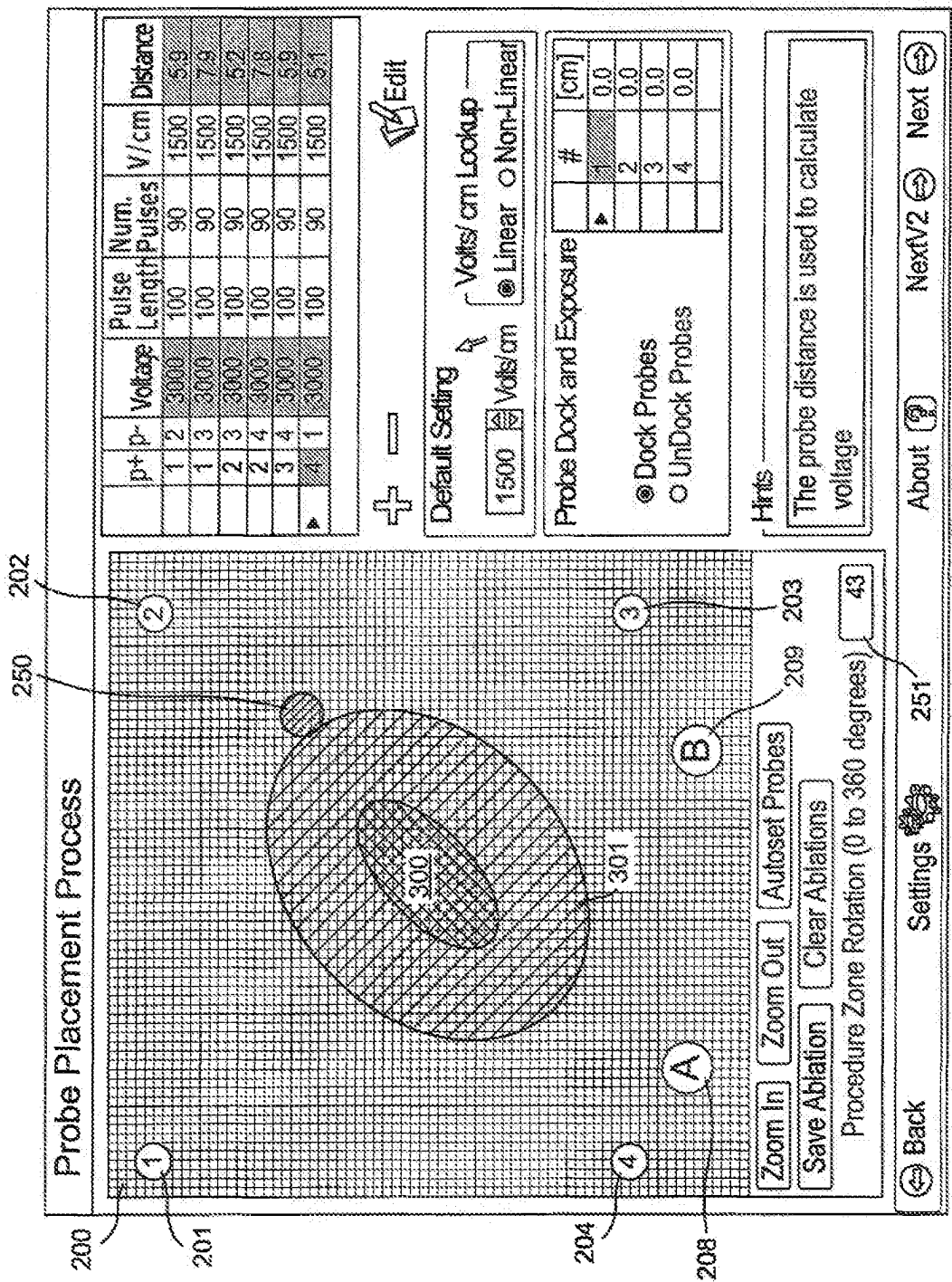


FIG. 11

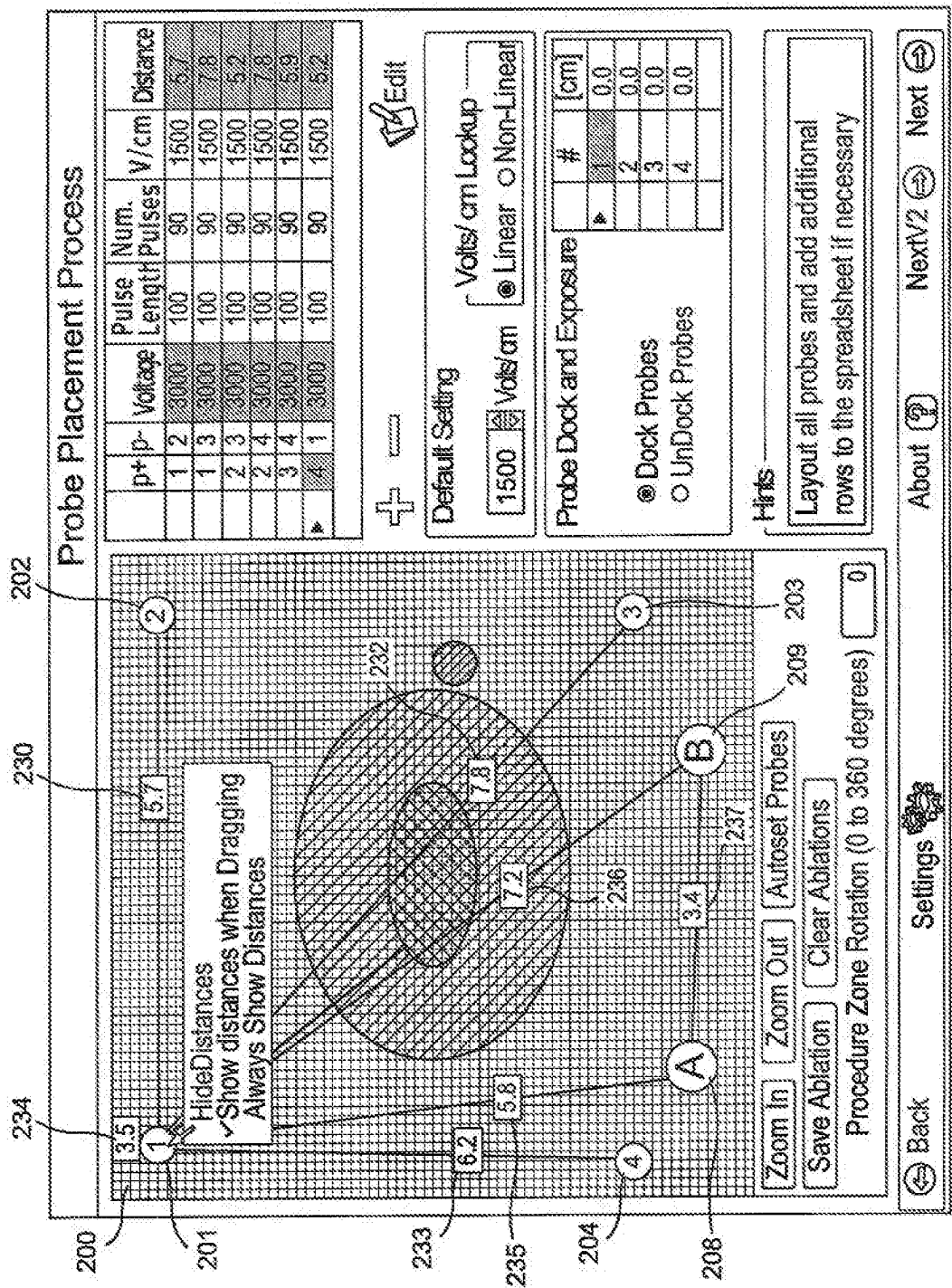


FIG. 12

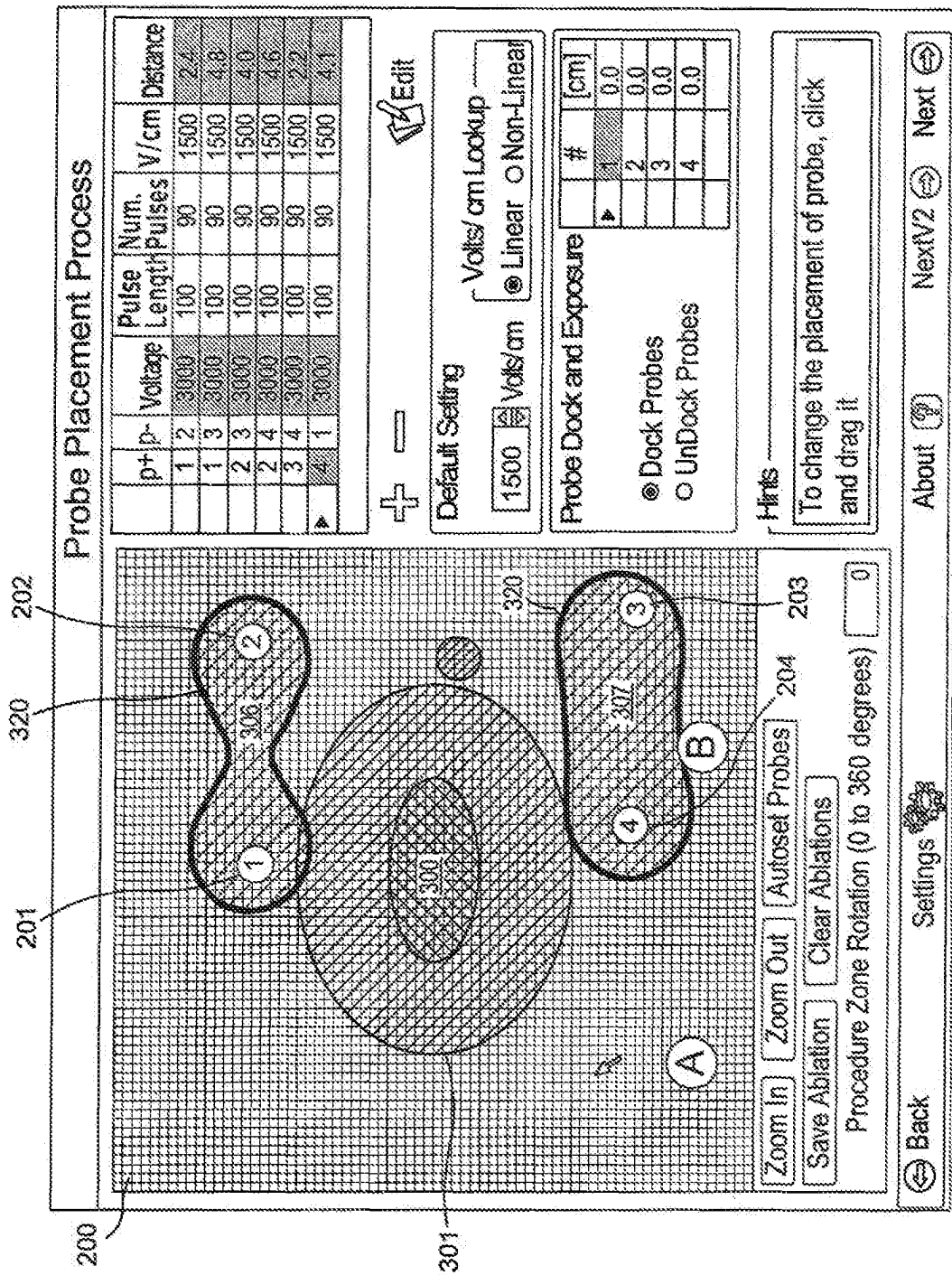


FIG. 13

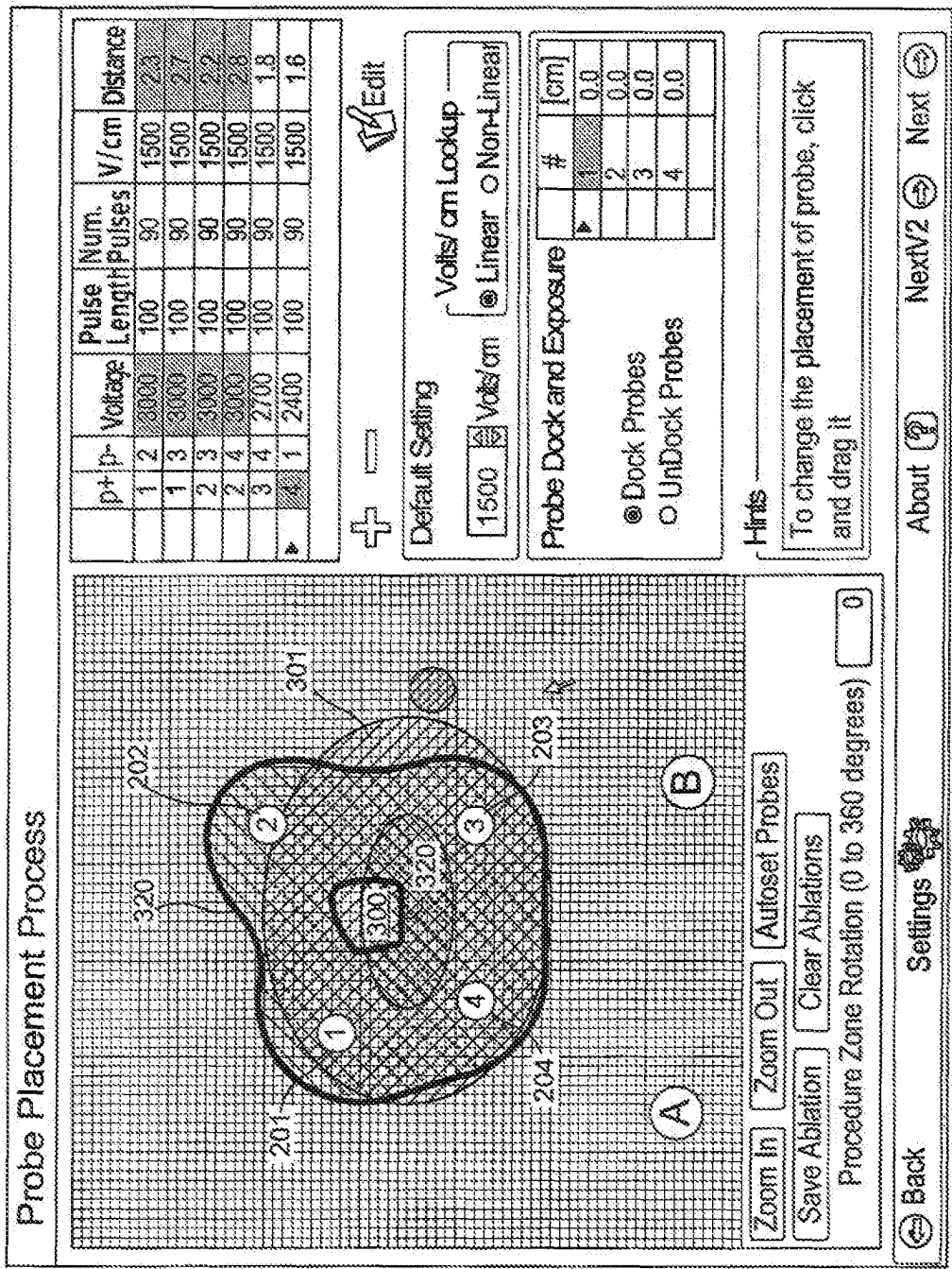


FIG. 14

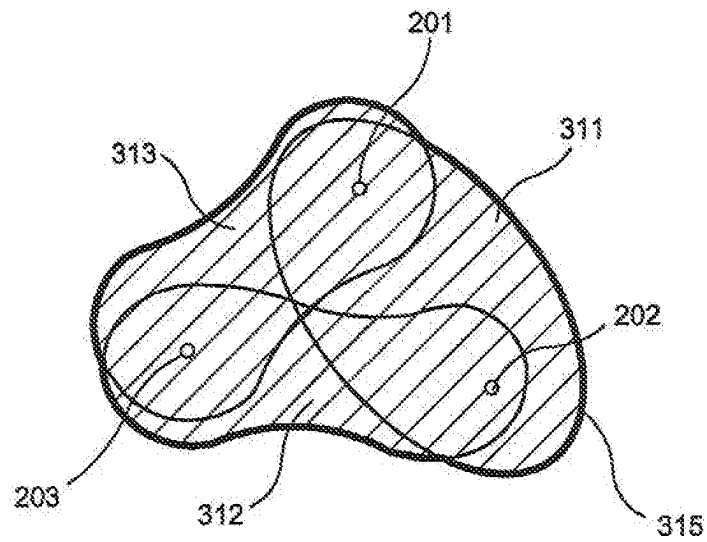


FIG. 15

$\frac{y}{x}$	-0010	-0008	-0006	-0004	-0003	-0002	-0001	0000	0001	0002	0003	0004	0005	0006	0007	0008	0009	0010
-0010	657	682	705	725	742	757	769	787	801	821	831	847	863	879	895	915	924	937
-0009	703	734	763	788	809	826	845	862	877	898	917	937	957	979	1000	1015	1032	1049
-0008	755	794	831	864	890	908	917	924	937	957	979	1000	1021	1042	1063	1085	1105	1127
-0007	814	864	912	955	980	1014	1024	1024	1021	1007	985	957	924	889	852	814	775	737
-0006	880	944	1009	1069	1118	1149	1160	1151	1128	1096	1057	1015	971	924	877	830	785	740
-0005	953	1039	1129	1215	1287	1330	1340	1319	1279	1230	1177	1122	1065	1008	950	892	837	784
-0004	1033	1147	1276	1409	1523	1589	1591	1541	1469	1393	1320	1248	1177	1105	1032	962	894	831
-0003	1114	1267	1455	1673	1883	2001	1970	1845	1706	1558	1409	1248	1085	924	763	602	441	280
-0002	1190	1388	1660	2038	2488	2785	2610	2260	1990	1815	1683	1588	1479	1361	1241	1127	1024	932
-0001	1248	1487	1852	2479	3685	5049	3828	2767	2289	2067	1945	1837	1702	1541	1376	1225	1094	983
0000	1276	1533	1945	2736	5016	#DIV/0!	5194	3104	2523	2338	2289	2217	2036	1784	1538	1330	1164	1030
0001	1266	1510	1883	2521	3744	5135	3959	2974	2632	2643	2853	2946	2610	2118	1719	1433	1235	1069
0002	1225	1433	1719	2118	2610	2946	2853	2643	2632	2674	3059	3135	3744	4521	5583	6510	7266	7883
0003	1164	1330	1538	1784	2036	2217	2289	2338	2523	3104	5194	#DIV/0!	5016	2736	1945	1533	1276	1097
0004	1094	1225	1376	1541	1702	1837	1945	2067	2289	2767	3828	5049	3685	2479	1852	1487	1248	1079
0005	1024	1127	1241	1361	1479	1588	1693	1815	1990	2260	2610	2785	2498	2038	1660	1388	1190	1042
0006	957	1040	1128	1220	1311	1400	1489	1580	1706	1845	1970	2001	1883	1673	1455	1267	1114	991
0007	894	962	1032	1105	1177	1248	1320	1393	1469	1541	1591	1589	1523	1409	1276	1147	1033	934
0008	837	892	950	1008	1065	1122	1177	1230	1279	1319	1340	1330	1287	1215	1129	1039	953	875
0009	785	830	877	924	971	1015	1057	1096	1128	1151	1160	1149	1118	1069	1009	944	880	818
0010	737	775	814	852	889	924	957	985	1007	1021	1024	1014	980	955	912	864	814	765

FIG. 16

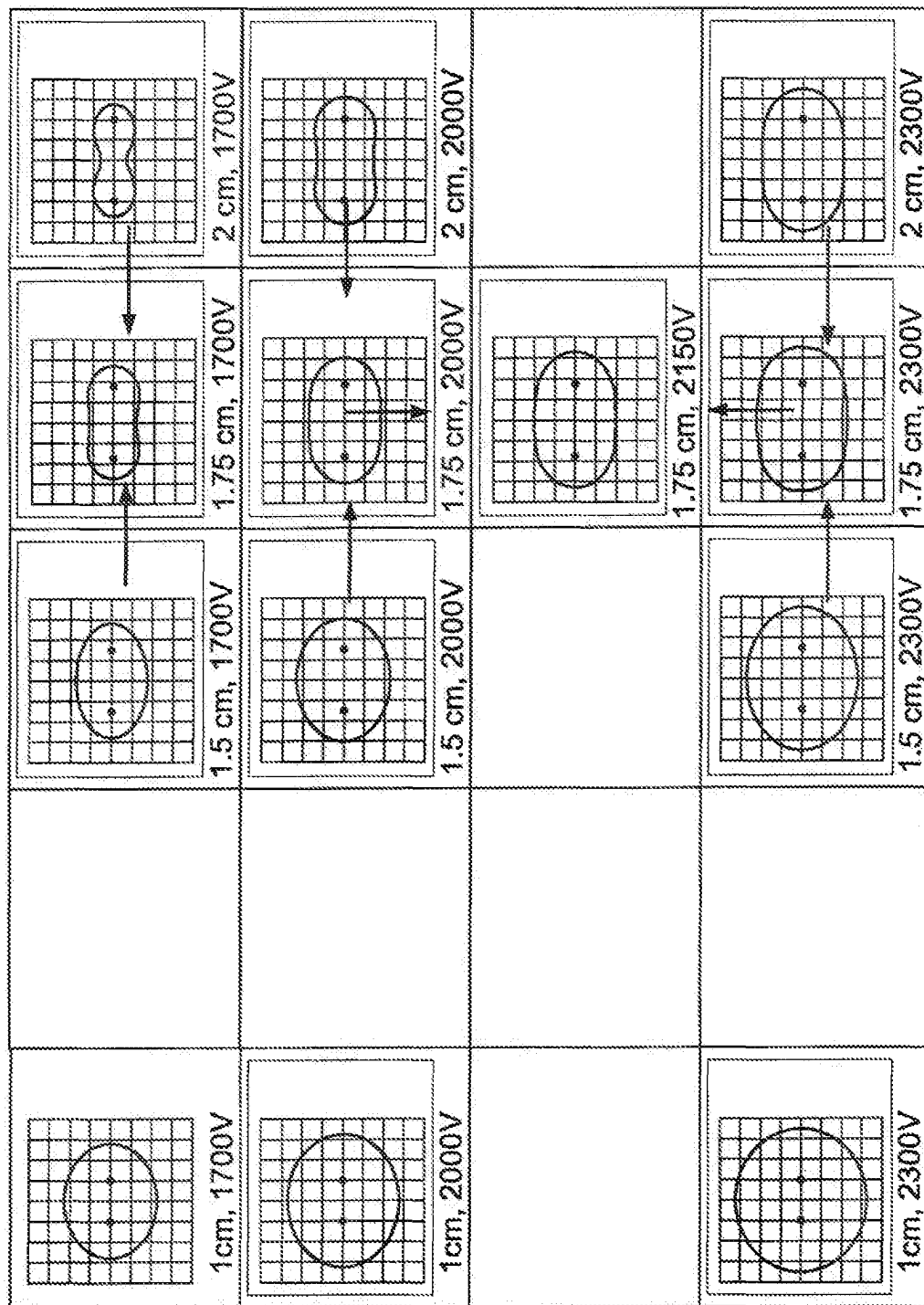


FIG. 17

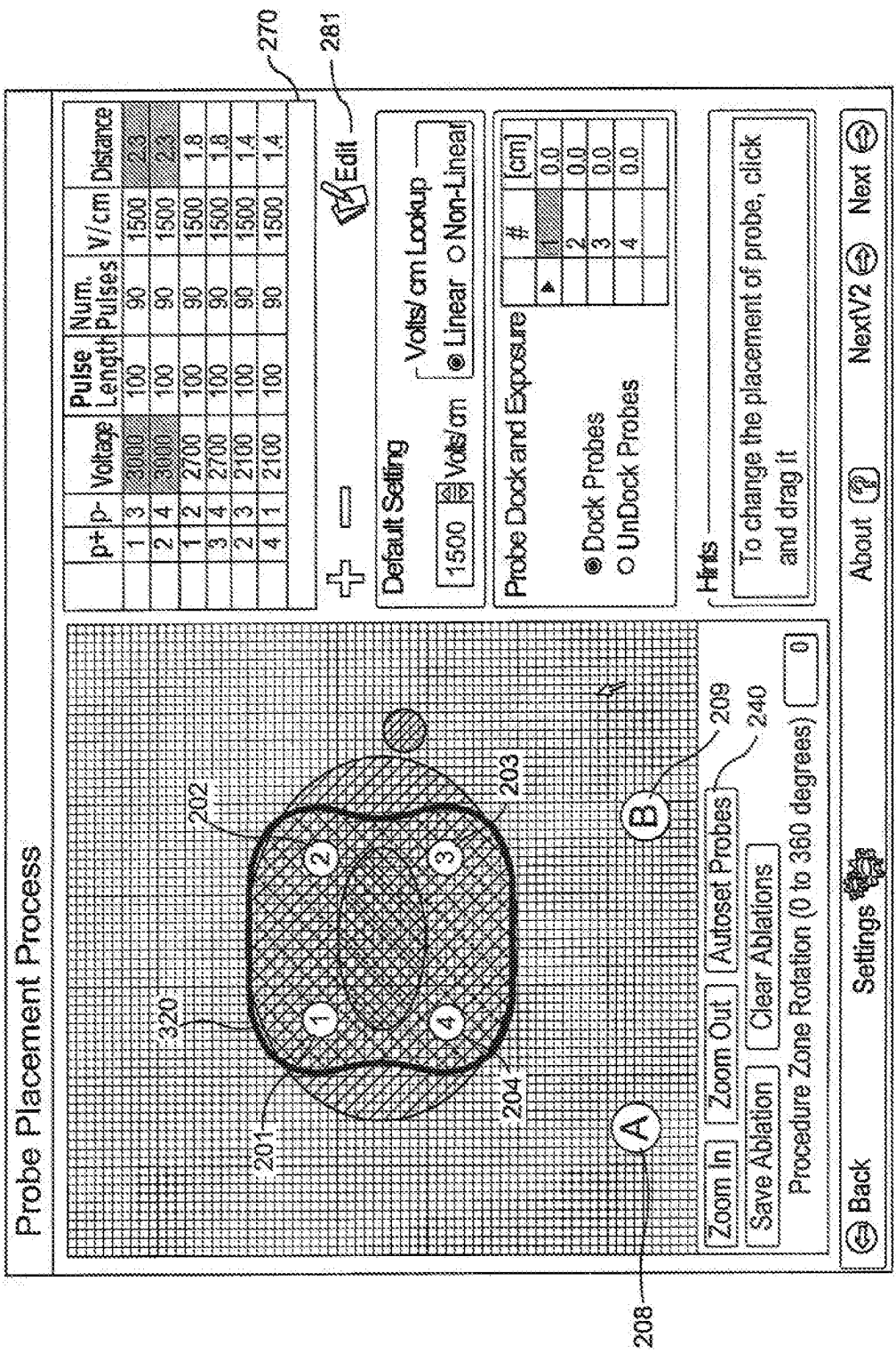


FIG. 18

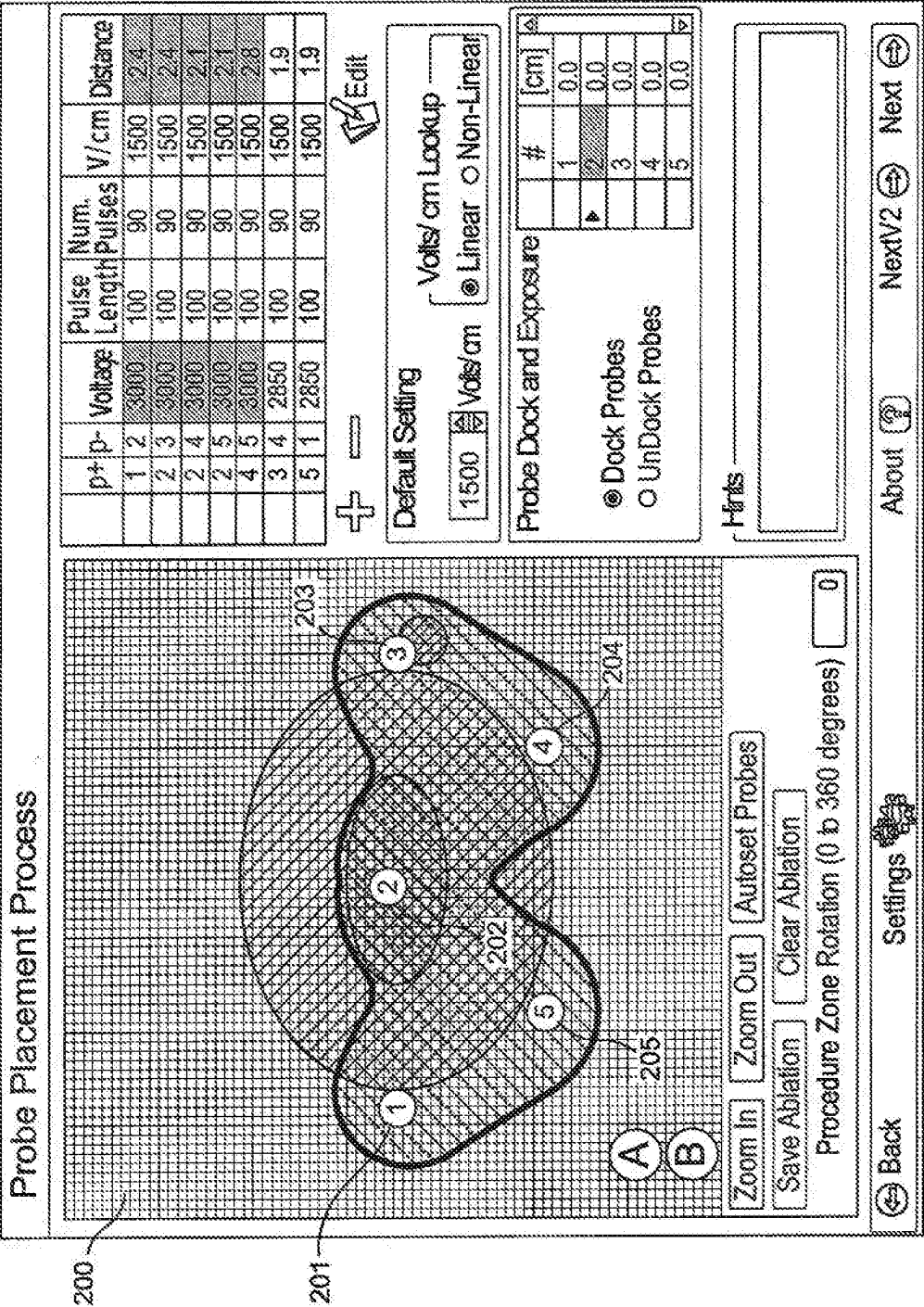


FIG. 19

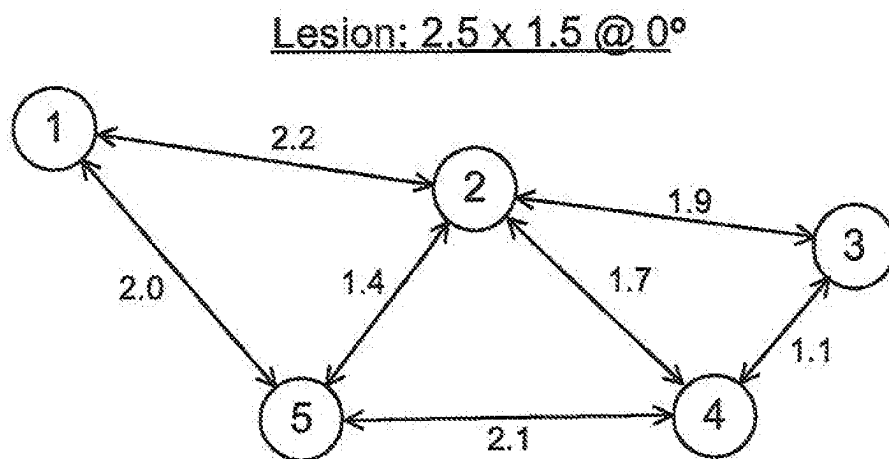


FIG. 20

333

Probe Distance Adjuster					
Locked	2	3	4	5	6
☐ 1	22.0			20.0	
☒ 2		19.0	17.0	14.0	
☐ 3			11.0		
☐ 4				21.0	
☐ 5					
☐ 6					

OK Cancel

FIG. 21

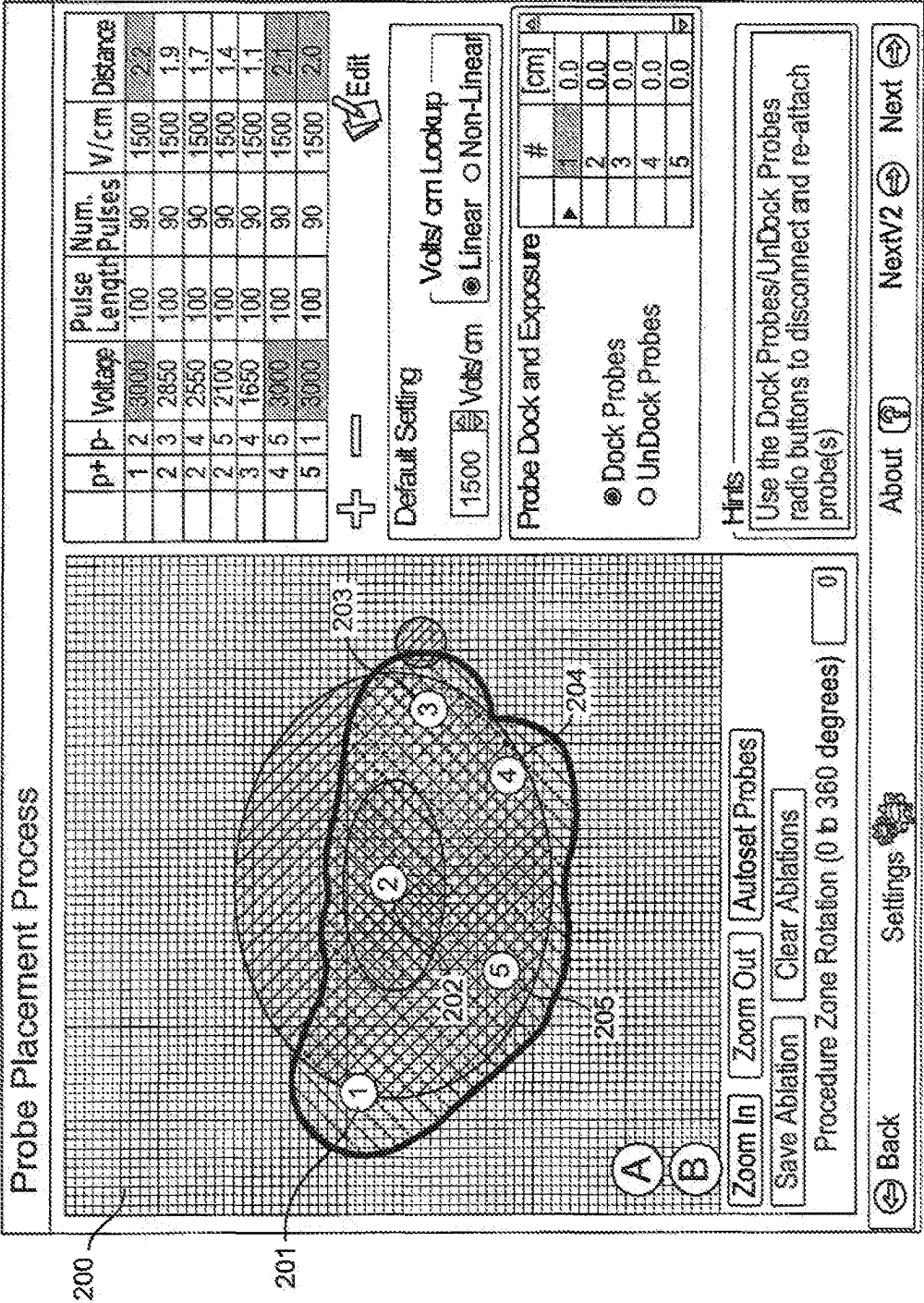


FIG. 22

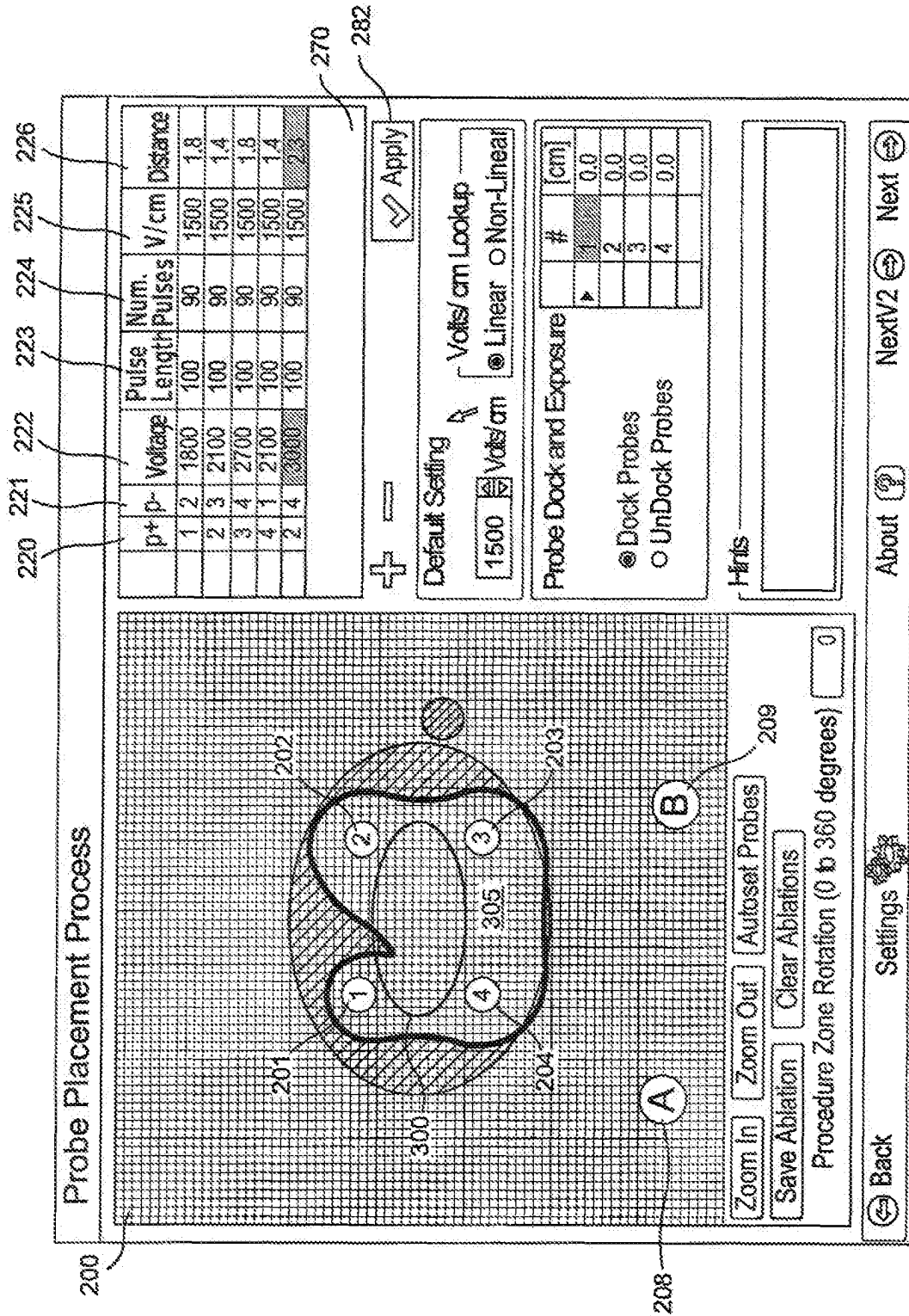


FIG. 23

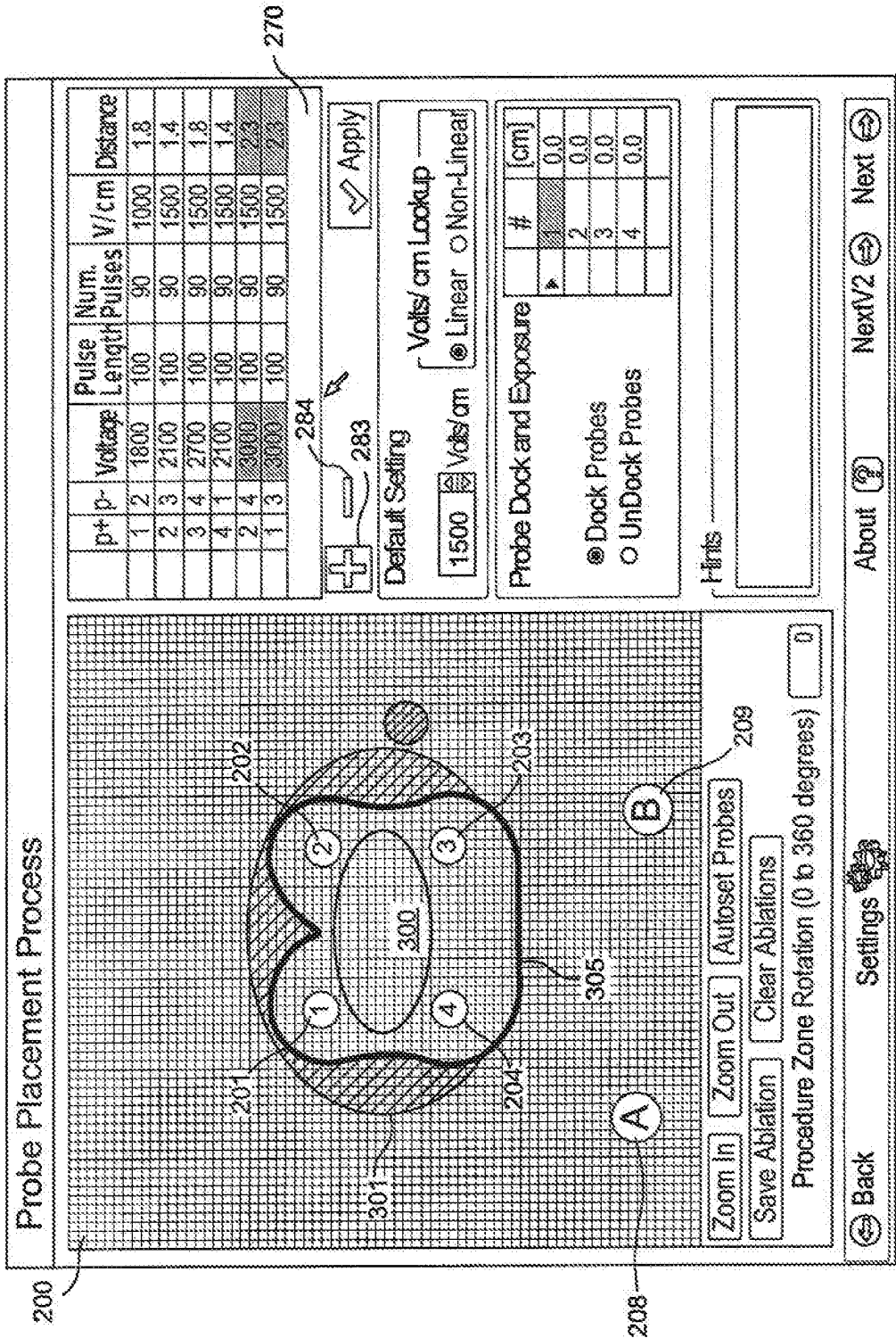


FIG. 24

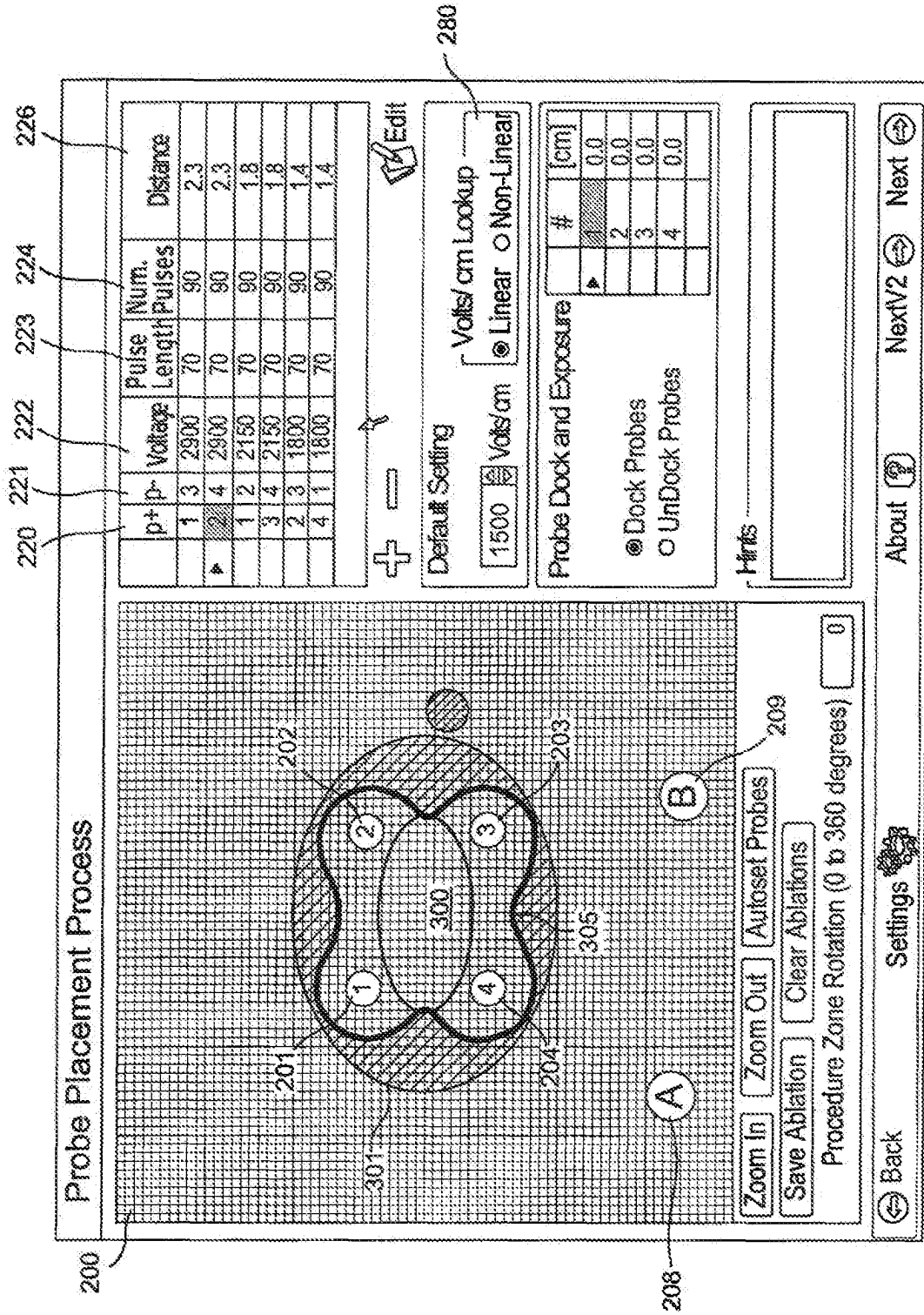


FIG. 25

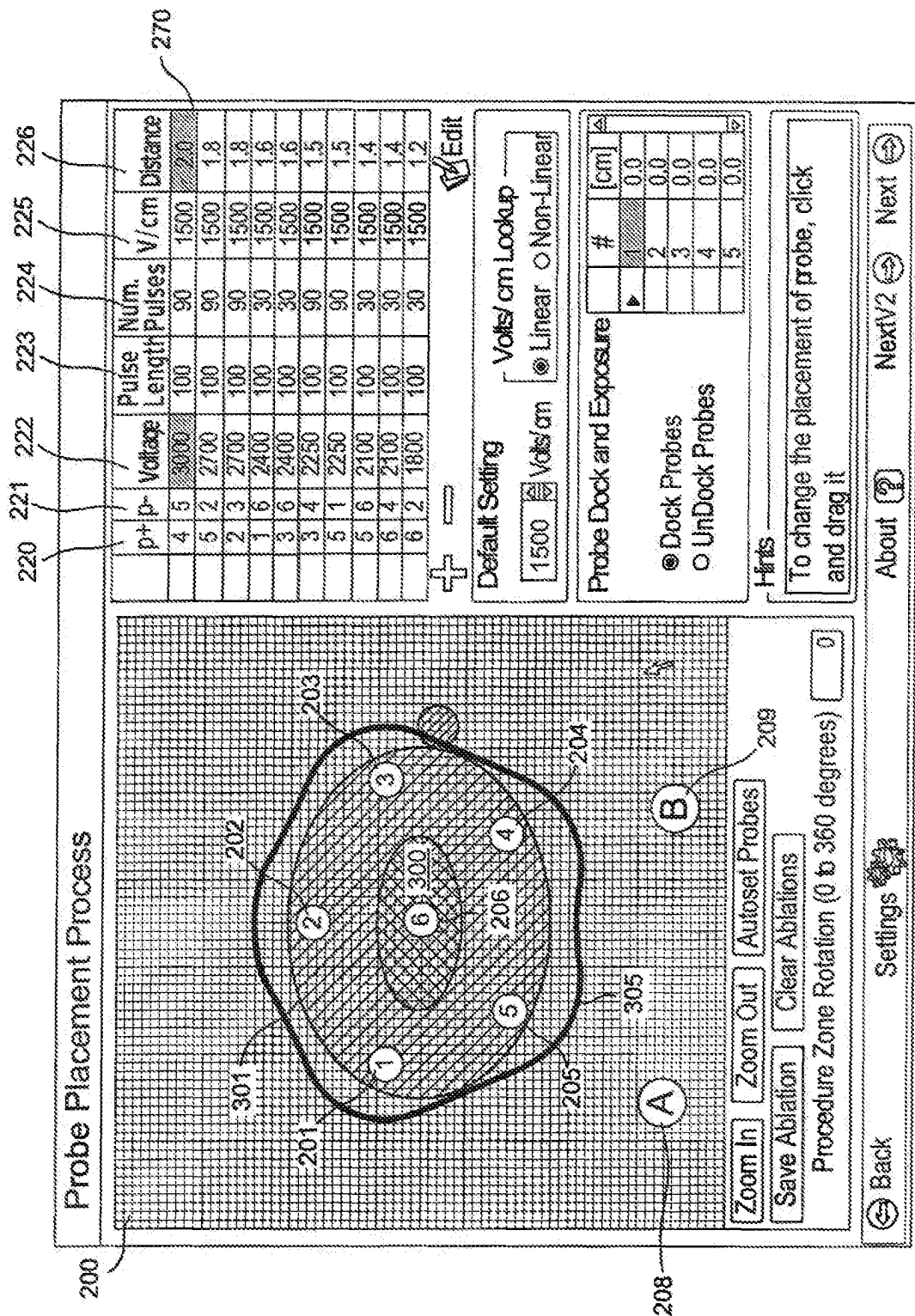


FIG. 26

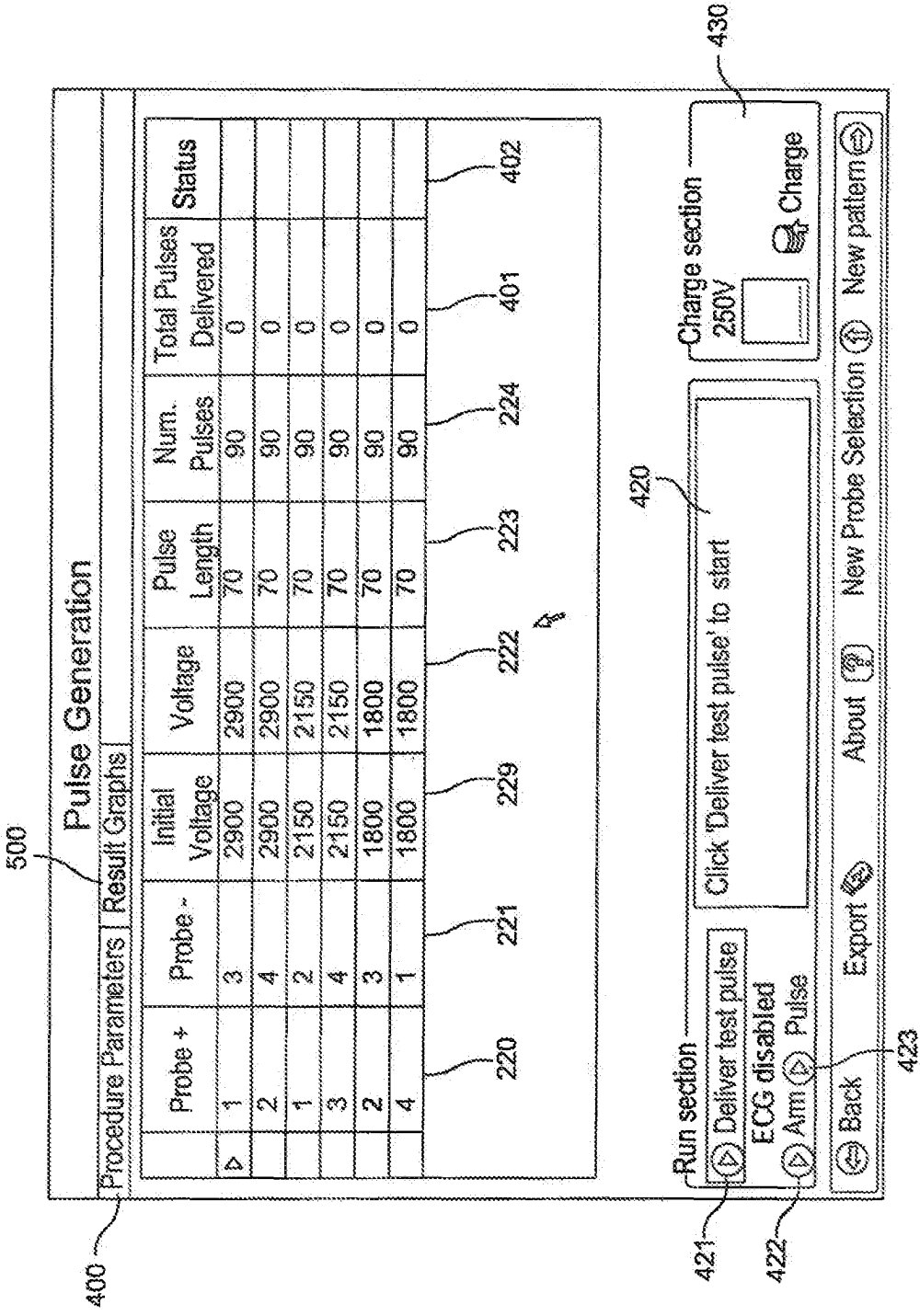


FIG. 27

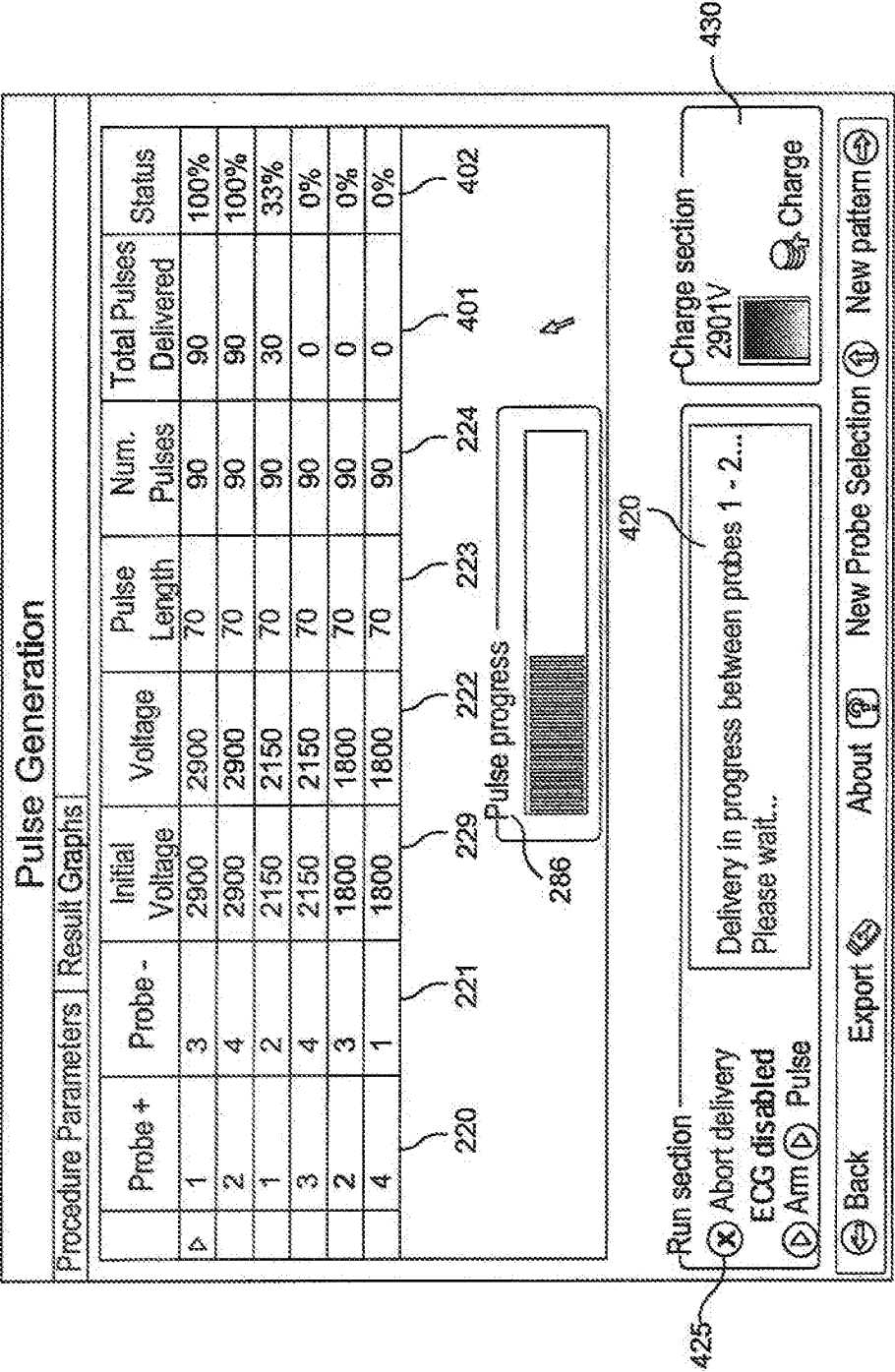


FIG. 28

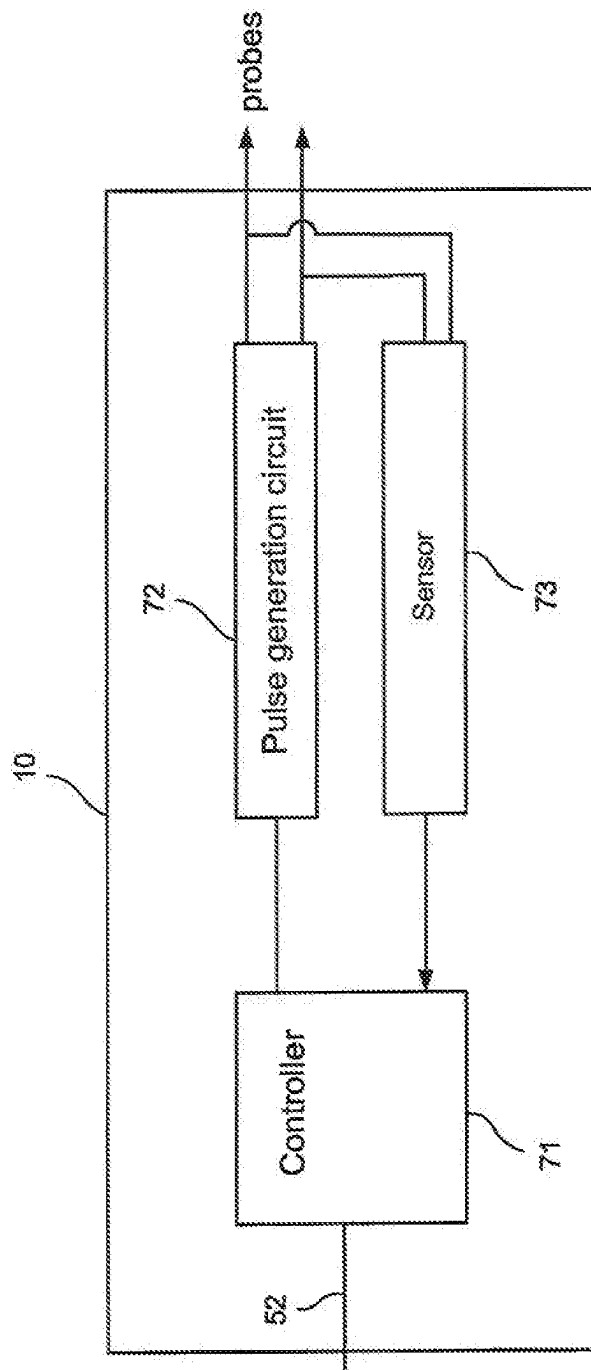


FIG. 29

Pulse Generation							
Procedure Parameters			Result Graphs				
Probe +	Probe -	Initial Voltage	Voltage	Pulse Length	Num. Pulses	Total Pulses Delivered	Status
1	3	3000	3000	100	90	90	Completed
2	4	3000	3000	100	90	20	High Current
1	2	2700	2700	100	90	90	Completed
3	4	2700	2700	100	90	90	Completed
2	3	2100	2100	100	90	90	Completed
4	1	2100	2100	100	90	90	Completed

220
221
229
222
223
224
401
402

426

427

Run section

Abort delivery
ECG disabled

Arm Pulse

Continue Procedure

Stop Procedure

Delivery completed

Charge section
2998V

Charge

Back

Export

About

New Probe Selection

New pattern

FIG. 30

Pulse Generation							
Procedure Parameters				Result Graphs			
Probe +	Probe -	Initial Voltage	Voltage	Pulse Length	Num. Pulses	Total Pulses Delivered	Status
1	3	3000	3000	100	90	90	Completed
2	4	3000	3000	100	90	20	High Current
1	2	2700	2700	100	90	90	Completed
3	4	2700	2700	100	90	90	Completed
2	3	2100	2100	100	90	90	Completed
4	1	Continue Procedure				90	Completed

? Adjust voltages for high current segments?

Yes No Cancel

428

Run section

☒ Abort delivery

☒ ECG disabled

☒ Arm ☐ Pulse

☐ Back

☐ Export

☐ About

☐ New Probe Selection

☐ New pattern

Delivery completed

Charge section

2998V

☐ Charge

☐ Continue Procedure

☐ Stop Procedure

FIG. 31

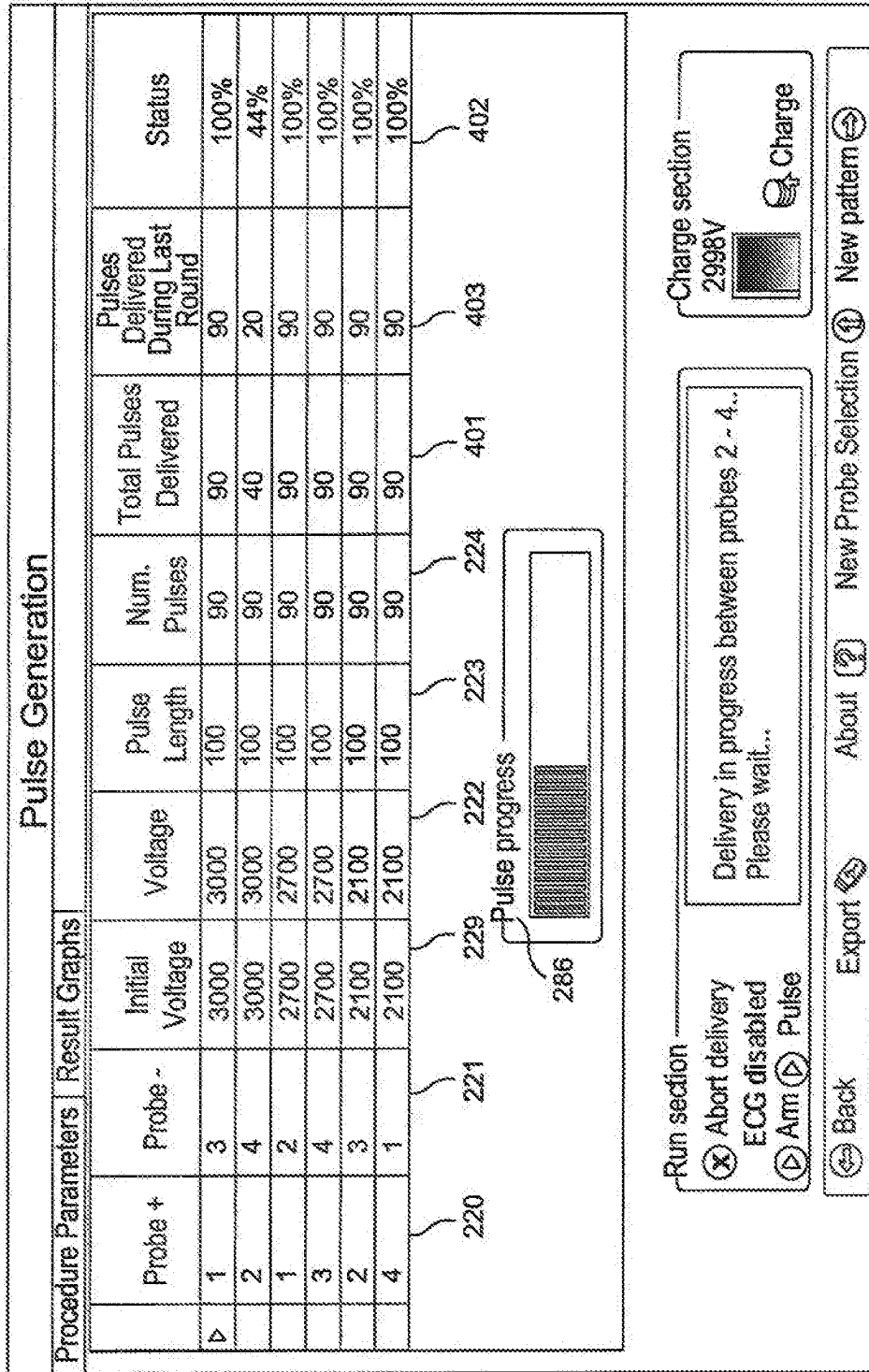


FIG. 32

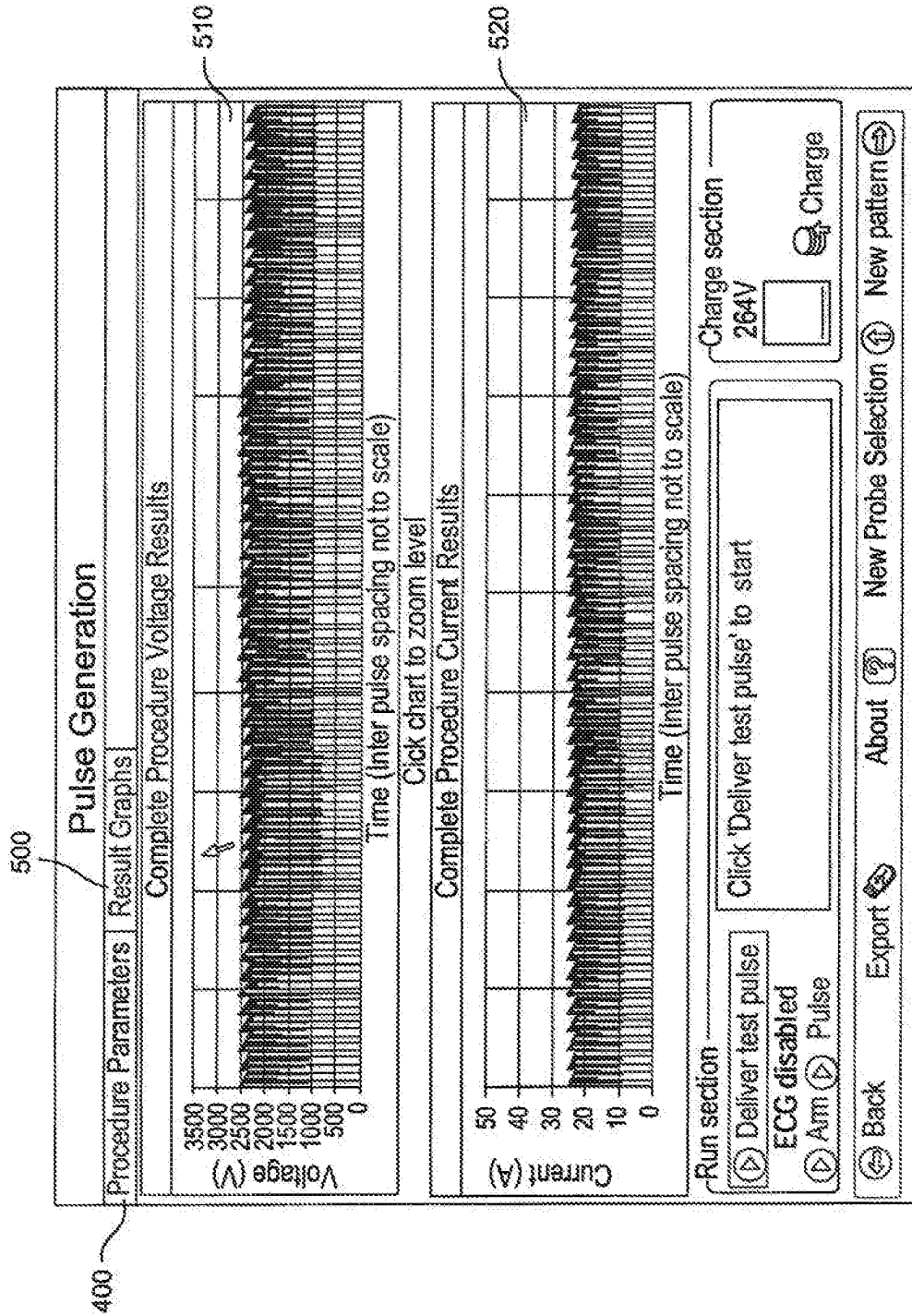


FIG. 33

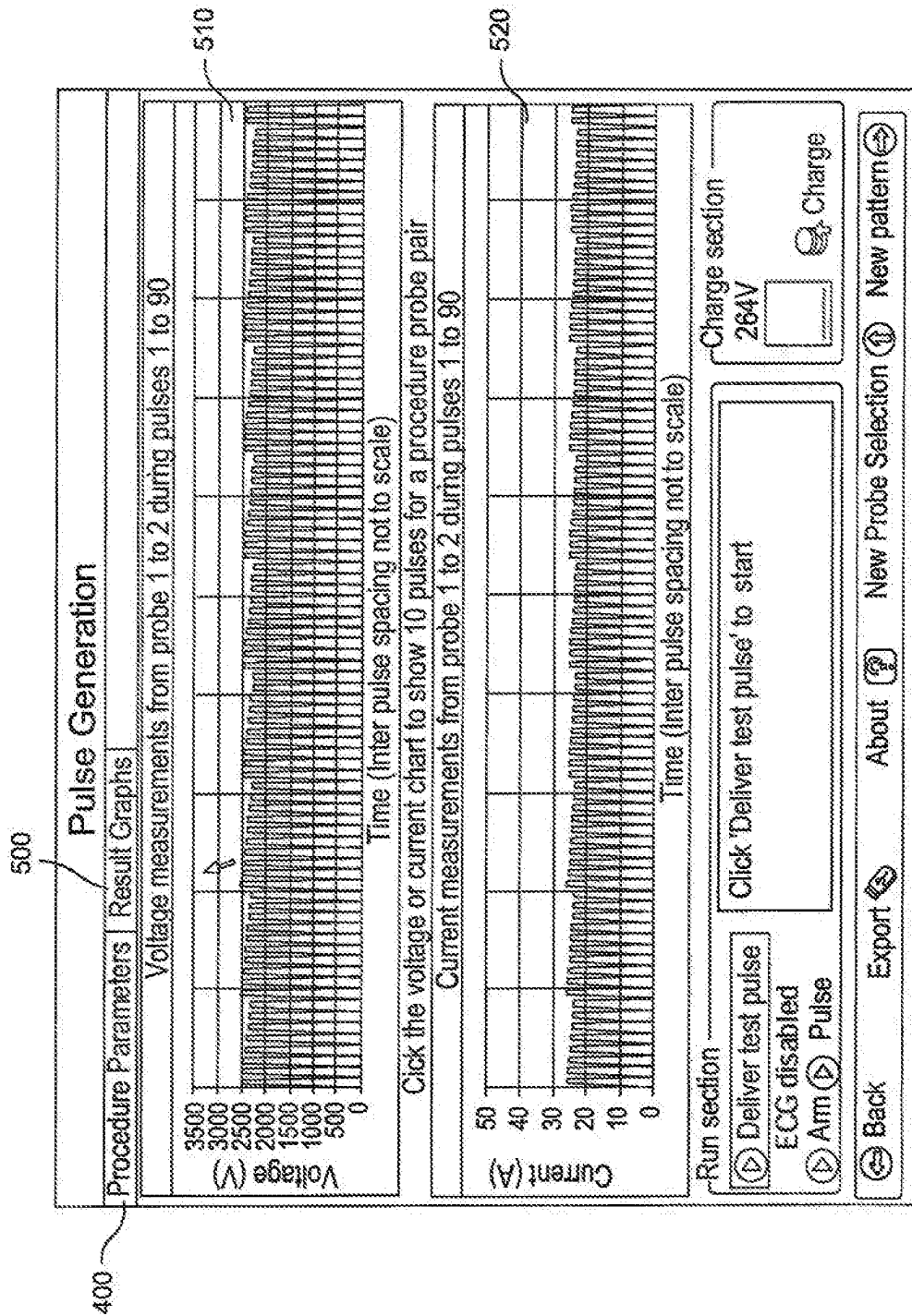


FIG. 34

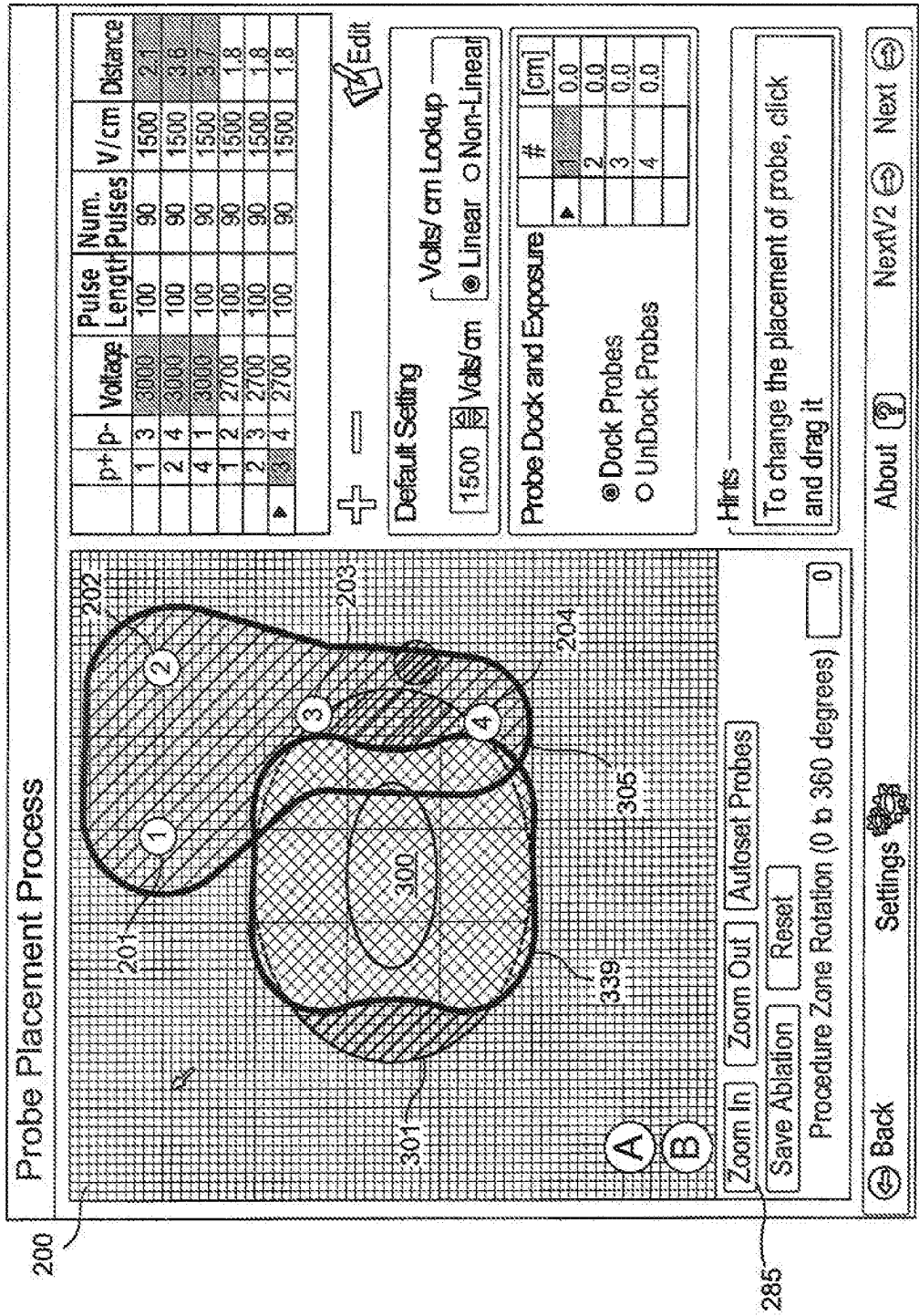


FIG. 35

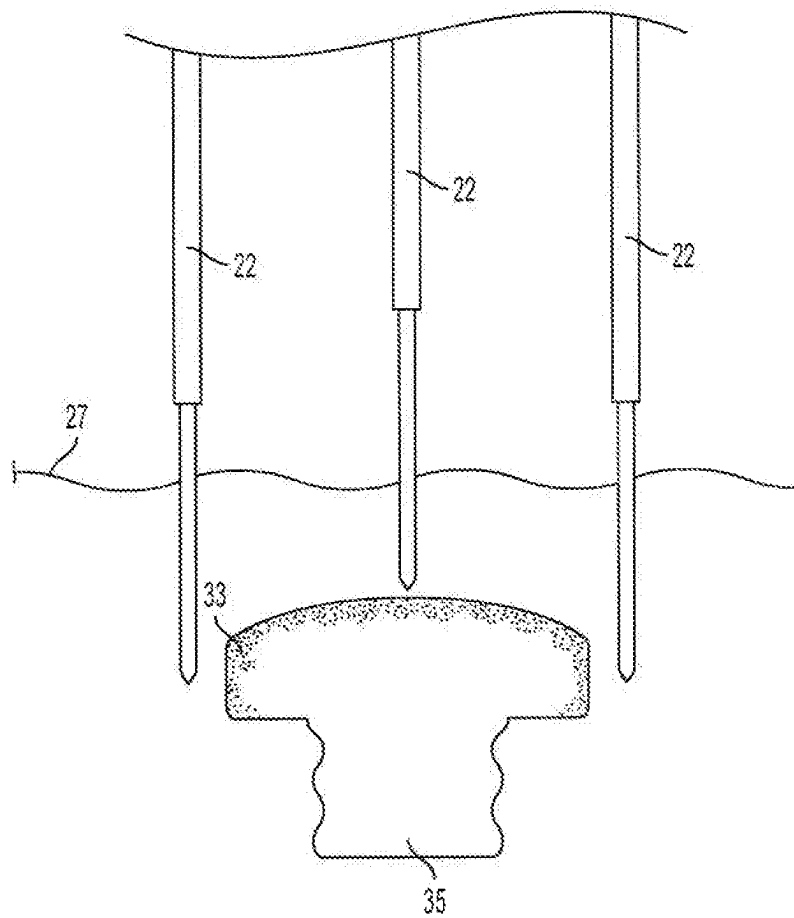


FIG. 36A

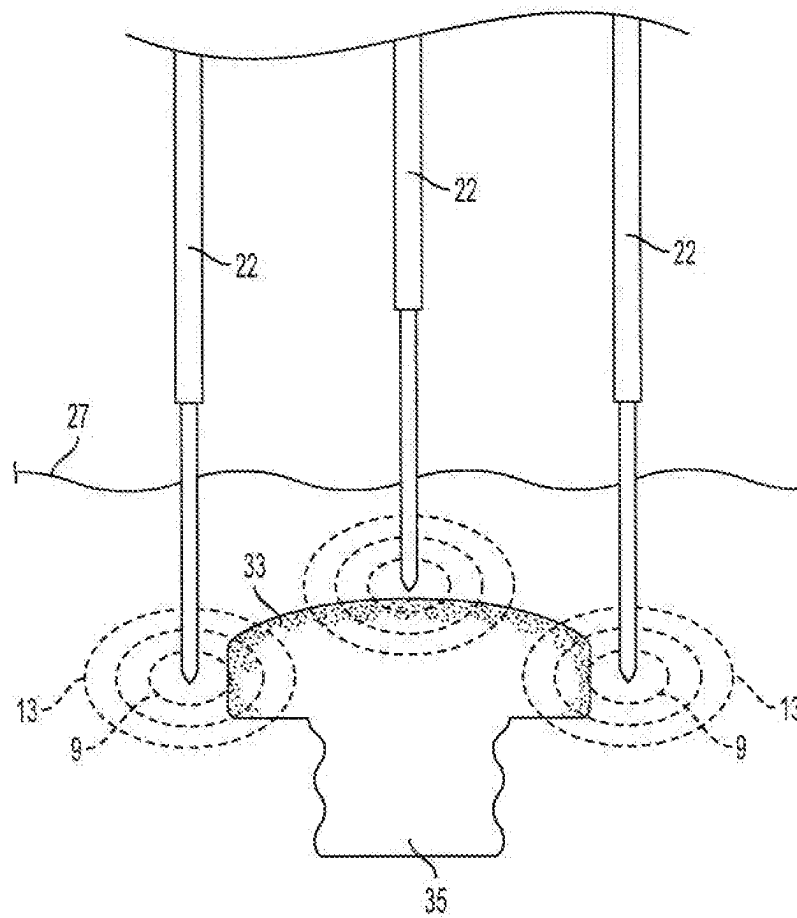


FIG. 36B

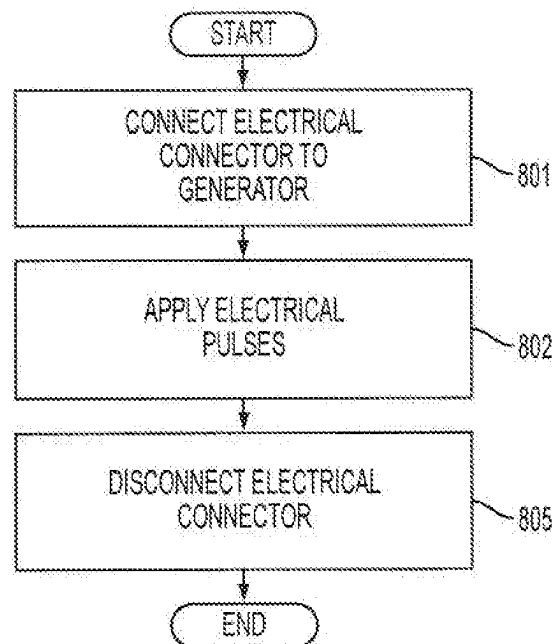


FIG. 37

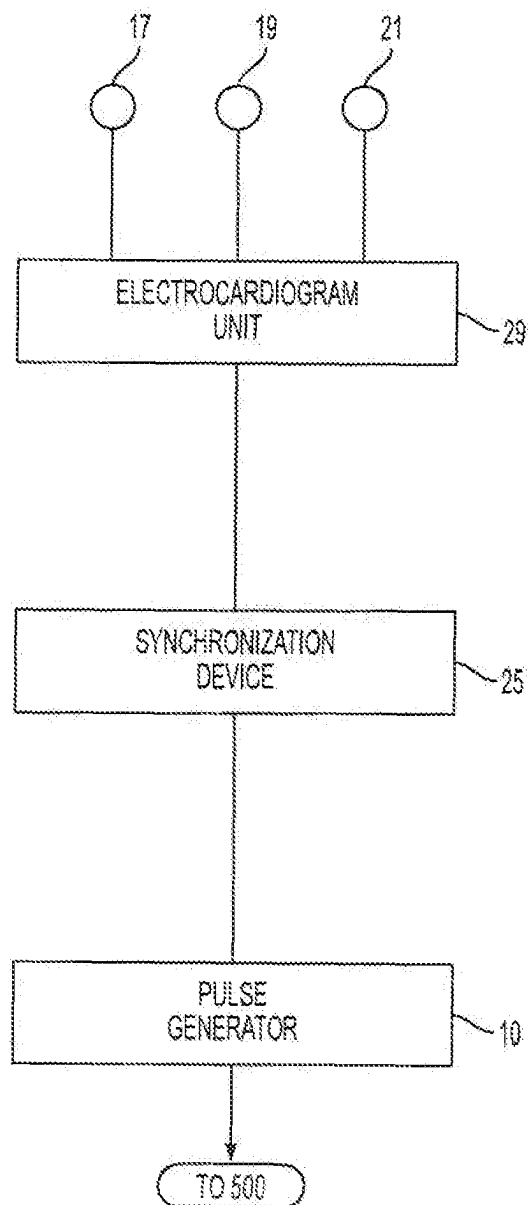


FIG. 38

METHODS OF STERILIZATION AND TREATING INFECTION USING IRREVERSIBLE ELECTROPORATION

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. application Ser. No. 17/073,039, filed Oct. 16, 2020; which is a Continuation of U.S. application Ser. No. 15/864,421, filed Jan. 8, 2018; which is a Continuation of U.S. application Ser. No. 14/302,678, filed Jun. 12, 2014, now U.S. Pat. No. 9,895,189, issued Feb. 20, 2014; which claims the benefit of U.S. Provisional Application No. 61/834,471, filed Jun. 13, 2013. This application is related to U.S. non-provisional application numbers Ser. No. 12/488,070, filed Jun. 19, 2009; Ser. No. 12/751,826, filed Mar. 31, 2010; and Ser. No. 12/751,854, filed Mar. 31, 2010. The content of all of which are hereby incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

The present invention relates to methods of treating infections in mammals using irreversible electroporation (IRE) to destroy the pathogenic microorganisms comprising the infection and methods of sterilization of implanted medical devices using IRE.

BACKGROUND OF THE INVENTION

Infections are caused by parasitic microorganisms such as viruses, bacteria or fungi which invade the host and rapidly reproduce. The body's natural immune system is designed to combat most infections. Some infections may become chronic, particularly if the host has a compromised immune system. Uncontrolled infections may cause serious health problems including death. Infections are frequently associated with wounds and may also occur in the locale of implanted devices such as ports, catheters, stents, artificial cartilaginous implants, orthopedic prosthetics, pacemakers, PICCs, prosthetic knee or hip implants, tooth implants, heart valves, spinal implants, and other types of plastic and/or metallic devices, to name a few. These medical devices can become infected after implanted into a patient. Long-term implants can be particularly susceptible to infection. Infection of implanted medical devices can constitute one of the most serious complications after surgery. Removing these infected medical devices can require lengthy procedures to remove and replace the old infected devices with a new device. Such procedures can involve steps such as removing the infected implant through surgical procedure, and/or administering antibiotics, waiting for any infection to heal, and then performing another surgical procedure to re-insert a new medical implant into the patient. In the case of cartilaginous implants, this may have to be repeated several times and may require administering not only a new cartilage implant but also numerous doses of antibiotics in order to enable the infection and the patient's skin to heal and then to reinsert a new artificial cartilage implant. Thus, multiple surgeries can be required. This can greatly increase procedure time and costs and can cause much discomfort to the patient. Pharmaceuticals such as antimicrobial agents and antibiotics can be used to fight infections in such cases, but there are drawbacks to the use of these agents including allergic reactions, negative interactions with other drugs, and ineffective treatment due to an increase in resistant

strains of these microorganisms. Sometimes a membrane can develop over infected areas, thereby creating an impassable barrier to antibiotics.

Despite the many problems with infections of implantable medical devices, little has been done to address the cost, pain, and increased procedure time and to present effective solutions for sterilizing and/or treating infections that occur as a result of implanted medical devices. Most procedures involve removing infected devices and/or sterilizing medical devices through external application of bactericidal solutions or temperature extremes such as thermal heating or freezing, UV light, RF, microwave, or other radiation measures.

Therefore, it is desirable to provide a cost-effective, painless, efficient method for treating infections by sterilizing infected implanted medical devices which methods can overcome the problems of traditional pharmaceutical treatment, such as, for example, resistance to systemic antibiotics. The methods provided herein use irreversible electroporation to sterilize implanted medical devices, thereby treating infection in patients, providing increased treatment efficacy, and eliminating or minimizing allergic reactions, which eliminates the chance of interactions with other drugs. This could also enhance patient quality of life, which would be very beneficial for patients having extreme arthritis, for example.

What is provided herein is a method of using irreversible electroporation (IRE) to treat parts of the human body that have been subject to infection. This method avoids surgical methods that require removal surgery, a waiting period, then replacement of an infected medical device. This method can greatly improve outcomes, particularly for devices where infection may be catastrophic, i.e., a prosthetic knee or hip implant. It can also greatly improve costs and improve the longevity of implanted medical devices which are susceptible to infection, such as, for example, implantable ports. Such infection can be on or within a medical device that is implanted within a patient, as described above. Alternatively, a patient's tissue can be infected due to some other type of infection, for example, gangrene. IRE can be used to treat such infections and/or simultaneously sterilize an implanted medical device, as described herein to solve the above-mentioned problems.

Electroporation is defined as a phenomenon that makes cell membranes permeable by exposing them to certain electric pulses. As a function of the electrical parameters, electroporation pulses can have two different effects on the permeability of the cell membrane. The permeabilization of the cell membrane can be reversible or irreversible as a function of the electrical parameters used. Reversible electroporation is the process by which the cellular membranes are made temporarily permeable. The cell membrane will reseal a certain time after the pulses cease, and the cell will survive. Reversible electroporation is most commonly used for the introduction of therapeutic or genetic material into the cell. Irreversible electroporation also creates pores in the cell membrane but these pores do not reseal, resulting in cell death.

Irreversible electroporation has recently been discovered as a viable alternative for the ablation of undesired tissue. See, in particular, PCT Application No. PCT/US04/43477, filed Dec. 21, 2004. An important advantage of irreversible electroporation, as described in the above reference application, is that the undesired tissue is destroyed without creating a thermal effect. When tissue is ablated with thermal effects, not only are the cells destroyed, but the connective structure (tissue scaffold) and the structure of blood

3

vessels are also destroyed, and the proteins are denatured. This thermal mode of damage detrimentally affects the tissue, that is, it destroys the vasculature structure and bile ducts, and produces collateral damage.

Irreversible and reversible electroporation without thermal effect to ablate tissue offers many advantages. One advantage is that it does not result in thermal damage to target tissue or other tissue surrounding the target tissue. Another advantage is that it only ablates cells and does not damage blood vessels or other non-cellular or non-living materials such as implanted medical devices.

Although the following examples discuss using the present invention and method to destroy various infectious cells, for example, such as implanted medical device-related bacteremia, that may substantially cover various implanted medical devices, persons of ordinary skill in the art will appreciate that the present devices and methods can be used to treat any undesirable cellular growth, including infectious cells, as well as to sterilize implanted medical devices.

SUMMARY OF THE DISCLOSURE

A method of treating an infection in patient that involves providing an ablation device, wherein the device comprises at least one electrode inserting the ablation device into a target tissue of the patient. The tissue at least partially surrounds an implanted medical device, and the outer surface of the medical device is at least partially covered by infectious cells. The method also involves positioning the at least one electrode in or near the implanted medical device and delivering electrical pulses to or near the implanted medical device sufficient to irreversibly electroporate the infectious cells.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates several components that can be used with the present invention to treat a patient using IRE or to sterilize an implanted medical device.

FIG. 2 is a schematic diagram of a treatment control computer of the present invention.

FIG. 3 is a screen shot of an "Information" screen of a treatment control module showing various input boxes.

FIG. 4 is a screen shot of a "Probe Selection" screen of the treatment control module showing a side view and top view of the bipolar probe and an example of the general shape of the treatment zone that can be generated by such a probe type. The "treatment zone", as defined herein, can comprise infected tissue cells.

FIG. 5 is a screen shot of a "Probe Selection" screen of the treatment control module showing a side view and top view of the two probe array and an example of the general shape of the treatment zone that can be generated by a two probe array.

FIG. 6 is a screen shot of a "Probe Selection" screen of the treatment control module showing a side view and top view of the three probe array and an example of the general shape of the treatment zone that can be generated by a three probe array.

FIG. 7 is a screen shot of a "Probe Selection" screen of the treatment control module showing a side view and top view of the four probe array and an example of the general shape of the treatment zone that can be generated by a four probe array.

FIG. 8 is a screen shot of a "Probe Selection" screen of the treatment control module showing a side view and top view

4

of the five probe array and an example of the general shape of the treatment zone that can be generated by a five probe array.

FIG. 9 is a screen shot of a "Probe Selection" screen of the treatment control module showing a side view and top view of the six probe array and an example of the general shape of the treatment zone that can be generated by a six probe array.

FIG. 10 is a screen shot of a "Probe Placement Process" screen of the treatment control module.

FIG. 11 is a screen shot of a "Probe Placement Process" screen of the treatment control module showing a rotation feature of the treatment control module.

FIG. 12 is a screen shot of a "Probe Placement Process" screen of the treatment control module showing an automatic measurement feature of the treatment control module.

FIG. 13 is a screen shot of a "Probe Placement Process" screen of the treatment control module showing examples of treatment zones that can be created between the electrodes.

FIG. 14 is a screen shot of a "Probe Placement Process" screen of the treatment control module showing an example of a combined treatment zone generated by a four probe array.

FIG. 15 illustrates an example of a three probe array defining three individual treatment zones, which combine to form a combined treatment region.

FIG. 16 is an example of a spreadsheet of the E-field values that are determined for x, y coordinates on the grid, as will be further described below in reference to Example 2.

FIG. 17 illustrates an example of a multi-dimensional lookup table and a method of interpolating treatment zones.

FIG. 18 is a screen shot of a "Probe Placement Process" screen of the treatment control module showing an automatic probe placement feature of the treatment control module.

FIGS. 19-22 are screen shots of the treatment control module illustrating an alternative embodiment of an automatic probe placement feature of the treatment control module.

FIGS. 23-25 are screen shots of a "Probe Placement Process" screen of the treatment control module showing several examples of how a user can edit and modify the treatment parameters.

FIG. 26 is a screen shot of a "Probe Placement Process" screen of the treatment control module showing an example of a treatment zone that is created by a six probe array after the "Autoset Probes" button has been depressed by a user.

FIG. 27 is a screen shot of a "Pulse Generation" screen of the treatment control module showing the status of the treatment parameters before the treatment procedure has been initiated by a user.

FIG. 28 is a screen shot of a "Pulse Generation" screen of the treatment control module showing the status of the treatment parameters during the treatment procedure.

FIG. 29 illustrates details of the generator shown in FIG. 1, including elements for detecting an over-current condition.

FIG. 30 is a screen shot of a "Pulse Generation" screen of the treatment control module showing the status of the treatment parameters after the treatment procedure.

FIG. 31 is a screen shot of a "Pulse Generation" screen of the treatment control module showing a dialogue box that pops up if the "continue procedure" button is pressed in the example.

5

FIG. 32 is a screen shot of a “Pulse Generation” screen of the treatment control module showing the status of the treatment parameters during the re-treatment procedure.

FIGS. 33-34 are screen shots of a “Pulse Generation” screen of the treatment control module showing examples of result graphs.

FIG. 35 is a screen shot of a “Probe Placement Process” screen of the treatment control module showing the probe placement grid after treatment has been delivered by the probes.

FIGS. 36A and 36B illustrate a method of treating infected tissue and/or sterilizing an implanted medical device, such as, for example, a cartilaginous implant.

FIG. 37 is a flowchart depicting the method steps for infectious cell ablation using irreversible electroporation methods described herein.

FIG. 38 illustrates a treatment setup for a patient for synchronization of the delivery of electroporation pulses with a specific portion of the cardiac rhythm.

DETAILED DESCRIPTION OF THE INVENTION

Throughout the present teachings, any and all of the one, two, or more features and/or components disclosed or suggested herein, explicitly or implicitly, may be practiced and/or implemented in any combinations of two, three, or more thereof, whenever and wherever appropriate as understood by one of ordinary skill in the art. The various features and/or components disclosed herein are all illustrative for the underlying concepts, and thus are non-limiting to their actual descriptions. Any means for achieving substantially the same functions are considered as foreseeable alternatives and equivalents, and are thus fully described in writing and fully enabled. The various examples, illustrations, and embodiments described herein are by no means, in any degree or extent, limiting the broadest scopes of the claimed inventions presented herein or in any future applications claiming priority to the instant application.

Disclosed herein are methods for treating infection in mammals using irreversible electroporation to destroy pathogenic microorganisms comprising the infection. In particular, the methods involve using a medical device to deliver electrical pulses to the treatment zone that comprises an implanted medical device and potentially infectious cells within a non-thermal irreversible electroporation range. A probe comprising at least one electrode is adapted to receive from a voltage generator a plurality of electrical pulses in an amount sufficient to cause destruction of cells comprising the infection. The number of pulses, pulse length, pulse amplitude can be used to irreversibly electroporate a target tissue. Also presented herein is a method for sterilizing an implanted medical device using irreversible electroporation.

One embodiment of the present invention is illustrated in FIGS. 1 through 35. The components that can be used with the present invention are illustrated in FIG. 1. One or more probes 22 can deliver therapeutic energy and are powered by a voltage pulse generator 10 that generates high voltage pulses as therapeutic energy such as pulses capable of irreversibly electroporating the tissue cells. In the embodiment shown, the voltage pulse generator 10 can include six separate receptacles for receiving up to six individual probes 22 which are adapted to be plugged into the respective receptacle. The receptacles are each labeled with a number in consecutive order. In other embodiments, the voltage pulse generator 10 can have any number of receptacles for receiving more or less than six probes.

6

In the embodiment shown, each probe 22 includes either a monopolar electrode or bipolar electrodes having two electrodes separated by an insulating sleeve. In one embodiment, if the probe includes a monopolar electrode, the amount of exposure of the active portion of the electrode can be adjusted by retracting or advancing an insulating sleeve relative to the electrode. See, for example, U.S. Pat. No. 7,344,533, which is incorporated by reference herein. The generator 10 is connected to a treatment control computer 40 having input devices such as keyboard 12 and a pointing device 14, and an output device such as a display device 11 for viewing an image of a target treatment area such as a target tissue 300 surrounded by a safety margin 301. The therapeutic energy delivery device 20 is used to treat a target tissue 300 inside a patient 15. In one aspect, the target tissue can comprise infectious cells and/or an implanted medical device comprising infectious cells. An imaging device 30 includes a monitor 31 for viewing the target tissue 300 inside the patient 15 in real time. Examples of imaging devices 30 include ultrasonic, CT, MRI and fluoroscopic devices as are known in the art.

The present invention includes computer software (treatment control module 54) which assists a user to plan for, execute, and review the results of a medical treatment procedure, as will be discussed in more detail below. For example, the treatment control module 54 assists a user to plan for a medical treatment procedure by enabling a user to more accurately position each of the probes 22 of the therapeutic energy delivery device 20 in relation to the target tissue 300 in a way that will generate the most effective treatment zone. The treatment control module 54 can display the anticipated treatment zone based on the position of the probes and the treatment parameters. The treatment control module 54 can display the progress of the treatment in real time and can display the results of the treatment procedure after it is completed. This information can be used to determine whether the treatment was successful and whether it is necessary to re-treat the patient.

For purposes of this application, the terms “code”, “software”, “program”, “application”, “software code”, “software module”, “module” and “software program” are used interchangeably to mean software instructions that are executable by a processor.

The “user” can be a physician or other medical professional. The treatment control module 54 executed by a processor outputs various data including text and graphical data to the monitor 11 associated with the generator 10.

Referring now to FIG. 2, the treatment control computer 40 of the present invention manages planning of treatment for a patient. The computer 40 is connected to the communication link 52 through an I/O interface 42 such as a USB (universal serial bus) interface, which receives information from and sends information over the communication link 52 to the voltage generator 10. The computer 40 includes memory storage 44 such as RAM, processor (CPU) 46, program storage 48 such as ROM or EEPROM, and data storage 50 such as a hard disk, all commonly connected to each other through a bus 53. The program storage 48 stores, among others, a treatment control module 54 which includes a user interface module that interacts with the user in planning for, executing and reviewing the result of a treatment. Any of the software program modules in the program storage 48 and data from the data storage 50 can be transferred to the memory 44 as needed and is executed by the CPU 46.

In one embodiment, the computer 40 is built into the voltage generator 10. In another embodiment, the computer

40 is a separate unit which is connected to the voltage generator through the communications link 52. In a preferred embodiment, the communication link 52 is a USB link.

In one embodiment, the imaging device 30 is a stand alone device which is not connected to the computer 40. In the embodiment as shown in FIG. 1, the computer 40 is connected to the imaging device 30 through a communications link 53. As shown, the communication link 53 is a USB link. In this embodiment, the computer can determine the size and orientation of the target tissue 300 by analyzing the data such as the image data received from the imaging device 30, and the computer 40 can display this information on the monitor 11. In this embodiment, the target tissue generated by the imaging device 30 can be directly displayed on the grid 200 of the monitor 11 of the computer running the treatment control module 54. This embodiment would provide an accurate representation of the target tissue on the grid 200, and may eliminate the step of manually inputting the dimensions of the target tissue in order to create the target tissue on the grid 200. This embodiment would also be useful to provide an accurate representation of the target tissue if the target tissue has an irregular shape.

The basic functionality of the computer software (treatment control module 54) will now be discussed in relation to the following example. It should be noted that the software can be used independently of the generator 10. For example, the user can plan the treatment in a different computer as will be explained below and then save the treatment parameters to an external memory device, such as a USB flash drive (not shown). The data from the memory device relating to the treatment parameters can then be downloaded into the computer 40 to be used with the generator 10 for treatment.

After the treatment control module 54 is initialized, it displays an "Information" screen with various input boxes as shown in FIG. 3. A keyboard or other input device 12, together with a mouse or other pointing device 14 (see FIG. 1) are used to input the data. Any data that is inputted into the input boxes can be saved into internal or external memory along with a record of the treatment as described below for future reference. The basic patient information can be inputted, such as a patient ID number in input box 100, the name of the patient in input box 101, and the age of the patient in input box 102. The user can enter clinical data, such as the clinical indication of the treatment in input box 114. The date of the procedure is automatically displayed at 111 or can be inputted by the user in another embodiment. The user can enter other case information such as the name of the physician in input box 112 and any specific case notes in input box 113.

The dimensions of the target tissue 300 are determined from viewing it on the monitor 31 of the imaging device 30 (see FIG. 1) such as an ultrasonic imaging device and using known methods to calculate the dimensions from the image generated from the imaging device 30. The dimensions of the target tissue 300 (length at input box 103, width at input box 104, and depth at input box 105) are inputted into the program. A safety margin is selected at input box 106 which will surround the entire target tissue 300 in three dimensions. According to the size of the safety margin that is selected, a target treatment region is automatically calculated and is displayed in boxes 107, 108, and 109 as shown. In one embodiment, the safety margin value may be set to zero. For example, when treating a benign tumor, a safety margin may not be necessary.

In the embodiment shown in FIG. 3, the user has indicated that the target tissue that will be treated has a length of 2 cm, width of 1 cm and a depth of 1 cm. With a user specified margin of 1 cm (which is a default margin setting), the target treatment region has a length of 4 cm, width of 3 cm and a depth of 3 cm.

The user can select the "ECG synchronization" option by clicking the circle in the box 110 in order to synchronize the pulses with an electrocardiogram (ECG) device, if such a device is being used during the procedure (FIG. 38). The other options available for treatment that are included in box 110 can include an option for "90 PPM" (pulses per minute) or "240 PPM". The user should select at least one of the three options provided in box 110. After all of the necessary data has been inputted, the user clicks on the "Next" button with a pointing device 14 to proceed to the next screen described below.

Further regarding the ECG synchronization option, if this circle is selected in window 110, the treatment control module 54 will test this functionality to verify that the system is working properly. The treatment control module 54 can automatically detect whether an error has occurred during the testing phase of the ECG feature. The detectable errors include, but are not limited to, "no signal" (such as no pulses for 3.5 seconds) and "noisy" (such as pulses occurring at a rate greater than 120 beats per minute for at least 3.5 seconds).

The treatment control module 54 can synchronize energy release with cardiac rhythm by analyzing cardiac output such as electrocardiogram results (or other cardiac function output) and sending synchronization signals to a controller of the pulse generator 10. The control module 54 is also capable of generating internal flags such as a synchronization problem flag and a synchronization condition flag to indicate to users on a graphic user interface a synchronization status, so that energy pulse delivery can be synchronized with the cardiac rhythm for each beat (in real-time) or aborted as necessary for patient safety and treatment efficiency.

Specifically, the control module 54 synchronizes energy pulses such as IRE (irreversible electroporation) pulses with a specific portion of the cardiac rhythm. The module uses the R-wave of the heartbeat and generates a control signal to the pulse generator 10 indicating that this portion of the heartbeat is optimal for release of IRE pulses. For clarity, the S wave would be an optimal time for delivery of an energy pulse, but due to the fact that the S wave ends nebulously in some cases, the R wave is used as an indicator to start timing of energy release.

More specifically, the synchronization feature of the control module 54 allows for monitoring of heart signals so as to ensure that changes, maladies, and other alterations associated with the heartbeat are coordinated such that pulses from the pulse generator 10 are released at the proper time, and that if the heartbeat is out of its normal rhythm, that the release of energy is either altered or aborted.

Next, the user can select the type of therapeutic energy delivery device according to the number of probes that the user believes will be necessary to produce a treatment zone which will adequately cover the target tissue 300 and any safety margin 301. The selection is made by clicking the circle next to each type of device, as shown in the "Probe Selection" screen, illustrated in FIGS. 4-9.

In one embodiment, a "Probes Selection Status" box 199 identifies which of the receptacles, if any, on the generator 10 have been connected to a probe by displaying the phrase "Connected" or the like next to the corresponding probe

number. In one embodiment, each receptacle includes an RFID device and a connector (not shown) for each probe which connects to the receptacle and includes a compatible RFID device, so that the treatment control module 54 can detect whether or not an authorized probe has been connected to the receptacle on the generator 10 by detecting a connection of the compatible RFID devices. If an authorized probe is not connected to a receptacle on the generator, the phrase "Not Connected" or the like will appear next to the probe number. In addition, the colors of each probe shown in the "Probes Selection Status" box 199 can be used to indicate whether or not each receptacle on the generator is connected to a compatible probe. This feature allows the user to verify that the requisite number of probes is properly connected to the generator 10 before selecting a probe type for the treatment procedure. For example, if the treatment control module 54 detects a problem with the probe connection status (e.g. selecting a three probe array when only two probes are connected to the generator), it can notify the user by displaying an error message.

The user can select which of the connected probes will be used to perform the treatment procedure, by clicking on the box next to the selected probes in the "Probes Selection Status" box 199. By default the treatment control module 54 will automatically select probes in ascending numerical order, as they are labeled.

Referring to FIG. 4, circle 120 is used to select a bipolar probe. FIG. 4 illustrates a side view 121 and top view 122 of the bipolar probe and an example of the general shape of the treatment zone that can be generated by such a probe type. The side view 121 shows an example of the general shape of the treatment zone that can be generated by an arrangement of two electrodes 123 separated by an insulation sleeve.

Referring to FIG. 5, circle 130 is used to select a two probe array. FIG. 5 illustrates a side view 131 and top view 132 of the two probe array and an example of the general shape of the treatment zone that can be generated by a two probe array. In the illustrated example, the exposed portion of each of the electrodes as shown is 20 mm in length and the two probes are spaced from each other by 15 mm.

Referring to FIG. 6, circle 140 is used to select a three probe array. FIG. 6 illustrates a side view 141 and top view 142 of the three probe array and an example of the general shape of the treatment zone that can be generated by a three probe array. In the illustrated example, the exposed portion of each of the electrodes as shown is 20 mm in length and each pair of the three probes are equally spaced from each other by 15 mm, as measured at three places (PLCS), meaning that there are three pairs (pairs 1-2, 2-3 and 1-3) where the spacing is equal to 15 mm.

Referring to FIG. 7, circle 150 is used to select a four probe array. FIG. 7 illustrates a side view 151 and top view 152 of the four probe array and an example of the general shape of the treatment zone that can be generated by a four probe array. In the illustrated example, the exposed portion of each of the electrodes as shown is 20 mm in length and each pair of the four probes are equally spaced from each other by 15 mm, as measured at four places (PLCS) along the perimeter.

Referring to FIG. 8, circle 160 is used to select a five probe array. FIG. 8 illustrates a side view 161 and top view 162 of the five probe array and an example of the general shape of the treatment zone that can be generated by a five probe array. In the illustrated example, the exposed portion of each of the electrodes as shown is 20 mm in length and

each pair of the five probes are equally spaced from each other by 15 mm, as measured at seven places (PLCS).

Referring to FIG. 9, circle 170 is used to select a six probe array. FIG. 9 illustrates a side view 171 and top view 172 of the six probe array and an example of the general shape of the treatment zone that can be generated by a six probe array. In the illustrated example, the exposed portion of each of the electrodes as shown is 20 mm in length and each pair of the six probes are equally spaced from each other by 15 mm, as measured at five places (PLCS) from the center probe. Each pair of the six probes are equally spaced from each other by 17 mm, as measured at 5 places (PLCS) along the perimeter.

Other probe type selection can include a "six probe array 10 mm" and "six probe array 15 mm", which refers to probe types utilizing a template which can be used to align a group of six needles in a fixed predetermined arrangement for treatment, wherein each pair of probes are equally spaced by 10 mm and 15 mm, respectively.

Other probe device types having seven or more probes can be used. The user can select a probe type having a number of probes 22 which will work most effectively to treat the specific size and shape of the target tissue 300 together with a safety margin 301.

After the user has selected a probe type on the "Probe Selection" screen, the user clicks on the "Next" button with a pointing device 14 to proceed to the "Probe Placement Process" screen described below.

FIG. 10 illustrates a "Probe Placement Process" screen of one aspect of the invention. The screen illustrated by FIG. 10 shows a target tissue 300 according to the dimensions which were inputted on the "Information" screen (see FIG. 3) along with a safety margin 301, if any, that was previously inputted. In the example depicted in FIG. 10, the target tissue 300 has a length of 2.0 cm and a width of 1.0 cm, and the device selected on the "Probe Selection" screen (see FIGS. 4-9) is a four probe array. The target tissue 300 is displayed near the center of an x-y grid 200 with the distance between two adjacent grid lines representing 1 mm. Each of the four probes 201, 202, 203, 204 is displayed in the grid 200 and each probe can be manually positioned within the grid by clicking and dragging the probe with the pointing device 14. Two fiducials 208, 209 labeled "A" and "B", respectively, are also displayed on the grid 200 and are used as a point of reference or a measure as will be described below.

The amount of longitudinal exposure of the active electrode portion for each probe that has already been manually adjusted by the user as explained above can be manually inputted in input box 210, which can be selected by the user according to the depth (z) of the target tissue. In this way, the treatment control module 54 can generate an estimated treatment zone according to the treatment parameters, and locations and depths of the probes. In one embodiment, a second x-z grid is displayed on the monitor 11 of the computer running the treatment control module 54. In one embodiment, the treatment control module 54 can automatically calculate preferred values for the amount of longitudinal exposure of the active electrode portions based on the size and shape of the target tissue. The depth (z) of the electric field image can be calculated analytically or with interpolation and displayed on the x-z grid. Because the distribution of the electric field (i.e., expected treatment region) between two monopolar electrodes may "dip in" along the boundary line (see, for example, the peanut shaped treatment region in FIG. 13 where the width of the region is smaller in the middle) depending on the electrode location and the applied voltage, it is beneficial to have an x-z grid included on the monitor. For example, if this "dip" of the

11

boundary line travels into, rather than surround, the target tissue region, then the targeted region may not be fully treated. As a default to ensure treatment of the entire target tissue region, the probe depth placement and the exposure length may be set unnecessarily higher to ensure erring on the safe side. However, this will potentially treat a much larger volume than needed, killing healthy surrounding tissue, which can be an issue when treating sensitive tissues such as the pancreas, brain, etc. By optimizing the treatment depth (z) together with the width (x) and height (y), this effect may be reduced, further enhancing procedural protocol and clinical outcome.

The probe dock status is indicated in box **210**, by indicating if the probes are “docked” or “undocked”. The “UnDock Probes” button allows the user to “unplug” the probes from the generator while the “Probe Placement Process” screen is displayed without causing error messages. In normal operation, the user plugs the probes into the generator on the “Probe Selection” screen, and then the probes are “authorized” as being compatible probes according to the RFID devices, as discussed above. When the user proceeds to the “Probe Placement Process” screen, the software requires that all the selected probes remain plugged into the generator, or else the software will display an error message (e.g. “Probe #2 unplugged”, etc.), and will also force the user back to the “Probe Selection” screen. However, sometimes users may want to perform another scan of the target tissue **300** or perform some other procedure while leaving the probes inserted in the patient. But, if the procedure cannot be performed near the generator, the probes are unplugged from the generator. If the user selects the “UnDock Probes” button, this will allow the probes to be unplugged from the generator without causing an error message. Then, after the user has performed the other procedure that was required, the user can re-attach the probes to the generator, and then select “Dock Probes” in input box **210**. In this way, the user will not receive any error messages while the “Probe Placement Process” screen is displayed.

There is a default electric field density setting (Volts/cm) which is shown in input box **211**. In the example, the default setting is 1500 Volts/cm. This number represents the electric field density that the user believes is needed to effectively treat the cells, e.g., ablate the tissue cells. For example, 1500 Volts/cm is an electric field density that is needed to irreversibly electroporate the tissue cells. Based on the number selected in input box **211**, the treatment control module **54** automatically adjusts the voltage (treatment energy level) applied between the electrodes, as shown in column **222**.

Box **280** allows a user to select between two different Volts/cm types, namely “Linear” or “Non-Linear Lookup”.

The default Volts/cm setting is “Linear”, in which case the Voltage that is applied between a given pair of electrodes, as shown in column **222**, is determined by the following formula:

$$\text{Voltage} = x \cdot d, \quad (1)$$

where x = the electric field density setting (Volts/cm) shown in column **225**, which is based on the value from box **211**, and

where d = the distance (cm) between the given pair of electrodes shown in column **226**.

Therefore, when “Linear” is selected, the Voltage that is applied between a given pair of electrodes is directly proportional to the Distance between the given electrode pair in a linear relationship.

12

If the user selects “Non-Linear Lookup” in box **280**, then the Voltage that is applied between the given pair of electrodes will be similar to the Voltage values for a “Linear” selection when a pair of electrodes are closely spaced together (e.g. within about 1 cm). However, as a pair of given electrodes are spaced farther from one another, a “Non-Linear Lookup” will produce lower Voltages between the given pair of electrodes as compared to the Voltage values for a “Linear” selection at any given distance. The “Non-Linear Lookup” feature is particularly useful for reducing “popping” during treatment. “Popping” refers to an audible popping noise that sometimes occurs, which is believed to be caused by a plasma discharge from high voltage gradients at the tip of the electrodes. The “Non-Linear Lookup” feature can also minimize any swelling of the tissue that might occur as a result of a treatment. The Voltage values used for the “Non-Linear Lookup” selection can be pre-determined based on animal experiments and other research. In one embodiment, different tissue types can each have their own “Non-Linear Lookup” table. In the example shown, the tissue being treated is prostate tissue.

The details of the treatment parameters are displayed in window **270**. The firing (switching) sequence between probes is listed automatically in window **270**. In the example, the firing sequence involves six steps beginning with between probes 1 and 2, then probes 1 and 3, then probes 2 and 3, then probes 2 and 4, then probes 3 and 4, and then probes 4 and 1. As shown, the polarity of each of the probes may switch from negative to positive according to step of the firing sequence. Column **220** displays which probe is the positive probe (according to a number assigned to each probe) for each step. Column **221** displays which probe is the negative probe (according to a number assigned to each probe) for each step. Column **222** displays the actual voltage generated between each probe during each step of the firing sequence. In the example, the maximum voltage that can be generated between probes is limited by the capabilities of the generator **10**, which in the example is limited to a maximum of 3000 Volts. Column **223** displays the length of each pulse that is generated between probes during each respective step of the firing sequence. In the example, the pulse length is predetermined and is the same for each respective step, and is set at 100 microseconds. Column **224** displays the number of pulses that is generated during each respective step of the firing sequence. In the example, the number of pulses is predetermined and is the same for each respective step, and is set at 90 pulses which are applied in a set of 10 pulses at a time. Column **225** displays the setting for Volts/cm according to the value selected at input box **211**. Column **226** displays the actual distance between the electrodes (measured in cm), which is automatically calculated according to the placement of each probe in the grid **200**.

FIG. **11** illustrates a rotation feature of the treatment control module **54**. The user can rotate the image of target tissue **300** on the grid **200** about its center in order to approximate the actual orientation of the target tissue **300** within the body of the patient **15** (see FIG. **1**), as shown by the imaging device **30**. To do so, the user can view the actual orientation of the target tissue **300** within the body of the patient **15** by viewing the monitor **31** of the imaging device **30** shown in FIG. **1**. While viewing, the user can rotate the target tissue **300** on the grid **200** in order to match the orientation of the target tissue **300** shown on the monitor **31** of the imaging device **30**. There are at least three ways to rotate the target tissue **300** on the grid **200**. The user can click on a tab **250** (to select the tab) with a pointing device

13

14 and drag the tab 250 to a new location which will rotate the target tissue 300. The user can alternatively click on any part of the safety margin 301 or the target tissue 300 and drag it to rotate the target tissue 300. Alternatively, the user can manually input the treatment zone rotation angle in input box 251, which represents the degree of rotation of the target tissue 300 measured from the horizontal "x" axis on the grid 200.

FIG. 12 illustrates an automatic measurement feature of the treatment control module 54. When the user clicks on a probe and drags it with the pointing device 14, the treatment control module 54 automatically and continuously displays the distance (cm) from that electrode 201 to each of the other electrodes 202 (distance displayed in box 230), 203 (distance displayed in box 232), 204 (distance displayed in box 233) as the probe is being dragged. The treatment control module 54 also displays the distance (cm) from that electrode 201 to the closest point on the outer surface of the target tissue 300 (distance displayed in box 234). The treatment control module 54 also displays the distance (cm) from that electrode 201 to fiducial "A" 208 (distance displayed in box 235), fiducial "B" 209 (distance displayed in box 236). The treatment control module 54 also displays the distance (cm) from fiducial "A" 208 to fiducial "B" 209 with the distance being displayed in box 237. This feature assists the user in placing the electrodes in preferred locations. This feature is especially beneficial if the imaging device 30 (see FIG. 1) allows the calculation of measurements as is known in the art.

FIG. 13 illustrates examples of the treatment zones that are automatically created between the electrodes by the treatment control module 54. The treatment control module 54 automatically calculates the treatment zones which are created and displays the area of the treatment zone. In a preferred embodiment, the monitor 11 of the generator 10 is in color and the color of the treatment zones 306, 307, the target tissue 300, and the safety margin 301 are all different to easily differentiate them from one another. In one embodiment, the target tissue 300 is yellow and the safety margin 301 is blue. In addition, the treatment control module 54 can be programmed to adjust the color of the target tissue 300 and/or the boundary line of the target tissue 300 if the treatment zones 306, 307 do not effectively cover the target tissue, which could otherwise result in a clinical failure.

In addition, the treatment control module 54 can be programmed to display a boundary line 320 that surrounds the areas of the treatment zones 306, 307 in a highlighted manner so that the outer boundaries of the treatment zones are readily identifiable. In one embodiment, the boundary line is a black line having sufficient thickness to provide a sharp contrast against the displayed target tissue and the grid. FIG. 14 illustrates a combined treatment region that is created by a four probe array 201, 202, 203, 204. The control module 54 displays a boundary line 320 to identify the outer boundary of the combined treatment zone. The control module 54 also displays a boundary line 320 to identify one or more inner boundaries of the combined treatment zone, if applicable. This allows the user to easily identify the presence of any incomplete treatment coverage areas of the target tissue 300. The inner boundary line 320 is especially useful to identify incomplete treatment areas that may have not otherwise been readily detectable on the computer screen. Since even a few surviving infectious cells can be detrimental in terms of recurrence of infection, this feature is especially important when treating an area with an implanted medical device having infectious cells.

14

The treatment control module can be programmed to calculate and display the area of the combined treatment regions on the grid 200 by one of the following three methods, although other methods can be used.

Each of the following methods determines a boundary line surrounding a treatment zone that is created between a pair of electrodes. By combining a plurality of treatment zones with each treatment zone being defined by a pair of electrodes, a combined treatment region can be displayed on the x-y grid. FIG. 15 illustrates three electrodes 201, 202, 203 defining three individual treatment zones 311, 312, 313, which combine to form a combined treatment region 315 which is shown with hatched lines.

As discussed above, the monitor can further include an x-z grid to illustrate the depth of the target tissue and the shape of treatment region. The shape of the treatment zone in the x-z grid will vary according to the selected amounts of electrode exposure for each probe and can be determined by one or more methods.

In one embodiment, the treatment boundary line that is created between two points on the x-y grid can be rotated about an axis joining the two points in order to generate the treatment region boundary line on the x-z grid. In this embodiment, several points may be selected along the exposed length of the active electrode portion for each probe at various depths (z). A three-dimensional combined treatment region can then be generated by determining the boundary line on the x-y grid between each individual pair of points and then rotating the boundary line along the axis joining each pair of points. The resulting boundary lines can be combined to create a three dimensional image that is displayed on the monitor.

The following is an alternate method for determining a boundary line on the x-z grid, thereby determining a three dimensional treatment region. This example describes a two probe array with the probes being inserted in a parallel relationship and with the probes having the same amount of exposed portions of the electrode. In this example, the exposed portions of each probe start at the same "uppermost" depth (z) and end at the same "lowermost" depth (z). First, a treatment zone boundary line is created in the x-y plane at the uppermost depth (z). Next, the treatment zone boundary line is repeatedly created stepwise for all subsequently lower depths (z), preferably evenly spaced, until the lowermost depth (z) is reached. The result is a 3-D volume (stacked set of treatment zone boundary lines) having a flat top surface and a flat bottom surface. Next, two new focus points are selected, with the first focus point positioned midway between the probe positions in the x-y grid and near the uppermost depth (z) of the exposed electrode. The second focus point is also positioned midway between the probe positions in the x-y grid, but near the lowermost depth (z) of the exposed electrode. Next, a treatment zone boundary line is created in the x-z grid using one of the methods described earlier. The actual placement of each focus point may be closer together, namely, not positioned in the uppermost and lowermost x-y planes defined by the exposed portions. The placement of each focus point should be selected so that the treatment zone boundary line that is created in the x-z grid closely matches the treatment zone boundary lines that were created in the uppermost and lowermost x-y grids. Next, the treatment zone boundary line that was created in the x-z grid according to the two focus points is rotated about the axis joining the two focus points. This creates the shapes for the upper and lower 3-D volumes which are added to the flat top surface and the flat bottom surface described above.

15

The above methods can be applied by persons of ordinary skill in the art to create 3-D treatment zones between exposed portions of electrodes even when the probes are not parallel to each other and even when the amount of the exposed portion varies with each probe.

Furthermore, there are situations where it is advantageous to show multiple boundary zones as a result of a therapy. For example, indicating which regimes undergo no change, reversible electroporation, irreversible electroporation, and conventional thermal damage is possible in accordance with the present invention. In addition, it is possible to output the entire distribution rather than just delineating boundaries. For example, the "Second Method" (as discussed below) can be used to determine the entire potential field or temperature distribution within the domain.

It has been shown repeatedly in the literature that tissue properties are highly variable between tissue types, between individuals, and even within an individual. These changes may result from differences in body fat composition, hydration levels, and hormone cycles. Due to the large dependence of IRE (irreversible electroporation) treatments on tissue conductivity, it is imperative to have accurate values. Therefore, to obtain viable conductivity values prior to treatment, a low amplitude voltage pulse is used between the electrode conductors and the resultant impedance/conductance is measured as a way to determine pertinent tissue property data such as the predicted current. The value determined may then be implemented when assessing field strength and treatment protocol in real time. For example, the resulting impedance or predicted current can be used to set the default electric field density.

As discussed in the background, one accurate numerical model based method for generating a treatment zone between a pair of treatment probes involves finite element analysis (FEA). For example, U.S. Patent Application Publication No. 2007/0043345, which is hereby incorporated by reference, discloses using FEA models to generate treatment zones between a pair of electrodes (the calculations were performed using MATLAB's finite element solver, Femlab v2.2 (The MathWorks, Inc. Natick, Mass.)).

Most engineering problems can be solved by breaking the system into cells where each corner of the cell or mesh is a node. FEA is used to relate each node to each of the other nodes by applying sets of partial differential equations. This type of a system can be coded by scratch, but most people use one of many commercial FEA programs that automatically define the mesh and create the equations given the model geometry and boundary conditions. Some FEA programs only work in one area of engineering, for example, heat transfer and others are known as multiphysics. These systems can convert electricity to heat and can be used for studying the relationships between different types of energy.

Typically the FEA mesh is not homogeneous and areas of transition have increased mesh density. The time and resources (memory) required to solve the FEA problem are proportional to the number of nodes, so it is generally unwise to have a uniformly small mesh over the entire model. If possible, FEA users also try to limit the analysis to 2D problems and/or use planes of symmetry to limit the size of the model being considered because even a modest 2D model often requires 30 minutes to several hours to run. By comparison, a 3D Model usually takes several hours to several days to run. A complicated model like a weather system or a crash simulation may take a super computer several days to complete.

Depending on the complexity of the FEA models that are required, the purchase price of the FEA modeling software

16

can cost several thousand dollars for a low end system to \$30 k for a non-linear multiphysics system. The systems that model the weather are custom made and cost tens of millions of dollars.

In one example, the steps which are required for generating a treatment zone between a pair of treatment probes using finite element analysis include: (1) creating the geometry of interest (e.g., a plane of tissue with two circular electrodes); (2) defining the materials involved (e.g., tissue, metal); (3) defining the boundary conditions (e.g., Initial voltage, Initial temperature); (4) defining the system load (e.g., change the voltage of the electrodes to 3,000 V); (5) determining the type of solver that will be used; (6) determining whether to use a time response or steady state solution; (7) running the model and wait for the analysis to finish; and (8) graphing the results.

As discussed above, using FEA is not at all practical for use in calculating and displaying a treatment zone that is created between a pair of treatment probes in accordance with the present invention because of the time required to run these types of analyses. For the present invention, the system should allow a user to experiment with probe placement and should calculate a new treatment zone in less than a few seconds. Accordingly, the FEA model is not appropriate for such use and it would be desirable to find an analytic solution (closed form solution), which can calculate the treatment zones with only simple equations, but which closely approximate the solutions from a numerical model analysis such as the finite element analysis. The closed loop solutions should preferably generate the treatment zone calculation in a fraction of a second so as to allow a physician/user to experiment with probe placement in real time.

According to the present invention, there are several closed loop (analytical model analysis) methods for estimating and displaying a treatment zone between a pair of treatment probes, which produce similar results to what would have been derived by a numerical model analysis such as FEA, but without the expense and time of performing FEA. Analytical models are mathematical models that have a closed form solution, i.e., the solution to the equations used to describe changes in a system can be expressed as a mathematical analytic function. The following three methods represent non-limiting examples of such alternative closed loop solutions.

The First Method

In mathematics, a Cassini oval is a set (or locus) of points in the plane such that each point p on the oval bears a special relation to two other, fixed points q_1 and q_2 : the product of the distance from p to q_1 and the distance from p to q_2 is constant. That is, if the function $\text{dist}(x,y)$ is defined to be the distance from a point x to a point y , then all points p on a Cassini oval satisfy the equation:

$$\text{dist}(q_1,p) \times \text{dist}(q_2,p) = b^2 \quad (2)$$

where b is a constant.

The points q_1 and q_2 are called the foci of the oval.

Suppose q_1 is the point $(a,0)$, and q_2 is the point $(-a,0)$. Then the points on the curve satisfy the equation:

$$((x-a)^2+y^2)((x+a)^2+y^2)=b^4 \quad (3)$$

The equivalent polar equation is:

$$r^4 - 2a^2 r^2 \cos 2\theta = b^4 - a^4 \quad (4)$$

The shape of the oval depends on the ratio b/a . When b/a is greater than 1, the locus is a single, connected loop. When

17

b/a is less than 1, the locus comprises two disconnected loops. When b/a is equal to 1, the locus is a lemniscate of Bernoulli.

The Cassini equation provides a very efficient algorithm for plotting the boundary line of the treatment zone that was created between two probes on the grid **200**. By taking pairs of probes for each firing sequence, the first probe is set as q_1 being the point (a,0) and the second probe is set as q_2 being the point (-a,0).

The polar equation for the Cassini curve was used because it provides a more efficient equation for computation. The current algorithm can work equally as well by using the Cartesian equation of the Cassini curve. By solving for r^2 from eq. (4) above, the following polar equation was developed:

$$r^2 = a^2 \cos(2\theta) + \sqrt{b^4 - a^4 \sin^2(2\theta)} \quad (5)$$

where a=the distance from the origin (0,0) to each probe in cm; and

where b is calculated from the following equation:

$$b^2 = \left[\frac{V}{\left[\ln(a)(595.28) + 2339 \right] \left(\frac{A}{650} \right)} \right]^2 \quad (6)$$

where V=the Voltage (V) applied between the probes;

where a=the same as from eq. (5); and

where A=the electric field density (V/cm) that is required to ablate the desired type of tissue according to known scientific values.

As can be seen from the mathematics involved in the equation, r can be up to four separate values for each given value for theta.

Example 1

If V=2495 Volts; a=0.7 cm; and A=650 V/cm;

Then $b^2=1.376377$

and then a cassini curve can be plotted by using eq. (5) above by solving for r, for each degree of theta from 0 degrees to 360 degrees.

A portion of the solutions for eq. (5) are shown in Table 1 below:

where $M=a^2 \cos(2\theta)$; and $L=\sqrt{b^4 - a^4 \sin^2(2\theta)}$

TABLE 1

Theta (degrees)	r = sqrt (M + L)	r = -sqrt (M + L)	r = sqrt (M - L)	r = -sqrt (M - L)
0	1.366154	-1.36615	0	0
1	1.366006	-1.36601	0	0
2	1.365562	-1.36556	0	0
3	1.364822	-1.36482	0	0
4	1.363788	-1.36379	0	0
5	1.362461	-1.36246	0	0
6	1.360843	-1.36084	0	0
7	1.358936	-1.35894	0	0
8	1.356743	-1.35674	0	0
9	1.354267	-1.35427	0	0
10	1.351512	-1.35151	0	0
11	1.348481	-1.34848	0	0
12	1.34518	-1.34518	0	0
13	1.341611	-1.34161	0	0
14	1.337782	-1.33778	0	0
15	1.333697	-1.3337	0	0

18

The above eq. (6) was developed according to the following analysis.

The curve from the cassini oval equation was calibrated as best as possible to the 650 V/cm contour line using two 1-mm diameter electrodes with an electrode spacing between 0.5-5 cm and an arbitrary applied voltage.

For this worksheet, q_1 and q_2 reference points (taken to be +/-electrodes) could be moved to locations along the x-axis to points of $(\pm a, 0)$. A voltage could then be selected, and an arbitrary scaling factor ("gain denominator") would convert this voltage to the corresponding "b" used in eq. (4). The worksheet would then plot the resulting Cassini oval, which has a shape progression with applied voltage beginning as two circles around the electrodes that grow into irregular ellipses before converging into a single "peanut" shape that ultimately becomes an ellipse expanding from the original electrode locations.

The Cassini oval creates a reasonable visualization that mimics the shape of numerical results for the field distribution. In order to understand which values or levels correspond to a desired electric field of interest, a calibration involving the b^4 term was necessary to develop the relationship between the analytical Cassini oval and the numerical results. This was done through a backwards calibration process defined as follows:

1. A reference contour was selected to correlate the analytical and numerical solutions. This was chosen to be when b/a=1, forming a lemniscate of Bernoulli (the point where the two ellipses first connect, forming " ∞ ").

2. A reference electric field density value was selected to be 650 V/cm

3. Numerical models were developed to mimic the x-y output from the Cassini oval for scenarios where $a=\pm 0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0, 2.25,$ and 2.5 cm.

4. Models were solved using trial and error to determine which voltage yielded the electric field contour of 650 V/cm in the shape of a lemniscate of Bernoulli

5. The determined voltage was placed into the Cassini oval electronic worksheet for the same electrode geometry and the "gain denominator" was adjusted until the shape from the cassini oval matched that from the numerical solution.

6. The determined gain denominators for all values of "a" were collected and a calibration plot was made and fitted with a logarithmic trendline of:

$$\text{Gain Denominator} = 595.28 \ln(a) + 2339; \quad R^2 = 0.993 \quad (7)$$

7. The calibration trendline function shown above was incorporated back into the Cassini Oval spreadsheet. At this point, the worksheet was capable of outputting a field contour of 650 V/cm for any electrode separation distance ($\pm a$) and applied voltage (V).

8. The calibration function was then scaled to a desired electric field contour input. This allowed the analytical solution to solve for any electric field for any given a separation distance and voltage. Since the Laplace equation is linear, scaling should provide a good estimate for how other fields would look.

Table 1 incorporates all the steps above to yield a single, calibrated Cassini Oval output that analytically predicts the electric field distribution; providing a quick and simple solution for the prediction of IRE (irreversible electroporation) treatment regions that may be adjusted in real-time. The inputs are the electrode location (as a given " $\pm a$ " distance from the origin along the x-axis), the applied voltage to the energized electrode, and the desired electric field to visualize. The resulting output is a contour repre-

19

senting a threshold where the entire area within it has been subjected to an electric field the one selected; and thus treated by IRE. It is important to remember that the analytical solution was calibrated for an electric field contour of 650 V/cm, and thus yields an accurate approximation for this value. Other field strength contours of interest still yield reasonable results that mimic the overall shape of the electric field. Overall, the analytical solution provided yields consistently good predictions for electric field strengths, and thus, treatment regions of IRE that may be used during treatment planning or analysis.

A similar algorithm for calibration has also been used for a bipolar electrode and the electric field contour has been mapped its length. For example, FIG. 4 illustrates an exemplary bipolar electrode.

In one example, the diameter of the probe is 0.065 cm, and the lengths of the two electrodes are respectively 0.295 cm and 0.276 cm, separated by an insulation sleeve of 0.315 cm in length. Adapting this scenario to the cassini oval presents some challenges because the distribution is now resulting from the two exposed cylinder lengths, rather than two distinct loci of points. This was solved by calibrating individual electric field contours for the same applied voltage and developing two equations that adjust the separation distance ($\pm a$) and gain denominator (GD) according to the equations:

$$a=7*10^{-9}*E^3-2*10^{-5}*E^2+0.015*E+6.1619; \quad R^2=0.9806 \quad (8)$$

$$GD=1.0121*E+1920; \quad R^2=0.9928 \quad (9)$$

where E is the electric field magnitude contour desired. These two equations may then be used to calibrate the cassini ovals into a satisfactory shape to mimic the electric field distribution, and thus treatment region accordingly.

The Second Method

Another closed loop method determines the E-field values (electric field density) for any x and y position on the grid based on the position of the probes, the diameter of the probes, and the voltage applied between the probes. To obtain the potential, temperature or field distribution, one can determine the analytical solution for a configuration.

Since the solution to the Laplace Equation is linear, analytical solutions can be scaled and super-imposed to determine the entire distribution. For example, if two electrodes are energized and two electrodes are set to ground, the solution can be determined by adding the solutions for the two-needle electrode configuration together.

For example, for a two-needle electrode configuration, the solution is an infinite series. This can be approximated using the following equation:

$$E = \frac{V_0}{2 \cdot \log(d/a)} \left(\frac{1}{|z - r_1|} + \frac{1}{|z - r_2|} \right) \text{ where,} \quad (10)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (11)$$

$$|z - r_1| = \sqrt{(x - x_1)^2 + (y - y_1)^2} \quad (12)$$

$$|z - r_2| = \sqrt{(x - x_2)^2 + (y - y_2)^2} \quad (13)$$

V_0 =the applied Voltage (V) between the probes

a =diameter of each of the probes in meters

d =distance between the probes in meters

(x_1, y_1) =the position of the first probe

(x_2, y_2) =the position of the second probe

20

The user can then select a contour line in V/cm (i.e. 650 V/cm) based on the type of tissue which is being treated. This contour line can be used to therefore plot a boundary line of the treatment zone between two probes.

Example 2

$$(x_1, y_1) = (-0.005 \text{ m}, 0 \text{ m})$$

$$(x_2, y_2) = (0.001 \text{ m}, 0.003 \text{ m})$$

$$V_0 = 1000 \text{ V}$$

$$a = 0.0010 \text{ m}$$

$$d = 0.006708 \text{ m}$$

Using eqs. (10-13) above, the E-field values are determined for x, y coordinates on the grid, as shown in the spreadsheet at FIG. 16.

This method can also be used to determine the E-field values for devices having two plate electrodes or two concentric cylinders.

The Third Method

As an alternate method of estimating the treatment zone in real time, a predetermined set of values that define the outer boundary of a plurality of predetermined treatment zones (determined by FEA, one of the above two methods or the like) can be stored in memory as a data table and interpolation can be used to generate an actual treatment zone for a particular treatment area (e.g., tumor area).

Interpolation is commonly used to determine values that are between values in a look up table. For example if a value half way between 5 and 10 in the first row of the lookup table (see Table 3 below) needs to be determined, a single interpolation (average of 5 and 10) is done to obtain 7.5. If a value between 15, 20, 25, and 30 needs to be determined, a double interpolation is done. A first interpolation is done between 15 and 20 to obtain 17.5 and between 25 and 30 to obtain 27.5. Then, a second interpolation is done between 17.5 and 27.5 to obtain 22.5.

TABLE 3

1	5	7.5	10
11	15	17.5	20
		22.5	
21	25	27.5	30

It is to be noted that the interpolation is not limited to finding the mid point between two points. Interpolation can be done on any point between two points. For example, interpolation can be done at 15% (i.e., 15% away from one point and 85% away from the other point) and 75% (i.e., 75% away from one point and 25% away from the other point).

Numerical techniques, such as Finite Element Analysis (FEA) which was described above, Finite Difference Methods, or Boundary Element Methods can be used to generate shapes that take into account multiple variables (applied voltage, electrode separation, desired field boundary, tissue specific constants, and the like). These shapes can be stored in a multidimensional array (i.e., a multi-dimensional lookup table) in either polar or Cartesian coordinates. When a specific treatment situation occurs, an interpolation between the known shapes as represented by the lookup table can be used to generate an estimate of an estimated treatment zone.

For example, FIG. 17 illustrates a multi-dimensional lookup table and a method of interpolating treatment zones. The multi-dimensional lookup table includes for each predetermined treatment zone a table or array of points that

21

represent a particular treatment zone. For example, the top left corner of FIG. 17 illustrates a lookup table for a predetermined treatment zone for 1 cm radius tumor area at 1700 Volts/cm electric field density for a pair of electrodes.

To treat a 1.75 cm radius tumor area at 1700 Volts/cm electric field density, the contour of the treatment zone is estimated by interpolating between two nearby zones (i.e., one for 1.5 cm radius tumor area at 1700 Volts and one for 2.0 cm radius tumor area at 1700 Volts).

To treat a 1.75 cm radius tumor area at 2150 Volts/cm electric field density, the contour of the treatment zone is estimated by double interpolation. First, the treatment zones for 1.75 cm, 2000 Volts and 1.75 cm at 2300 Volts are determined. Then, the treatment zone for 1.75 cm at 2150 Volts is determined based on the interpolation results (i.e., estimated zone for 1.75 cm at 2000 Volts and estimated zone for 1.75 cm at 2300 Volts).

Automatic Probe Placement Feature

Now referring to FIG. 18, this figure illustrates an automatic probe placement feature of the treatment control module 54. If the user clicks on the "Autoset Probes" button 240 with the pointing device 14, the treatment control module 54 will automatically position the probes 201, 202, 203, 204 in the most efficient way in order to treat the target tissue 300. FIG. 18 illustrates the position of the probes 201, 202, 203, 204 after the "Autoset Probes" button 240 has been depressed with the pointing device

The automatic placement feature of the treatment control module 54 is further discussed below. This feature is carried out by the following algorithm. The algorithm functions to most efficiently place a given number of probes, which is based on the type of device which is selected in FIGS. 4-9 as discussed above (ranging from 2 to 6 probes,) in an optimal pattern to cover the defined treatment area (i.e., the combined target tissue 300 and the safety margin 301). The algorithm assumes that the combined target tissue 300 and safety margin 301 form a generally elliptical shape. This elliptical shape is determined from the dimensions of the target tissue zone (length and width), together with the desired safety margin which were inputted by the user (see FIG. 3).

The algorithm uses the following formulas to calculate the most efficient placement of each of the probes on the grid 200. The algorithm calculates the (x_i, y_i) location of each probe i on the grid 200, relative to (0,0) origin, using the following two formulas:

$$x_i = \epsilon_j * a * \cos(\theta_i + \varphi) \quad (14)$$

$$y_i = \epsilon_j * b * \sin(\theta_i + \varphi), \quad (15)$$

where, a =the major axis of the elliptical shape (cm) that is selected at FIG. 3;

b =the minor axis of the elliptical shape (cm) that is selected at FIG. 3; and;

φ =the rotational angle (degrees) of the ellipse as shown on treatment screen (see input box 251 at FIG. 11)

θ_i =the angular offset (degrees) for each probe according to Table 2.1 below:

TABLE 2.1

Total number of probes in the device	Angular Offsets (θ_i) for each probe, referenced from zero degrees (positive x-axis) of the grid
2 probes	0° and 180°
3 probes	90°, 210°, and 330°
4 probes	45°, 135°, 225°, and 315°

22

TABLE 2.1-continued

Total number of probes in the device	Angular Offsets (θ_i) for each probe, referenced from zero degrees (positive x-axis) of the grid
5 probes	90°, 162°, 234°, 306°, and 18°
6 probes	90°, 162°, 234°, 306°, and 18° PLUS the 6 th probe at the center of the grid (0, 0)

ϵ_j =the ratio of (the probe placement radius) to (the total radius to the edges of the target tissue), according to Table 2.2 below:

TABLE 2.2

Total number of probes in the device	Probe placement ratios (ϵ_j)
2 probes	$\epsilon_2 = 0.70$
3 probes	$\epsilon_3 = 0.70$
4 probes	$\epsilon_4 = 0.65$
5 probes	$\epsilon_5 = 0.65$
6 probes	$\epsilon_6 = 0.65$

The above algorithm is based on the following assumptions:

Treatment zone center is at (0,0) or will be translated to (0,0) for calculations.

Treatment zone area may or may not be adequately covered depending on size and number of probes to be deployed.

A fixed angular array of probe placements is used, with the exception of 6 probes in which the last probe is placed in the center of the target tissue at (0,0). (see Table 2.1)

A predetermined firing sequence is used according to the total number of probes. (see Table 2.3 below)

An array of ϵ_j for $j=2, 3, \dots, 6$ is used to determine the ratio of the probe placement radius from the edges of the target tissue. (see Table 2.2) The numbers are determined empirically for best-fit. Alternatively, these values can be represented as functions rather than fixed numerical values for each number of probes.

A default electric field density between probes is 1500 volts/cm which can be changed by the user. The actual voltage value between probes is adjusted based on the default electric field density. For example, if the default is set at 1500 volts/cm, the actual treatment voltage for a pair of probes that are 1.5 cm apart is 2250V.

TABLE 2.3

Total number of probes in the device	Firing Sequence of Probe Treatment Pairs, identified by polarity and specific probe number
2 probes	(1 treatment pair) (+) 1, (-) 2
3 probes	(3 treatment pairs) (+) 1, (-) 2 (+) 2, (-) 3 (+) 3, (-) 1
4 probes	(5 treatment pairs) (+) 1, (-) 2 (+) 2, (-) 3 (+) 3, (-) 4 (+) 4, (-) 1 (+) 2, (-) 4

TABLE 2.3-continued

Total number of probes in the device	Firing Sequence of Probe Treatment Pairs, identified by polarity and specific probe number
5 probes	(8 treatment pairs) (+) 1, (-) 2 (+) 2, (-) 3 (+) 3, (-) 4 (+) 4, (-) 5 (+) 5, (-) 1 (+) 2, (-) 5 (+) 1, (-) 3 (+) 4, (-) 1
6 probes	(10 treatment pairs) (+) 1, (-) 2 (+) 2, (-) 3 (+) 3, (-) 4 (+) 4, (-) 5 (+) 5, (-) 1 (+) 1, (-) 6 (+) 6, (-) 2 (+) 3, (-) 6 (+) 6, (-) 4 (+) 5, (-) 6

Example 3

A device having 3 probes is used to treat a target tissue where:

a=2.0 cm; b=1.0 cm; and $\varphi=0$ degrees

Using Table 2.1, $\theta_1=90^\circ$, $\theta_2=210^\circ$, and $\theta_3=330^\circ$

Using Table 2.2, $\epsilon_3=0.70$

Therefore, when using the “Autoset Probes” feature, and eqs. (14) and (15) above, the (x,y) locations on the grid for each probe are calculated as follows: Probe #1

$$x_1 = \epsilon_j * a * \cos(\theta_j + \varphi) = 0.70 * 2.0 \text{ cm} * \cos(90 \text{ degrees}) = 0$$

$$y_1 = \epsilon_j * b * \sin(\theta_j + \varphi) = 0.70 * 1.0 \text{ cm} * \sin(90 \text{ degrees}) = 0.70 \text{ cm}$$

Probe #2

$$x_2 = \epsilon_j * a * \cos(\theta_j + \varphi) = 0.70 * 2.0 \text{ cm} * \cos(210 \text{ degrees}) = -1.21 \text{ cm}$$

$$y_2 = \epsilon_j * b * \sin(\theta_j + \varphi) = 0.70 * 1.0 \text{ cm} * \sin(210 \text{ degrees}) = -0.35 \text{ cm}$$

Probe #3

$$x_3 = \epsilon_j * a * \cos(\theta_j + \varphi) = 0.70 * 2.0 \text{ cm} * \cos(330 \text{ degrees}) = 1.21 \text{ cm}$$

$$y_3 = \epsilon_j * b * \sin(\theta_j + \varphi) = 0.70 * 1.0 \text{ cm} * \sin(330 \text{ degrees}) = -0.35 \text{ cm}$$

Using Table 2.3, the firing sequence and respective polarity of the three probes will proceed as follows:

(3 treatment pairs)

(+) Probe #1, (-) Probe #2

(+) Probe #2, (-) Probe #3

(+) Probe #3, (-) Probe #1

In another embodiment, the automatic probe placement feature can be executed by the treatment control module 54 to reposition the probes on the grid 200 according to distance measurements which are taken from the actual position of the probes after they have been inserted into the patient.

The user is allowed to enter any or all specific distance measurements taken between any pairs of treatment probes, and may also specify which probes may be repositioned on the grid 200 by the treatment control module 54 and which

may not. The treatment control module 54 then finds the minimal error in the positions of the probes that best match the positions seen on the imaging software by the user.

It is very difficult with several probes to place them exactly on the treatment grid 200 at the proper distances that are measured on a CT or similar scan. Often times, two, three, or four probes should be moved or rotated as a group to maintain proper distances between the other probes on the treatment grid 200. This can be a frustrating, time-consuming, and error-prone method of ensuring that the probe locations on the treatment grid 200 mirror the actual probe locations in the patient's body. The positions and distances of the probes are critical in treatment planning and delivery. Furthermore, in one embodiment, the probes may only be placed at exact 1 mm locations on the treatment grid 200 so that they can easily be moved to “snap” to the grid 200, which makes the optimal placements of the individual probes even more difficult.

The main code of the software for this feature involves a “solver” algorithm which performs an iterative search based on the starting positions of the probes and the distances desired as input by the user. Some probes may be specified as “Locked” meaning that their positions are fixed relative to the grid 200. The solver moves all probes in a 1 mm×1 mm array in all possible positions and calculates the root mean square (RMS) error of the distances between the new probe locations and the desired probe locations on the grid 200. The probe positions within each probes' bounding 1 mm box that offer the minimum RMS error to the total solution are taken as the “next” iteration of the algorithm. The solver then takes this new location and re-iterates to find a new, better set of positions on the grid 200. The iterations continue until no improvement in the RMS error of the solution is found, at which point the solver quits and returns the optimal new positions that were found.

This distance placement feature will be used by the user to directly input the probe distances and cause the optimal positions of the probes based on these distances to be displayed on the treatment grid 200 with a minimum of effort and error. This will allow better treatment planning and better treatments. The distance placement feature works best when the user places the probes in “approximately” the correct starting positions on the grid 200 before running the solver algorithm.

This distance placement feature is illustrated by way of an example which is shown in FIGS. 19-22. FIG. 19 illustrates five probes 201, 202, 203, 204, 205 which form a five probe array to treat a target tissue. The five probes have been placed on the grid 200 by a user to plan the treatment of the target tissue on the “Probe Placement Process” screen. Next, the user actually inserts the five probes into the patient according to the planned locations. However, it is very difficult to actually physically place the probes at the exact same respective locations shown on the grid 200. For example, certain anatomical structures of the patient may prevent the optimal placement of the probes, e.g., the location of the target tissue with respect to the location of the patient's ribs, etc. After the user has placed the five probes in the patient, distance measurements are taken as shown in FIG. 20. These measurements represent the actual position of the five probes in the patient. One way to measure the distances between probes is to use the imaging device 30 such as an ultrasonic imaging device which allows the user to select any two point on the display device 31 to automatically measure the distance as is well-known in the art.

Next, the user clicks on a “Probe Distance Adjuster” button or the like on the screen. FIG. 21 illustrates an

25

example of a pop-up window 333 that appears which includes input boxes for entering the measurement distances taken by the user. As discussed above, the user can select which probes to “lock” on the grid 200, which will fix the location of those probes relative to the grid 200. In the present example, the user has “Locked” the position of the second probe (labelled #2) 202. After the measurement distances have been inputted into the pop-up window 333, the user clicks on the “OK” button to execute this automatic probe placement feature.

The treatment control module 54 then automatically adjusts the placement of the probes on the grid 200 which have not been “Locked” to best match the distance measurements taken. FIG. 22 shows the placement of the five probes 201, 202, 203, 204, 205 on the grid 200 after the program has been executed.

Referring back to the example shown in FIG. 18, the combined treatment region 315 does not fully cover the safety margin 301 by using a four probe device in the example. FIG. 18 illustrates the four probe device after the “Autoset Probes” button has been depressed. It should be noted that at any time, the user may move any of the probes in the grid 200. When a probe is moved on the grid 200, the treatment control module 54 automatically updates the voltage (treatment energy level) calculation in column 222 based on the distance between the probes and continuously displays the distance between the probe being moved and the other probes (see FIG. 12). The treatment control module 54 also automatically recalculates the size and boundary line 320 of the treatment zones in real time when a probe is moved on the grid 200. Also, when the maximum voltage is achieved (e.g. 3000 Volts), that number as well as its corresponding distance value are highlighted (in different color, for example, relative to the other voltage values and distance values) in column 222 to alert the user.

FIG. 26 illustrates the same target tissue 300 being treated by a six probe device after the “Autoset Probes” button has been depressed. The six probe device does a better job at covering the entire safety margin 305 surrounding the target tissue 300. With the six probe device, more rows of data appear in window 270 than compared with the four probe device discussed above because additional treatment pairs are executed. Because of the additional probes, the distance is smaller between respective probes to cover a similar ablation area. This is reflected in column 226 of window 270. This is also reflected in column 222 of window 270 which displays the voltage generated during each step of treatment. As discussed earlier, the example assumes that the maximum capability of the generator 10 is 3000 Volts. It is preferred to stay below the maximum capability of the generator if possible. Column 222 shows that with the six probe device the power delivered during each step is below 3000 Volts.

Adjusting Treatment Parameters

The treatment control module 54 allows a user to manually edit some of the numbers in window 270 in order to tailor the treatment. To edit the numbers in window 270, the user first clicks on the “Edit” icon 281 with the pointing device 14, as shown in FIG. 18. After clicking on the “Edit” icon 281, in one embodiment the treatment control module 54 can change the colors of the particular boxes in window 270 which are able to be edited. For example, as shown in FIG. 23, the treatment control module 54 can display the boxes which are able to be edited (columns 220, 221, 223, 224, 225) in a white color and can display the boxes which cannot be edited (columns 222, 226) in a grey color. Once the user has determined which box(es) to edit, the data in an

26

individual box can be edited by clicking on that particular box with the pointing device 14. After the box has been clicked with the pointing device 14, the value can be edited by either manually deleting and typing in new data with the keyboard 12, or by adjusting the number up or down by clicking on the up arrow or down arrow that appears in the box with the pointing device 14.

In the example shown in FIG. 23, the Volts/cm between probes “1” and “2” has been adjusted down from 1500 Volts/cm to 1000 Volts/cm. If a change is made to the data in window 270 which would affect the shape of the combined treatment zones 305, then the treatment control module 54 automatically adjusts the depiction of the treatment zones 305 shown in grid 200 to reflect this. In the example shown in FIG. 23, the area of the projected combined treatment region 305 has been diminished between probes “1” and “2” as shown. The ability to edit the treatment parameters as described can be particularly useful to a user in certain situations. For example, a user can edit the treatment parameters in order to avoid areas that should be preserved, such as a location of a nerve or the like. Once the user has completed making any edits of the boxes in window 270, if any, the edits are saved to the treatment control module 54 by clicking on the “Apply” icon 282 with the pointing device 14.

The treatment control module 54 allows a user to manually add additional rows or delete rows from the window 270 in order to tailor the treatment. To add rows in window 270, the user first clicks on the “+” icon 283 with the pointing device 14, as shown in FIG. 24. After clicking on the “+” icon 283, an additional row will appear at the bottom of the list in window 270, which will indicate an additional pair of probes in which treatment will occur. In the example shown in FIG. 24, an additional row has been added which indicates that treatment will occur between probe “1” and probe “3”. Referring to the grid 200, this additional treatment pair indicates the diagonal treatment across the target tissue. It should be noted that a diagonal treatment was already present between probe “2” and probe “4” as indicated in window 270. However, by adding an additional diagonal treatment, overlapping with the other treatment zones, in combination with other edits to the boxes in window 270, as described above, the user can tailor the shape of the combined treatment region 305. Whenever a row is added or deleted, the treatment control module 54 automatically updates the anticipated combined treatment zone displayed in grid 200. By comparing the grid 200 in FIG. 23 to the grid 200 in FIG. 24, the effect of adding the additional treatment row between probe “1” and probe “4” can be visually understood and appreciated. To delete rows in window 270, the user first selects which row is to be deleted by clicking to the left of the selected row with the pointing device 14. Next, the user clicks on the “-” icon 284 with the pointing device 14, as shown in FIG. 24, thereby deleting the selected row from window 270.

As discussed earlier, the user can select between a “Linear” or “Non-linear Lookup” for determining how the treatment control module 54 will calculate the actual voltage (column 222) that will be applied between each pair of probes. FIG. 25 illustrates the result when the “Non-linear Lookup” circle is selected in box 280, when compared to FIG. 18 which illustrates the same number of probes, the same probe placement, and the same Default Setting (1500 V/cm), but instead when the “Linear” circle is selected in box 280.

After the user is satisfied with the positioning of the probes of the device and the other settings according to the

features discussed above, the user clicks on the “Next” button with a pointing device **14** to proceed to the “Pulse Generation” screen described below.

FIG. **27** illustrates the status of the treatment before the treatment has been initiated. The following steps describe how the treatment is administered once the user has reached the “Pulse Generation” screen illustrated by FIG. **27**.

In FIG. **27**, the treatment control module **54** asks the user to “Click ‘Deliver test pulse’ to start” in window **420** in order to start the test signal (pulse) sequence. After the user presses the “Deliver test pulse” button **421** with the pointing device **14**, the control module **54** charges the pulse generator **10** to the test pulse voltage. When the generator **10** is charged, the control module **54** applies a test pulse for each probe pair through the generator **10**. For a 4 probe treatment, for example, a test pulse is applied through pairs 1-2, 1-3, 2-3, 2-4, 3-4 and 4-1 as shown in FIG. **11**.

In one embodiment, this test pulse voltage is approximately $\frac{1}{10}$ to $\frac{1}{5}$ the maximum treatment voltage but no lower than 200 volts and no higher than 500 volts. (It should be noted that in a preferred embodiment, valid treatment voltages are between 500 to 3000 volts.) In the embodiment shown, a test pulse of 400 volts is used for each pair of electrodes. From the test pulse, the treatment control module **54** then checks the current through a sensor **73** (see FIG. **29**) for each probe pair to determine whether a treatment current will be too low (e.g., below approximately 300 milliamperes) or too high (e.g., approximately 45 amperes or more). Based on the current, the resistance R or conductance ($1/R$) of the tissue is calculated by the module **54**. Then, the voltage to be used to actually treat the tissue (see column **222** in FIG. **23**, for example) is divided by the resistance to obtain the treatment current draw to be used in the treatment.

If the treatment current draw is determined to be too low (e.g., below 300 milliamperes), the system will give the user the option to “Proceed to Treatment” for each pair that was too low in current. If the current is determined to be too high (e.g., 45 amperes or more of threshold maximum current draw), the control module **54** will indicate an error in the display device **11** and the user should change the treatment voltage and/or re-position the mis-aligned probes to reduce the current.

The treatment control module **54** generally applies one test pulse for every pair listed in the treatment spreadsheet although more than one pulse can be applied to each pair. For example, if the user sets up treatment between pairs (1 to 2), (1 to 3) and (2 to 3) there will be three test pulses, one for each pair. There is no therapeutic value in the test pulse. The test pulse only checks the setup before full therapeutic treatment is applied. Each test pulse is intended to ensure that two conditions are met with each test pulse: first, that there is a valid connection between the selected treatment pairs, and second, that the current will not exceed the maximum output capability of the generator **10** (see FIG. **1**).

Another reason for the administration of a “test pulse” is to ensure that the patient is properly anesthetized. Prior to treatment, the patient is administered general anesthesia in combination with a paralytic agent. If the patient is not paralyzed with anesthesia, then a noticeable muscle contraction will occur during administration of the “test pulse”. Since the test pulse is at approximately 10% to 20% of the therapeutic level, any muscle contraction displayed by the patient is not as much as it would be if full energy was applied. The user should be trained to watch for muscle movement during the test pulse. In one embodiment, the treatment control module **54** can display a window which asks the user to confirm that there is no muscle movement

being displayed by the patient by selecting an answer with the pointing device **14**. In this embodiment, the treatment control module **54** will not continue to the next step unless the user presses a button with the pointing device to indicate that the patient did not display any muscle contraction during the test pulse. Irreversible electroporation (IRE) requires that a paralytic agent is given as well as the normal anesthesia. These agents tend to have a short half-life and it is easy for the patient to be under medicated at the time of treatment. If the patient is under-medicated, it is possible that the patient could be injured from the severe muscle contraction that would occur from a full power treatment without a muscle blockade. The energies delivered by IRE are similar to a defibrillation pulse and the muscle contraction would also be similar.

After these steps are completed, the system charges to the full therapeutic treatment voltage (as shown in window **430**) and waits for instructions from the user to begin treatment. In a preferred embodiment, a user is required to press both foot pedals of a double foot pedal device (not shown) in order to activate treatment (the first pedal is used to arm the generator **10**, the second pedal is used to fire or start the treatment). This provides a type of safety check and prevents accidental activation of the treatment. For illustration purposes, the screen shown in FIG. **27** uses two buttons **422**, **423** instead of a double foot pedal device. Accordingly, the user will click on the “Arm” button **422** with the pointing device **14** to arm the probes. Then, the user will click on the “Pulse” button **423** with the pointing device **14** to initiate the treatment.

As shown in FIG. **28**, after the treatment has been initiated, the treatment control module **54** controls the generator **10** and administers a series of pulses according to the predetermined instructions illustrated in columns **220**, **221**, **222**, **223**, **224**. During each step of the treatment, column **401** illustrates the number of pulses that have been delivered in real time until the total number of predetermined pulses has been delivered. Column **402** displays the status percentage of the treatment for each pair of electrodes. The treatment process runs until each step of the probe firing sequence has been accomplished. Audible beeps are generated during the treatment to track the operation of the generator **10**. Window **430** displays the status of the charge of the generator **10** during operation. Window **286** displays the total “Pulse progress” of the treatment. Window **420** displays further details of the treatment progress.

The treatment control module **54** can include a feature that prevents the generator from exceeding its maximum current limit by reading the current every ten pulses and reducing the voltage by a predetermined percentage (e.g., 5% or 10%) if it approaches the maximum limit of the generator.

FIG. **29** illustrates one embodiment of a circuitry to detect an abnormality in the applied pulses such as a high current, low current, high voltage or low voltage condition. This circuitry is located within the generator **10** (see FIG. **1**). A USB connection **52** carries instructions from the user computer **40** to a controller **71**. The controller can be a computer similar to the computer **40** as shown in FIG. **2**. The controller **71** can include a processor, ASIC (application-specific integrated circuit), microcontroller or wired logic. The controller **71** then sends the instructions to a pulse generation circuit **72**. The pulse generation circuit **72** generates the pulses and sends electrical energy to the probes. For clarity, only one pair of probes/electrodes are shown. However, the generator **10** can accommodate any number of probes/electrodes (e.g., 6 probes as shown in FIG. **4**). In the

embodiment shown, the pulses are applied one pair of electrodes at a time, and then switched to another pair. The pulse generation circuit 72 includes a switch, preferably an electronic switch, which switches the probe pairs based on the instructions received from the computer 40. A sensor 73 such as a sensor can sense the current or voltage between each pair of the probes in real time and communicate such information to the controller 71, which in turn, communicates the information to the computer 40. If the sensor 73 detects an abnormal condition during treatment such as a high current or low current condition, then it will communicate with the controller 71 and the computer 40 which may cause the controller to send a signal to the pulse generation circuit 72 to discontinue the pulses for that particular pair of probes.

The treatment control module 54 can further include a feature that tracks the treatment progress and provides the user with an option to automatically retreat for low or missing pulses, or over-current pulses (see discussion below). Also, if the generator stops prematurely for any reason, the treatment control module 54 can restart at the same point where it terminated, and administer the missing treatment pulses as part of the same treatment.

In other embodiments, the treatment control module 54 is able to detect certain errors during treatment, which include, but are not limited to, "charge failure", "hardware failure", "high current failure", and "low current failure".

The following discussion relates to an example of a "high current failure". Referring to FIG. 30, a "high current" failure occurred during treatment between probe "1" and probe "2". As can be seen in column 401, the total number of pulses that were delivered between probe "1" and probe "2" was 20 instead of 90. This is because a "high current" condition occurred sometime after the 20th pulse was delivered. The treatment control module 54 was able to react to this error by discontinuing the remainder of the pulses between probe "1" and probe "2".

During treatment energy stored on capacitors acts like a constant voltage source. It is not an ideal source and there is some drift of the applied voltage but it is close. What tends to occur during IRE treatment is as the cells porate, the intracellular fluid moves to the extracellular space. Since the intra cellular fluid is more conductive than the bulk tissue the overall resistance of the system decreases. Given the approximately constant voltage source when resistance goes down current goes up ($V=IR$). During treatment the system constantly monitors the energy being delivered. If the voltage is too high or too low the therapy is aborted because the primary variable controlling poration is the voltage applied and the geometry the voltage is applied to. The system also monitors the current delivered and ensures that for patient safety reasons and for hardware reliability the maximum current capabilities of the system are not exceeded. Low currents are also detected as a sign of poor connection to the patient.

Any time current flows the tissue will heat up. For IRE the system is trying to deliver as much energy as it can without significant thermal effects. If the current was allowed to run uncontrolled, then thermal damage could occur. Also the components in the system would fail at some point if the system allowed unlimited amounts of current to flow.

After the treatment control module 54 has completed treatment for all probe pairs, column 402 displays whether the treatment was successful for each step of the treatment process by indicating a checkmark, or other indicia, if the step was successful and a lightning bolt, or other indicia, if the step encountered an error. In the example shown in FIG.

30, the treatment between probe "1" and probe "2" indicates a "High Current" error in column 402 as discussed above. The treatment control module 54 tracks which pairs of probes have failed and automatically asks the user whether to "Continue Procedure" (by pressing button 426) or "Stop Procedure" (by pressing button 427). In this example, then the user chooses one of the following three options. 1) Accept the treatment as is (by pressing button 427). For example, if 89 of 90 pulses were properly delivered that may be acceptable. 2) Use the automated reduce and reapply option in the treatment control module 54 (by pressing button 426). This will lower the voltage and the corresponding current and will provide some level of treatment. 3) Reposition the probes to be further apart. This will increase the resistance of the system.

If the user clicks on the "Continue Procedure" button 426, then as shown in FIG. 31, a dialogue box 428 will automatically pop up, which asks the user whether to "Adjust voltages for high current segments?" The user answers by clicking "Yes", "No", or "Cancel". If the user clicks on the "Yes" button, then the treatment control module 54 will automatically reduce the treatment voltage by a predetermined percentage (e.g. 5% which can be set or changed by the user). If the user clicks on the "No" button, then the treatment control module 54 will keep the treatment voltage the same. Next, the treatment control module 54 will go back to the arm ready state. The user can then activate the treatment to re-treat only the missing pulses. This high-current detection and reapplication feature is particularly advantageous because (1) the software remembers which pairs have failed so that the user does not have to remember and (2) the treatment control module 54 is able to accurately keep track of which pulses were unsuccessful so that only the missing pulses are re-administered.

FIG. 32 illustrates the "Pulse Generation" screen during re-treatment for the example. Again the pulse progress bar 286 is displayed and the status column 402 is updated again in real time. After the re-treatment has completed, the user will verify that the status of each probe pair is complete by checking column 402.

At any time, the user can click on the "Result Graphs" tab 500 to view the complete voltage (V) results of the treatment vs. time, and the complete current (A) results of the treatment vs. time. FIG. 33 illustrates such result graphs according to the results of the treatment from the example.

In the embodiment shown, a plurality of sets of pulses are applied, and more specifically 9 sets of 10 pulses per set are applied with each pulse having a pulse duration of 100 microseconds.

In the illustrations of FIGS. 33-34, the inter pulse spacing is not to scale. The inter pulse spacing is calculated based on the pulses per minute (PPM) which was selected on the "Information" screen (see FIG. 3). The time between sets of pulses is about 3.5 seconds and can be a function of how long the capacitors need to charge. In another embodiment, the time between sets of pulses is less than 3.5 seconds or is completely eliminated.

The user can click the chart to change the zoom level of the result graphs. FIG. 34 illustrates the graph results after the user has clicked the chart to zoom in on the results. The user can further zoom in by clicking the voltage or current chart to show 10 pulses for a treatment probe pair. The graph results shown in FIGS. 33-34 are the results of a demonstration mode of the treatment control module 54. It should be understood that during a real world treatment, the graph

31

results would be less uniform. The shape of the pulses can also be used as an indicator of the degree of poration of the cells.

FIG. 35 is a screen shot of a "Probe Placement Process" screen of the treatment control module showing the probe placement grid 200 after treatment has been delivered by a four probe array. The treated area 339 is saved by the module 54 in the memory 44 and is shown in a highlighted contrasting manner to the displayed target region 301 and the treatment region 305 defined by the probes. In FIG. 35, the treated area 339 is filled in with cross-hatched lines that are marked in any other way for distinguishing this area on the grid 200. In one embodiment, the treated area 339 is displayed in a different color to more easily distinguish it. This feature allows a user to plan for additional treatment of the target tissue 300 which is surrounded by the safety margin 301. This feature is especially useful when the target tissue (treatment target area) requires more than one round of treatment to effectively cover the entire area.

Although the present treatment method has been discussed in relation to irreversible electroporation (IRE), the principles of this invention can be applied to any other method where therapeutic energy is applied at more than one point. For example, other methods can include reversible electroporation, supraporation, RF ablation, cryo-ablation, microwave ablation, etc. "Supraporation" uses much higher voltages and currents, in comparison to electroporation, but with shorter pulse widths.

In addition to the example parameters described above, specific electro-medical applications of this technology include reversible electroporation as well as irreversible electroporation. This could include reversible or irreversible damage to the external cell membranes or membranes of the organelles, or damage to individual cellular structures such as mitochondrion so as to affect cellular metabolism or homeostasis of voltage or ion levels. Example embodiments for reversible electroporation can involve 1-8 pulses with a field strength of 1-100 V/cm. Other embodiments altering cellular structures adversely involve generators having a voltage range of 100 kV-300 kV operating with nano-second pulses with a maximum field strength of 2,000 V/cm to and in excess of 20,000 V/cm between electrodes. Certain embodiments can involve between 1-15 pulses between 5 microseconds and 62,000 milliseconds, while others can involve pulses of 75 microseconds to 20,000 milliseconds. In certain embodiments the electric field density for the treatment is from 100 Volts per centimeter (V/cm) to 7,000 V/cm, while in other embodiments the density is 200 to 2000 V/cm as well as from 300 V/cm to 1000 V/cm. Yet additional embodiments have a maximum field strength density between electrodes of 250 V/cm to 500 V/cm. The number of pulses can vary. In certain embodiments the number of pulses is from 1 to 100 pulses. In one embodiment, as described herein, between about 10 pulses and about 100 pulses can be applied at about 2,000 V/cm to about 3,000 V/cm with a pulse width of about 10 μ sec to about 50 μ sec. After applying these pulses, a predetermined time delay of from about 1 seconds to about 10 minutes can optionally be commenced in order that intra-cellular contents and extra-cellular contents of the target tissue cells can mix. This procedure can be repeated, as necessary, until a conductivity change is measured in the tissue. Following this step, about 1 pulse to about 300 pulses of about 2,000 V/cm to about 3,000 V/cm can be applied with a pulse width of about 70 psec to about 100 μ sec to widely ablate the tissue. This last step can be repeated until a desired number of ablation pulses is delivered to the tissue, for example, in the range of

32

about 10 pulses to about 300 pulses, more particularly, about 100 pulses. In other embodiments, groups of 1 to 100 pulses (here groups of pulses are also called pulse-trains) are applied in succession following a gap of time. In certain embodiments the gap of time between groups of pulses can be from about 0.5 second to about 10 seconds.

Therapeutic energy delivery devices disclosed herein are designed for tissue destruction in general, such as resection, excision, coagulation, disruption, denaturation, and ablation, and are applicable in a variety of surgical procedures, including but not limited to open surgeries, minimally invasive surgeries (e.g., laparoscopic surgeries, endoscopic surgeries, surgeries through natural body orifices), thermal ablation surgeries, non-thermal surgeries, as well as other procedures known to one of ordinary skill in the art. The devices may be designed as disposables or for repeated uses.

In yet another embodiment, the electrodes can be adapted to administer electrical pulses as necessary in order to reversibly or irreversibly electroporate the cell membranes of infectious cells located along the outer surface of an implanted medical device within a treatment zone, thereby treating a patient and sterilizing the medical device. By varying parameters of voltage, the number of electrical pulses, and pulse duration, the electrical field will either produce irreversible or reversible electroporation of the infectious cells that cover the implanted medical device. The pulse generator of the present invention can be designed to deliver a range of different voltages, currents and duration of pulses as well as number of pulses. Typical ranges include but are not limited to a voltage level of between 100-3000 volts, a pulse duration of between 20-200 microseconds (more preferably 50-100 microseconds), and multiple sets of pulses (e.g. 2-5 sets) of about 2-25 pulses per set and between 10 and -500 total pulses. The pulse generator can administer a current in a range of from about 2,000 V/cm to about 6,000 V/cm. The pulse generator can provide pulses which are at a specific known duration and with a specific amount of current. For example, the pulse generator can be designed upon activation to provide 10 pulses for 100 microseconds each providing a current of 3,800 V/cm $\pm$ 50% $\pm$ 25%, $\pm$ 10%, $\pm$ 5%. Plastic and/or other types of metallic devices can be sterilized along the outer surface of the device, as electric field lines from the electrodes can travel along the outer surface of devices. The electroporation treatment zone is defined by mapping the electrical field that is created by the electrical pulses between two electrodes.

As illustrated in FIGS. 36A and 36B, a method of treating infectious cell growth 33 on an implanted medical device, such as a cartilaginous implant 35, using IRE is disclosed. Although three (3) probes are illustrated in the method, any number of suitable probes can be used to deliver the electrical pulses. When electrical pulses are administered within the irreversible parameter ranges, as described above, permanent pore formation occurs in the cellular membrane, resulting in cell death of the targeted infectious cells 33. In another aspect, by proactively administering the electrical pulses according to a predetermined schedule, infectious cell growth 33 on an implanted medical device can be prevented altogether. Alternatively, electrical pulses may be administered within a reversible electroporation range. Temporary pores will form in the cellular membranes of infectious cells.

The voltage pulse generator 10 can be configured to generate electrical pulses between electrodes 22 in an amount which is sufficient to induce irreversible electroporation of infectious cells that may be present on or near an implanted medical device without creating a clinically significant thermal effect to the surrounding tissue or critical

33

structures. Specifically, the electrical pulses can create permanent openings in infectious cells of smooth muscle cells, for example, thereby invoking cell death without creating a clinically significant thermal effect. The infectious cells 33 will remain in situ and can be subsequently removed by natural body processes. These infectious cells can be located anywhere along the implantable medical device. Research has also shown that growths of infectious cells can enhance implanted medical device-related bacteremia by providing an interface for adherence and colonization. These pathogens may then produce a "biofilm" which is impenetrable to systemic antibiotics leading to a cause of implanted medical device dysfunction, subsequent removal, and the attendant increases in morbidity and mortality. These biofilms can also lead to chronic microbial infection, inflammation, and tissue necrosis. These infections are frequently due to gram-positive bacteria, such as *Staphylococcus epidermidis* or *Escherichia coli* in the case of catheter-associated or other plastic-types of medical devices and *Staphylococcus aureus* in the case of metallic implants. However, other types of bacteria could be present as well, such as, but not limited to, *Streptococcus mutans*, *Streptococcus mitis*, *Streptococcus salivarius*, and *Enterobacter aerogenes*. Initial infection is always linked to the ability of the bacteria to adhere to material surfaces. At first the bacterial adhesion is reversible, but then later becomes irreversible. Over time, the individual microorganisms and bacterial matrix formation can develop a high resistance to antibiotics.

As illustrated in FIG. 36B, as electrical pulses are delivered to the target implanted device, a series of electrical field lines can be generated. The strongest (defined as volts/cm) electrical field is nearest to the electrodes 22 and is depicted by gradient line 9. As the distance away from the electrodes 22 increases, the strength of the electrical field decreases. Outer gradient line 13 represents the outer perimeter of electric field gradient. As an example, any infectious cells 33 or other bio-film growth on the surface of the implantable medical device 35 within the outer perimeter 13 will undergo cell death by irreversible electroporation. Because the voltage pulse generation pattern from the generator does not generate damaging thermal effect, and because the voltage pulses only ablate living cells, the treatment does not damage blood, blood vessels or other non-cellular or non-living materials such as the implantable device itself.

Referring again to FIG. 36B, the pulse parameters that characterize the field gradient line can be adjusted to vary the treatment zone according to the location of the infectious cells growth and/or infectious cells to be destroyed. Furthermore, in some embodiments of the invention, the electrodes can be positioned at any location necessary to destroy any such infectious cells 33 that have grown around or on the indwelling implanted medical device 35. For example, the electrodes 22 can be positioned at a proximal section of the implanted device for treating infectious cells that have grown around the implanted device, at a distal section of the implanted device, or surrounding the implanted medical device. In addition, the electrodes 22 can be positioned to destroy infectious cells 33 that have grown near the insertion site of the implanted device.

In another aspect of the invention, by periodically administering the electrical pulses according to a predetermined schedule, infectious cell growth on the implanted device can be prevented altogether. As an example, the formation of infectious cells may occur as early as 24 hours after implanted medical device implantation. Application of electrical pulses applied to the implanted medical device at

34

regular intervals post-implantation may be effective in preventing infectious cell growth during the implanted medical device implantation period.

FIG. 37 illustrates the procedural steps associated with performing irreversible or reversible electroporation treatment using any of the probe devices described herein. After the infectious cell formation 33 has been detected and the location determined using ultrasound or fluoroscopic imaging, electrode probes 22, as illustrated in FIG. 36A, are inserted into the patient's skin 27 such that the probes can be near or in contact with the infectious cell growth on the implanted medical device. The probes 22 are then positioned relative to the infectious cells' 33 location as previously described. The electrodes 22 can be deployed and positioned outside of the implanted medical device 35 as shown in FIGS. 36A and 36B. Electrical connectors from each probe are connected (801) to generator 10 using an extension cable. This completes an electrical circuit between the electrodes 22 and the generator 10. Electrical pulses are applied across the electrodes 22 in the desired pattern (802) to electroporate the smooth muscle cells of the infectious cells 17. Electrical pulses are then applied across the electrodes 22 creating field gradient lines 9 and 13, for example, as illustrated in FIG. 36B, sufficient to non-thermally electroporate the infectious cells. As illustrated in FIG. 37, after treatment, the extension cable is disconnected from the electrical connector (805) and the probes are removed from the tissue. Non-thermal death of the infectious cells 33 will occur within the first twenty-four hours after electroporation treatment followed by a cellular breakdown of the infectious cells.

In one embodiment, the electroporation pulses can be synchronously matched to specifically repeatable phases of the cardiac cycle to protect cardiac cellular functioning. See, for example, U.S. Patent Application No. 61/181,727, filed May 28, 2009, entitled "Algorithm For Synchronizing Energy Delivery To The Cardiac Rhythm", which is fully incorporated by reference herein. This feature is especially useful when the electroporation pulses are delivered in a location that is near the heart. This can be especially useful if a user is attempting to treat infection near an implanted medical device such as a pacemaker. FIG. 38 illustrates a treatment setup for a patient for synchronization of the delivery of electroporation pulses with a specific portion of the cardiac rhythm. Electrocardiogram (ECG) leads 17, 19, 21 are adapted to be attached to the patient for receiving electrical signals which are generated by the patient's cardiac cycle. The ECG leads transmit the ECG electrical signals to an electrocardiogram unit 29. The electrocardiogram unit 29 can transmit this information to a synchronization device 25 which can include hardware or software to interpret ECG data. If the synchronization device 25 determines that it is safe to deliver electroporation pulses, it sends a control signal to a pulse generator 10. The pulse generator 10 is adapted to connect to the electrical connector for delivering electroporation pulses. Each of the synchronization device 25 and pulse generator 10 can be implemented in a computer so that they can be programmed.

The present invention affords several advantages. Infectious cell growths 33 are destroyed without having to remove the implanted medical device 35 from the patient. The treatment is minimally-invasive and highly efficacious. Because irreversible electroporation does not create thermal activity, the implanted medical device 35 is not damaged by the treatment. Infectious cells growths 33 are treated quickly, and the implanted medical devices 35 can be

maintained according to a predetermined schedule to insure that they are routinely treated with IRE to remove the infected cells 33.

Although the irreversible electroporation device and method has been described herein for use with cartilaginous implanted medical devices, it should be understood that the irreversible electroporation device can be used with any type of implanted medical device 35, such as, but not limited to, ports, catheters, stents, artificial cartilaginous implants, orthopedic prosthetics, pacemakers, PICCs, hip implants, tooth implants, heart valves, spinal implants, and other types of plastic and/or metallic devices, to name a few. While the embodiments shown use pulses that cause IRE, persons of ordinary skill in the art will appreciate that other types of pulses can be used for the destruction of the infectious cell growths 33. In particular, ultrashort sub-microsecond pulses (pulses of less than 1 microsecond in duration) can be used to induce apoptosis that cause damage to the intracellular structures such as a cell nucleus.

In yet another embodiment, the method of IRE treatment described herein could be used to treat one or more parts of a patient's body that are infected due to some other infection, such as, for example, gangrene. Gangrene is particularly problematic in patients with poor blood flow, such as diabetic patients, particularly in certain parts of the body, such as the extremities (i.e., hands, feet, nose, ears, etc.). Gangrene is a type of necrotizing (flesh-destroying) bacterial infection. Typically, a physician prescribes antibiotics to a patient to treat gangrene. However, poor circulation can prevent certain drugs from reaching these extremities, thereby making patients more susceptible to infection in these areas. Gangrene often spreads so quickly that it can't be stopped by antibiotics alone. In some cases, the gangrene can cause lesions in the skin. At most, antibiotics may help slow the infection and clear small, unnoticed pockets of bacteria. A surgeon may still have to debride or cut away the infected flesh and a margin around it to prevent further spread. If debridement isn't enough, amputation is the next step. During this procedure the patient is put in a hyperbaric chamber if one is available. This is a sealed metal tank pumped full of oxygen under high pressure. The pressure forces oxygen into the tissues, stopping the spread of anaerobic bacteria. All of these methods have pitfalls. For example, patients can build up a resistance to antibiotics, as described above. Surgery to excise infected flesh can be expensive, time-consuming, and most importantly, painful for the patient. As an alternative, IRE methods of treatment, as described above using any of the pulse parameters disclosed herein, could be used in such tissues to effectively kill gangrene tissue, allowing healthy tissue to regenerate, thereby eliminating the need for amputation.

In yet another aspect, IRE could be used to treat infected cells present on the patient's tissue scaffold. A patient's own natural tissues could be used as a scaffold. The IRE electrical pulses can effectively de-cellularize the tissue scaffold in a desired target tissue zone of infectious cells 33, while leaving the tissue scaffold intact. After IRE is performed on or near the tissue scaffold, then the patient's tissue can naturally re-cellularize onto the patient's natural tissue scaffold after IRE has been performed. This method allows vascular, neural, and other critical structures to remain intact and essentially unharmed. However, even if the IRE pulses target collateral structures, this method would still be a safe alternative to other forms of cellular ablation or decellularization such as physical, chemical, or enzymatic methods of decellularization. Such treatments can be very harsh and can degrade the tissue scaffold as well as the vasculature and

neural structures. Furthermore, such methods are often insufficient by themselves to achieve decellularization alone, and often have to be combined with each other to achieve the desired result. These methods can also compromise the structure of the extracellular matrix (ECM).

In contrast, the use of IRE to decellularize infected tissue can provide a controlled, precise way to destroy cells of a tissue or organ, while leaving the underlying extracellular matrix intact, including the vascular and neural structures as well as other critical features intact. IRE can also prevent scar tissue formation as well as promote a beneficial immune response. The method could involve treating in vitro or in vivo, a tissue comprising infectious cells located on an underlying scaffold with electrical pulses of sufficient duration and power to kill infectious cells of the tissue, while leaving the extracellular matrix and other vascular and neural structures intact. As used herein, the term "intact" means a state of being whereby an element is capable of performing its original function to a substantial extent.

In one aspect, any of the IRE methods and parameters described herein can be used to deliver electrical pulses to the infectious cells present on the tissue scaffold. Any suitable frequency of pulsing that allows for irreversible electroporation without significant thermal damage to the tissue is acceptable. In one embodiment, the electrodes 22 described herein can be placed into or near the vicinity of the tissue scaffold to cause irreversible electroporation of the infectious cells on or near the tissue scaffold. The electrical pulses irreversibly permeabilize the membranes of the infectious cells, thereby invoking cell death. The length of time of the electrical pulses, the voltage applied, and the resulting membrane permeability are all controlled within desired ranges. The debris left by the infectious cells may be left in situ and may be removed by natural processes, such as the body's own immune or circulation system.

In another exemplary embodiment, IRE could be used to ablate cysts in the abdomen. A cyst is an overgrowth of the epithelium, comprising specialized cells that reside on the surface of tissue, such as organs and glands. Once formed, they often become detached from the point of origin and circulate, although they may also become lodged between tissue structures and cause pressure. They generally appear as sacs or lumps surrounded by a thin membrane and consist of fluid or semi-solid material. While most cysts are benign, the development of an abdominal cyst may signal an underlying disease.

There are several types of abdominal cysts. One of the most common is an ovarian cyst, which forms on ovarian follicles. Even though most ovarian cysts are benign, they can cause pain and bleeding. Standard treatment to remove cysts involves surgical removal if they become too large. Mesenteric cysts are another type of abdominal cyst that may indicate impaired lymphatic functioning. These types of cysts develop in the mesentery, the area of the peritoneum that encompasses the gastrointestinal tract and extends from the duodenum to the rectum. In addition, these cysts may involve any of the organs of the retroperitoneum, such as the bladder and kidneys. Omental cysts generally occur in the anterior abdominal wall in the regions of the stomach and colon. These types of cysts can usually be removed without having to resection the stomach or colon. While most abdominal cysts are not dangerous, they can sometimes grow from the size of a pea to a grapefruit over time and can sometimes present complications. Abdominal cysts can also cause pain, unexplained bleeding, bowel obstruction, and distension of the stomach or other organs, should be investigated without delay. Typical treatment modalities to

37

remove cysts include surgical measure. However, surgery can be time-consuming, painful, and expensive.

In one aspect, IRE treatment methods, as described herein, can be used as a minimally invasive and safe procedure to ablate and remove benign abdominal cysts as well as abdominal cancer cells. IRE methods can be more efficient, cheaper, and less painful compared to surgical procedures and would leave critical structures intact, such as vascular structures, etc. Even if the collateral structures are affected, using non-thermal IRE as a treatment method for removal of abdominal cysts would provide a safe treatment alternative to thermal or surgical removal methods that can be time-consuming, costly, and painful for the patient or the use of drugs that can have harmful side effects.

Another example of infected tissue that can be treated using IRE is infected bone, or osteomyelitis. Bone infections can be extremely difficult to treat. Typically, bone infections can be treated using surgical procedures. The bone can be accessed by variety of procedures, such as through the skin. After the bone is surgically cleaned out, the remaining bone defect(s) is treated with a large dose of antibiotics via a non-resorbable bone cement to eradicate any bacterial cells in the bone and bloodstream. After this, subsequent surgery is required for removal and replacement with a bone graft or an absorbable mix of synthetic bone substitute. After all of these bone cleaning and replacements, the bone is typically not strong enough to bear weight. Bone rebuilding techniques can involve bone grafting or bone transport. Antibiotic treatment is then administered through an intravenous catheter. These treatment procedures have the attendant disadvantages mentioned above.

Instead of using the above-described extensive, painful, and expensive procedures, IRE can be used to treat bone infection. In one aspect, sufficient electrical pulse parameters can be selected, as described herein above, to irreversibly electroporate infected cells that are present within or along bone. In one aspect, the electrical probes described herein can be inserted into a target tissue surrounding an infected bone, and sufficient electrical pulse parameters could be selected to adequately irreversibly electroporate an infected bone. In one embodiment, an outer layer of bone could be treated to remove infected cells. When infected tissue of a bone is irreversibly electroporated, such target bone tissue could include muscle and/or vessels which could be acutely necrosed. However, in time, the critical cellular and/or vascular structures could grow back so that no long term harmful consequences would occur.

IRE could also be used to sterilize implanted medical devices using any of methods comprising any of the electrical pulse parameters described herein. Implanted medical devices such as metallic devices could be sterilized using the device itself as a conductor. In one aspect, the metallic device could be comprised of stainless steel, cobalt-chromium-molybdenum alloys, pure titanium, or titanium alloys. The implanted medical device, functioning as a conductor, could be placed between two electrodes, for example, thereby effectively functioning as another electrode. In another embodiment, a plurality of electrodes could be used and placed surrounding the implanted medical device. A practitioner can position the electrodes a predetermined distance from the implant so that the implanted medical device can function like a return electrode. When IRE is performed, the electric field lines 9 through 13 could travel from each electrode to the outer surface of the implant, thereby allowing the implant to effectively become part of the sterilization.

38

In summary, the use of IRE to sterilize such medical devices and/or treat infected tissue can be an effective alternative to RF, cryo, or other scar-prone surgical procedures. This method can be used proactively to ensure that deep-seated infection will not occur post-operatively. This would prevent tissues from being permanently scarred. This method is beneficial because after IRE is performed, healthy tissue will grow instead of scar tissue. The use of this procedure could also provide better long term results for patients and would eliminate the need to have patients endure multiple surgeries and lengthy healing times.

The above disclosure is intended to be illustrative and not exhaustive. This description will suggest many modifications, variations, and alternatives may be made by ordinary skill in this art without departing from the scope of the invention. Those familiar with the art may recognize other equivalents to the specific embodiments described herein. Accordingly, the scope of the invention is not limited to the foregoing specification.

What is claimed:

1. A method of treating infected tissue of a patient comprising:

placing at least one electrode near a treatment site that comprises the infected tissue;

activating a generator to generate electrical pulses through the at least one electrode;

wherein the electrical pulses are sufficient to non-thermally ablate the infected tissue while leaving intact a natural tissue scaffold of the patient within the treatment site;

wherein the treatment site further comprises a pre-placed return electrode, wherein the pre-placed return electrode includes a medical device having a conductive material.

2. The method of claim 1, wherein the infected tissue at least partially surrounds the pre-placed medical device.

3. The method of claim 1, wherein the pre-placed medical device comprises at least one of: a catheter, an artificial cartilaginous implant, a pacemaker, an implantable port, a stent, orthopedic prosthetic, PICC, implantable valve, spinal implant, tooth implant, and a filter.

4. The method of claim 1, wherein an outer surface of the pre-placed medical device is at least partially covered by infectious cells.

5. The method of claim 4, wherein the infectious cells comprise bacterial cells selected from the group comprising: bacteremia, *Staphylococcus epidermidis*, *Escherichia coli*, *Staphylococcus aureus*, *Streptococcus Mutans*, *Streptococcus Mitus*, *Streptococcus Salivarius*, and *Enterobacter Aero-genes*.

6. The method of claim 1, wherein the electrical pulses are sufficient to irreversibly electroporate the infected tissue.

7. The method of claim 1, wherein the electrical pulses are sufficient to promote an immune response in the patient.

8. The method of claim 1, wherein the electrical pulses are sufficient to promote re-cellularization on the tissue scaffold.

9. The method of claim 1, wherein the electrical pulses reduce a likelihood of scar tissue forming within the treatment site.

10. The method of claim 1, wherein the electrical pulses decrease a likelihood of causing irreversible damage to blood vessels within the treatment site.

11. The method of claim 1, wherein the infected tissue comprises bone.

12. The method of claim 1, wherein the infected tissue comprises gangrene.

39

13. A method of decellularizing infected tissue comprising:

placing a probe near a treatment site comprising the infected tissue, wherein the probe comprises at least one electrode;

activating a generator that is operatively coupled to the probe to generate electrical pulses to be delivered through the at least one electrode that are sufficient to non-thermally ablate cells of the infected tissue leaving intact an underlying extracellular matrix of the infected tissue, wherein the electrical pulses are sufficient to irreversibly electroporate the infected tissue;

wherein the treatment site further comprises a pre-placed return electrode including a pre-placed medical device having a conductive material, wherein the infected tissue at least partially surrounds the pre-placed medical device.

14. The method of claim 13, wherein the electrical pulses are sufficient to promote an immune response in a patient.

15. The method of claim 13, wherein the electrical pulses decrease a likelihood of causing irreversible damage to blood vessels within the treatment site.

16. A method of treating a patient with an infected tissue site comprising:

40

placing at least two electrodes near the infected tissue site comprising an infected tissue including a prostate tissue;

activating a generator operatively coupled to the at least two electrodes to deliver electrical pulses between the at least two electrodes, wherein the electrical pulses are configured to:

non-thermally ablate the infected tissue;

leave intact a patient specific natural tissue scaffold within the infected tissue site; and

promote an immune response within the patient;

wherein the infected tissue site further comprises a return electrode including an implanted conductive medical device.

17. The method of claim 1, wherein the at least one electrode comprises at least two electrodes.

18. The method of claim 1, wherein the pre-placed return electrode includes an implanted medical device and the infected tissue at least partially surrounds the implanted medical device.

19. The method of claim 1, wherein the infected tissue comprises a tumor.

* * * * *